

Vector Field Guidance for Autonomous Vehicles

Tammineni Harinarayana

Vector Field Guidance for Autonomous Vehicles

*Thesis submitted to
Indian Institute of Technology Kharagpur
for the award of the degree*

of

Doctor of Philosophy

by

Tammineni Harinarayana

under the guidance of

Prof. Sikha Hota

(Department of Aerospace Engineering)



**DEPARTMENT OF AEROSPACE ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
August 2023**

©2023 T Harinarayana. All rights reserved.

To my family

CERTIFICATE OF APPROVAL

Certified that the thesis entitled **Vector Field Guidance for Autonomous Vehicles**, submitted by **Tammineni Harinarayana** to the Indian Institute of Technology Kharagpur, for the award of the degree of Doctor of Philosophy has been accepted by the external examiners and that the student has successfully defended the thesis in the viva-voce examination held today.

Prof. Naba Kumar Peyada
Dept. of Aerospace Engineering
IIT Kharagpur
(Member of the DSC)

Prof. Susmita Bhattacharyya
Dept. of Aerospace Engineering
IIT Kharagpur
(Member of the DSC)

Prof. Sourav Patra
Dept. of Electrical Engineering
IIT Kharagpur
(Member of the DSC)

Prof. Sikha Hota
Dept. of Aerospace Engineering
IIT Kharagpur
(Member of the DSC)

Prof. Manoranjan Sinha
Dept. of Aerospace Engineering
IIT Kharagpur
(Chairman DSC)

Prof.
(External Examiner)



Department of Aerospace Engineering
Indian Institute of Technology Kharagpur
Kharagpur, India-721302

Certificate

This is to certify that this thesis entitled **Vector Field Guidance for Autonomous Vehicles** submitted by **Tammineni Harinarayana**, to the Indian Institute of Technology Kharagpur, is a record of bona fide research work carried out under our supervision and is worthy of consideration for award of Ph.D of the Institute.

Prof. Sikha Hota

Department of Aerospace Engineering
Indian Institute of Technology Kharagpur
India - 721302.

Declaration

I certify that

- a. the work contained in the thesis is original and has been done by me under the guidance of my supervisor;
- b. the work has not been submitted to any other institute for any other degree or diploma;
- c. I have followed the guidelines provided by the Institute in preparing the thesis;
- d. I have conformed to ethical norms and guidelines while writing the thesis;
- e. whenever I have used materials (data, models, figures and text) from other sources, I have given due credit to them by citing them in the text of the thesis, and giving their details in the references, and taken permission from the copyright owners of the sources, whenever necessary.

Tammineni Harinarayana

Acknowledgment

I am deeply grateful to my advisor, Prof. Sikha Hota, for her invaluable guidance and encouragement during my Ph.D. journey. Working under her supervision has been a privilege.

I extend my heartfelt thanks to the members of my Doctoral Scrutiny Committee (DSC): Prof. Naba Kumar Peyada and Prof. Susmita Bhattacharyya from the Department of Aerospace Engineering, and Prof. Sourav Patra from the Department of Electrical Engineering. Their valuable suggestions and critical feedback greatly contributed to my research. I also thank Prof. M. Sinha, the Head of the Department of Aerospace Engineering, for his support.

I would like to thank to my colleagues, Nikhil Kumar Singh, Soumitra Ray, Sayantan Pal, and Biswajit Tripathy, for their friendly and helpful demeanor.

I am thankful to all my friends for sharing memorable moments with me.

I am truly grateful to the wonderful and supportive family of my supervisor, whose positive vibes and encouragement have been a source of inspiration throughout my journey.

Finally, my deepest appreciation goes to my family, especially my mother Krishnaveni, father Indradhyumnudu, wife Divya Spandana, brother Khagesh, and sisters Vani and Nirmala, for their unwavering love and continuous support throughout my academic journey.

Abstract

This thesis presents novel algorithms on waypoint following, target tracking, obstacle avoidance, collision avoidance and formation control for unmanned vehicles using Lyapunov Guidance Vector Field (LGVF) framework. The primary objective is to enhance the applicability of different autonomous systems consisting of Unmanned Aerial Vehicles (UAVs) and/or Unmanned Ground Vehicles (UGVs) in dull, dirty and dangerous missions.

The first part of the thesis develops C^2 continuous paths with bounded curvature based on the LGVF framework for two-dimensional (2D) waypoint following problems. Faster convergence is ensured as the generation of these paths are based on optimal Dubins paths which are C^1 continuous in nature. To begin with the algorithm is developed for the straight line path convergence and then it is extended to follow a sequence of waypoints by joining any two consecutive waypoints with straight line segments. Then a more challenging version of the algorithms is developed to address the waypoint following in three-dimensional (3D) space. It offers practical solutions to widen the usage of UVs in day to day life, such as automated crop spraying, where coverage area is crucial. The effectiveness of the algorithms is demonstrated through numerical simulations and comparisons with the existing techniques available in the literature. The algorithm is also successfully implemented on a quadrotor in an indoor environment, showcasing its potential for real-world applications.

Next, a guidance algorithm is presented for standoff target tracking by UAVs, employing the Lyapunov guidance vector field framework. Normalized even and odd functions are introduced as circulation and convergence terms for faster convergence. The algorithm generates continuous curvature paths maintaining the bounds on the curvature to make it applicable for fixed-wing UAVs. The effect of wind has also been considered for real-world outdoor applications. Simulation results demonstrate the effectiveness of the proposed guidance law by comparing it with the similar guidance algorithms existing in the literature.

Finally, a singularity-free vector field guidance approach for a swarm of UAVs has been solved for waypoint-following problems in environments filled with static and dynamic obstacles. A novel behavioral-based balance formation topology is introduced, distributing unmanned vehicles evenly within a desired formation circle using a guidance vector field method. The proposed algorithm exhibits asymptotic stability. Its faster convergence compared to existing approaches is demonstrated through comparison with the Barycentric coordinate-based control approach. The proposed formation approach is advantageous as it relies on relative position measurements in local coordinate frames and does not require inter-vehicle communication. Additionally, a proximity-based obstacle avoidance technique is developed, ensuring safe trajectories for each vehicle without encountering singularities. The proposed methodologies are validated through numerical simulations using MATLAB.

To sum up, this thesis provides comprehensive insights into the application of the Lyapunov Guidance Vector Field framework in various unmanned systems, offering practical solutions for waypoint following, target tracking, and obstacle avoidance, collision avoidance, and formation flying. The numerical simulations and real-world implementations highlight the effectiveness and suitability of the proposed algorithms for real-world applications.

Keywords: Lyapunov Guidance Vector Field (LGVF), Unmanned Aerial Vehicles (UAVs), Waypoint-following, Standoff target tracking, Obstacle avoidance, Formation control.

Contents

Abstract	xiii
1 Introduction	1
1.1 Motivation	2
1.2 Literature Review	3
1.2.1 Path Planning	3
1.2.2 Obstacle Avoidance	5
1.2.3 Target Tracking	6
1.2.4 Cooperative Control	8
1.3 Objectives of the Thesis	10
1.4 Thesis Organization	10
2 Waypoint Following in 2D Plane	13
2.1 Introduction	13
2.2 Problem Definition	14
2.3 Proposed Lyapunov Vector Field	16
2.3.1 Straight Line Following	16
2.3.2 Switching Waypoint Segments	18
2.4 Stability Analysis	22
2.5 Curvature Analysis	24
2.6 Simulation Results	26
2.7 Summary	31
3 Waypoint Following in 3D Space	33
3.1 Introduction	33
3.2 Problem Definition	33
3.3 Proposed Lyapunov Vector Field	35

3.4	Simulation Results	37
3.5	Experimental Results	39
3.6	Summary	40
4	Standoff Target Tracking	43
4.1	Introduction	43
4.2	Problem Definition	44
4.3	Background	47
4.4	Proposed Lyapunov Vector Field	48
4.4.1	Target Tracking	51
4.4.2	Proposed Control Parameter, c	53
4.5	Dynamic Model of a UAV for Path Following	55
4.5.1	Autopilot Design	59
4.5.1.1	Longitudinal Autopilot	59
4.5.1.2	Lateral Autopilot	60
4.6	Simulation Results	61
4.6.1	Stationary Target Tracking	61
4.6.2	Moving Target Tracking	67
4.6.3	Path following using dynamic model	68
4.6.4	Wind Absent Case	68
4.6.5	Wind Case	68
4.7	Experimental Results	70
4.8	Summary	75
5	Coordinated Waypoint Following by Group of UAVs	77
5.1	Introduction	77
5.2	Problem Definition	78
5.3	Balanced Formation Topology	80
5.4	Proposed Guiding Vector Field	82
5.4.1	Stability	85
5.4.2	Speed Profile:	86
5.5	Obstacle Avoidance	89
5.5.1	Static Obstacle Avoidance	89
5.5.2	Moving Obstacle Avoidance	94
5.6	Waypoint Following	95
5.7	Comparative Analysis	96
5.8	Simulation Results	100

CONTENTS

5.9 Summary	105
6 Conclusion	107
6.1 Future Scopes	107
References	109

List of Abbreviations

UAV	Unmanned Aerial Vehicle
UV	Unmanned Vehicle
UGV	Unmanned Ground Vehicle
2D	Two-dimensional
3D	Three-dimensional
LOS	line-of-sight
LGVF	Lyapunov Guidance Vector Fields
WMR	Wheeled Mobile Robot
APF	Artificial Potential Field
BCB	Barycentric Coordinate Based
LQR	Linear Quadratic Regulator
NURBS	Non-Uniform Rational Basis Spline
B-spline	Basis Spline

List of Figures

2.1	Waypoint Following Problem	14
2.2	Straight line following	17
2.3	Straight line following	19
2.4	Dubins Path for transition	19
2.5	Convergence (f_1) and circulation (f_2) terms for Dubins path	20
2.6	Convergence (f_1) and circulation (f_2) terms of the proposed approach	20
2.7	Three waypoints geometry	21
2.8	Guidance vector field for transition from one waypoint segment to the next	23
2.9	Curvature profile	26
2.10	Quadrotor indoor lab setup	27
2.11	Experimental and numerical results of waypoint following	27
2.12	curvature profile	28
2.13	X, Y, Z positions and Yaw angle profiles	28
2.14	Experimental and numerical results of waypoint following	29
2.15	curvature profile	29
2.16	X, Y, Z positions and Yaw angle profiles	30
3.1	Illustration of waypoint following in 3D scenario	34
3.2	Illustration of waypoint following in 3D scenario	36
3.3	Waypoint following in 3D space	38
3.4	Experimental result of waypoint following in 3D	39
3.5	X, Y and Z position profiles	40
4.1	Scenario of target tracking	44
4.2	Velocity triangle	45
4.3	Dubins path with zero initial orientation angle	52

4.4	Variation of the convergence and circulation terms for optimal convergence	52
4.5	Optimal path of converging from inside standoff circle	54
4.6	Circulation and convergence functions in the proposed guidance law	55
4.7	Longitudinal autopilot	60
4.8	Longitudinal autopilot	61
4.9	Lateral Autopilot	61
4.10	Attitude hold controller	62
4.11	Stationary target tracking when the UAV starts from outside of the standoff circle	63
4.12	Stationary target tracking	64
4.13	Trajectories for different initial orientation angles	65
4.14	Stationary target tracking	66
4.15	Stationary target tracking in variable wind condition for different initial poses	67
4.16	Moving target tracking	69
4.17	Stationary target tracking using dynamic model of the UAV	70
4.18	State and control variables for standoff target tracking (a) heading angle, (b) altitude profile, (c) airspeed profile, (d) bank angle, (e) throttle input, (f) elevator input, (g) rudder input, and (h) aileron input.	71
4.19	Stationary target tracking in windy condition using dynamic model of the UAV	72
4.20	State and control variables for standoff target tracking (a) heading angle, (b) altitude profile, (c) airspeed profile, (d) bank angle, (e) throttle input, (f) elevator input, (g) rudder input, and (h) aileron input.	73
4.21	Architecture of control flow for autonomous operation by the ground robot	74
4.22	Circle following by the ground robot	74
4.23	Circle following by the ground robot	75
5.1	Waypoint following by formation of three agents	78
5.2	Balanced formation topology	81
5.3	$\chi_{ic} - \phi_i$ vs $\phi_i - \chi_o$ profile	84
5.4	Vector diagram	87
5.5	Geometry of obstacle avoidance	89

LIST OF FIGURES

5.6	On the obstacle boundary	90
5.7	Angle difference between the proposed virtual LOS angle and LOS angle to the obstacle, $(\theta' - \gamma)$ vs. Angle difference between the virtual LOS angle and LOS angle to the obstacle, $(\theta - \gamma)$, for different values of k_3	91
5.8	Profile of $\text{sech}(k_4(l - r_o))$ vs. l , for $k_4 = 4/r_o$, $2/r_o$ and $1/r_o$ ($r_o = 30\text{ m}$)	91
5.9	Obstacle avoidance	92
5.10	Obstacle avoidance	93
5.11	Vehicle trajectories for moving obstacle avoidance	94
5.12	Case 1: Three vehicles in a triangular formation of $r_f = 40\text{ m}$. . .	96
5.13	Case 2: Three vehicles in a triangular formation of $r_f = 30\text{ m}$. . .	96
5.14	Case 3: Three vehicles in a triangular formation of $r_f = 20\text{ m}$. . .	97
5.15	Case 1: Distance profiles of all the vehicles from their neighboring vehicles for Barycentric coordinate based control algorithm [40] . . .	97
5.16	Case 1: Distance profiles of all the vehicles from their neighboring vehicles for the proposed algorithm	97
5.17	Case 2: Distance profiles of all the vehicles from their neighboring vehicles for Barycentric coordinate based control algorithm [40] . . .	98
5.18	Case 2: Distance profiles of all the vehicles from their neighboring vehicles for the proposed algorithm	98
5.19	Case 3: Distance profiles of all the vehicles from their neighboring vehicles for Barycentric coordinate based control algorithm [40] . . .	98
5.20	Case 3: Distance profiles of all the vehicles from their neighboring vehicles for the proposed algorithm	99
5.21	Waypoint following by the formation of three non-holonomic robots in an obstacle filled environment. Snapshots of each robot and moving obstacles at 7 instants are shown with time gap of 50 sec . .	101
5.22	Distances of all robots from the virtual robot	101
5.23	Distances of all robots from the other robots	102
5.24	Control inputs of virtual robot	102
5.25	Control inputs of robot 1	102
5.26	Control inputs of robot 2	103
5.27	Control inputs of robot 3	103
5.28	Heading angle comparison	104

List of Tables

4.1	Parameter Values	58
4.2	UAV Coefficients	59
4.3	Specifications of QBot 2.	72
5.1	Control parameters	104

Introduction

Unmanned vehicles (UVs) are recently being used extensively for several civilian and military applications to execute dull, dirty, and dangerous missions like search and rescue operations during disasters [1], forest fire monitoring [2], border surveillance [3] bombing targets and many more. Commercial applications of unmanned vehicles have also got momentum due to fast development in cost-effective technologies. Very precise mission objectives, such as, power line inspection [4], package transport [5], crop monitoring [6] etc, are achieved efficiently by employing unmanned vehicles. Many active researches are going on to meet the challenges of using unmanned vehicles more efficiently and economically; one of them is to include autonomy in unmanned systems. Autonomous flight is more effective in comparison with the tele-operated UAVs, mainly in long-duration missions. As for long-duration operations, radio signals communication between the human operator and the vehicle are mostly unreliable. In such a scenario an autonomous flight is reliable. Also, accuracy, precision, and efficiency of mission can be improved significantly using autonomous flights. An autonomous vehicle is equipped with necessary data processing units, sensors, and communication systems to process localization, state estimation, motion control operations to execute missions autonomously without any human intervention. The localization system is required to find out the global position of a vehicle in a map. State estimation is the process

of finding the pose of a vehicle. The motion controller is responsible for achieving the desired position or velocity.

In many of the above-said applications, an unmanned vehicle has to go from one waypoint to another waypoint with specified orientations to complete the mission. In those scenarios, path planning plays an essential role in generating one or more feasible and safe paths between waypoints. Several optimal path planning algorithms using Dubins paths and some continuous curvature paths have been discussed in the literature. After planning the path, it is required to track these paths by the vehicle, which is termed as path following. If the trackable path is explicitly a function of time it will be named as trajectory tracking. Typical hierarchical architecture of path following controller consists of outer-loop (guidance) and inner-loop controllers. The guidance loop generates desired transnational velocities, which should be followed by the inner-loop controller by generating appropriate lateral acceleration commands.

Along with the path following application several other guidance algorithms have been developed in the literature for obstacle avoidance, cooperative tasks, and target tracking *etc.* For successful autonomous real-life flight, the guidance law should be simple, robust to wind disturbance and should consider the dynamic constraints of the vehicle. In the next section a brief literature survey of guidance algorithms is presented. This thesis develops guidance laws for waypoint following, standoff target tracking, obstacle avoidance, collision avoidance and formation control by non-holonomic autonomous vehicles.

1.1 Motivation

Various waypoint following or target tracking applications, like, crop spraying, land mapping, traffic monitoring and surveillance, require a unmanned vehicle to accurately follow paths consisting of straight line segments and circular arcs. For example, for spraying in agriculture fields UAVs need to track straight-line segments while maintaining a constant speed. UVs need to loiter in circles to track a suspicious target. Similarly UVs need to travel paths made of straight and curve sections to achieve their missions like dropping medicines at particular points of remote villages or monitoring general gathering, broadcasting of sports,

1.2 Literature Review

monitoring forest area etc.

This motivates us to address straight line and circular path convergence problems. The concept is also extended for waypoint following problems which was earlier addressed in the literature [7] with the help of Dubins paths [8] of C^1 continuity. It is well known that following C^1 type paths is difficult and if it is followed then tracking error will be significant. To address this shortcoming, this thesis develops C^2 smooth paths taking care of the dynamic constraints of the vehicles, including bank angle constraints and ensures obstacle free optimal convergence.

To the best of authors' knowledge there has been no such work existing in the literature for generating singularity free C^2 continuous path with curvature bounds for waypoint following by a UAV or a swarm of UAVs considering the presence of both static and dynamic obstacles in the environment. This motivates us also to find solutions for a swarm of UAVs to follow waypoints in three dimensional (3D) dynamic environment to enhance the success of surveillance and reconnaissance missions.

1.2 Literature Review

1.2.1 Path Planning

Path following controllers steer a vehicle to follow a geo-referenced path. There exist a wide range of techniques in the literature to deal with the decoupled guidance and control based path following. The basic philosophy of these techniques is, first to develop the error dynamics of the concerned path and then the guidance law is designed by ensuring that the system is stable to converge to the desired path. Here the error indicates the lateral distance error and the course angle error for two-dimensional (2D) path following problems, and for 3D path following problems, the flight path angle error is also included. Different path planning algorithms for unmanned ground vehicles as well as for UAVs operating with constant altitude were presented in ([9], [10]). Rysdyk [11] proposed a good helmsman behavior based guidance algorithm where the Serret-Frenet frame was considered on the desired path and in the frame the cross-track error and the course angle error dynamics were formulated. The good helmsman behavior dictates the de-

sired course to nullify the cross-track and the course angle errors simultaneously. Park *et al.* [12] presented a simple pure pursuit based nonlinear guidance law and robust stability was analysed for the case of the control signal (lateral acceleration in this case) getting saturated. Ratnoo *et al.* [13] proposed a LQR controller using linearized error dynamics for path following problems. Osborne and Rysdyk [14] presented another good helmsman behavior-based guidance law, where the wind disturbance problem was tackled by using lookup tables of proximate distances corresponding to wind estimation data. Kothari *et al.* [15] used both line-of-sight and pure pursuit philosophy to generate the error dynamics and then a linear controller was designed using the linearized model of the error dynamics. Andersen and Kristiansen [16] presented a guidance law for the 3D path following using quaternions instead of using Euler angles. There, two unit quaternions were formulated; one was for convergence and the second one was for following the path and then these were added with moulding functions to assure a smooth convergence. Tanaka *et al.* [17] formulated error dynamics in the Serret-Frenet frame in 3D and for stabilization a fuzzy-based controller was developed.

In vector field-based techniques for path following, the desired path is considered as the equilibrium set or zero level set and convergence to the desired path is assured by making the equilibrium set stable. In the vector field guidance case, the integral curves of the vector field give the convergence trajectories to the desired equilibrium set. The review paper [10] concluded that the vector field guidance gives the less cross-sectional error and requires less control effort. The Lyapunov vector field [18, 19, 20, 21] and guiding vector field [22, 23, 24, 25], algorithms can consider the bounded lateral acceleration constraints. Lawrence *et al.* [21] proposed a Lyapunov vector field guidance law for circular vector fields and which was further extended to waypoint following problems using online switching between circular vector fields. Nelson *et al.* [26] constructed the vector fields to converge to straight lines and circles in the presence of wind. The method is simple and effective, but this method is only applicable to paths composed of straight lines and circular arcs. A similar approach for curved paths following was presented in [27]. Liang *et al.* [28] proposed a vector field guidance using Helmholtz theory where any guidance vector was represented as a sum of conservative and solenoidal fields. The extension of this work in 3D was presented in [29] and the maximum

1.2 Literature Review

bounds on tuning parameters for global feasibility was considered. Rezende *et al.* [30] proposed a 3D guidance vector field as a sum of the two surface vector fields. Stability analysis for path following using 3D vector fields was addressed by Yao and Cao [31].

1.2.2 Obstacle Avoidance

For instance, consider an unmanned vehicle is tracking a planned path and on the way it meets a dynamic obstacle. In such cases, it requires to generate a new path online avoiding obstacles. The re-planned path should also have features like computationally efficient, optimal, and feasible. Some of the well-known approaches of obstacle avoidance are grid search algorithms [32, 33], road map based algorithms [34, 35], optimal control algorithms [36, 37], artificial potential field algorithms [38, 39]. Except artificial potential field approaches remaining all the approaches are computationally complex and also after the path planning it requires the help of some path smoothening techniques for making it practically feasible for implementation.

Khatib [38] proposed an artificial potential field (APF) approach for collision avoidance of robot manipulators where the goal point is modeled as an attractive force and the obstacle is modeled as a repulsive force. APF approach has more potential in online trajectory planning due to its inherent naturality and simplicity. The main drawback of the APF approach is the singularity of field occurs when an obstacle is exactly on the line between the vehicle and the goal point. Kim and Khosla [39] proposed a harmonic potential function to deal with the local minima problem. Here, harmonic functions are derived as a solution of the Laplace equation with Dirichlet boundary conditions to avoid singularities in the field except at the goal and the source points. Munasinghe *et al.* [40] presented an intelligent technique by modelling an obstacle as a velocity dipole to deal with the presence of moving obstacles effectively.

Inspired by APF approaches, Li and Bui [41] and Connolly *et al.* [42] proposed guidance algorithms using the concept of fluid mechanics. Poisson's equations are used in [41] to develop an ideal fluid to deal with the local minima or singularities. There, streamlines served as obstacle free paths. Waydo and Murray [43] devel-

oped streamline functions satisfying the Laplace equation for obstacle avoidance problems. Similar work was presented in [44] for 3D obstacle avoidance problems. The main advantages of streamline based approaches are that the resultant paths are smooth and practically realizable.

Another approach similar to APF is the artificial velocity vector field method. In the APF approach a negative gradient is required to find the local minima in the map, which is not required in the case of vector field methods. Wilhelm and Clem [45] proposed a gradient vector field as a sink flow and an obstacle was modeled as a circular field. For optimal path generation parametric tuning was done using lookup tables. The generated paths were smooth and suitable for curvature constrained vehicles. Zhu *et al.* [46] studied different scaling functions for modeling obstacles in an artificial vector field. Marchidan and Bakolas [47] proposed collision avoidance vector fields using velocity modulation functions to generate constant velocity fields.

Some vector field guidance algorithms [45],[47],[48] can deal with obstacle-avoidance and bounds in lateral acceleration to generate safe trajectories. Mainly these studies focus on modifying the actual vector field by combining it with the obstacle vector fields to realize both path-following and obstacle avoidance. In these cases, there is a possibility of cancellation of the main vector field by the obstacle field leading to singularities in those points [45]. At these singular points, the vehicles do not get any feasible speed and heading commands, which may cause failure of the mission.

1.2.3 Target Tracking

Target tracking while maintaining a constant standoff distance from the target is called as standoff target tracking. It can be accomplished by a fixed-wing UAV by loitering around the target in a circular path. Usually the speed of the UAV will be more than the speed of the target as fixed-wing UAVs need to maintain forward motion to avoid stalling.

Several guidance laws for standoff target tracking were developed in the past. Lawrence [18] proposed a Lyapunov vector field guidance law to make UVs loiter in a circular path above a target. In general Lyapunov functions are constructed

1.2 Literature Review

to check the stability but, in that work, the Lyapunov function is chosen first considering the desired circle as a zero level set, and stable vector fields are generated by ensuring that the time derivative of the Lyapunov function is negative definite. Rysdyk [49] introduced good helmsman behavior-based guidance law, where the Helmsman behaviour indicated functional relationship between the cross track error and the course angle error. Morgan *et al.* [50] used super critical Hopf bifurcation concept to generate trajectories for target tracking. Frew *et al.* [51] added phase keeping guidance to Lyapunov vector field guidance for coordinate standoff target tracking. Hongda *et al.* [52] proposed Tangent plus Lyapunov vector field guidance to achieve fast convergence to a standoff circle. It was shown there that the tangent vector field method outperformed the Lyapunov vector field of [18] if there existed a tangent to the circle. Ma *et al.* [53] proposed an inverse kinematic guidance with fast estimation scheme for vision based tracking. Park [54] proposed a simple, global asymptotically stable relative side bearing angle based guidance law for circling of UAVs over a target.

For better performance of target tracking the desired converging trajectories should be feasible and optimal. Pothen and Ratnoo [20] generated a curvature constrained Lyapunov vector field where the curvature of the vector field was within the maximum curvature limits and it had faster convergence compared to [18]. Oh *et al.* [55] added adaptive sliding mode concept to the existing tangent plus Lyapunov vector field guidance for accounting the effect of remodeled disturbances in the autopilot loop. Shun *et al.* [56] proposed a fast convergence Lyapunov vector field guidance by using an off-line search algorithm.

As discussed above a wide variety of vector fields can be generated by the Lyapunov vector field framework for path following, target tracking and obstacle avoidance [45, 46] applications. Constructing a simple vector field to guide vehicles for faster convergence to a desired standoff circle considering maneuverability constraints of a vehicle is still a challenging area as in many applications UAVs are initially deployed sufficiently far apart from the desired circle but the life of the battery of UAVs is limited. Hota and Ghose [57] and Alok and Hota [58] solved the optimal path planning problem for converging to a straight line and a circle by using geometrical arguments but the curvature profile of the path was not continuous.

1.2.4 Cooperative Control

Complex missions can be accomplished by operating a group of agents in a reliable, economical, and efficient way. In many of the multi-agent operations, UAVs are required to fly in a desired geometric formation to enhance mission success. In these formation flight control problems, the general objective is to find the control inputs that adjust the spatial position of each vehicle and to direct them to move as a single-vehicle [59].

The control strategies of formations are classified into centralized control and distributed control based on control commands to each vehicle. In the centralized control scheme [60], each vehicle receives its control commands from the ground station or from the leader vehicle, whereas in the distributed control scheme, each vehicle generates its own command [61]. Centralized control strategies are less reliable as losing communication with the control station may lead to the complete failure of the mission. Under the distributed control scheme there are mainly two approaches in literature : i) virtual structure-based approach ([62],[63]) and ii) behavior-based approach ([64],[65]). In the virtual structure-based approach, each vehicle is specified with the desired position around a reference point and the vehicles act like a rigid body. In the behavior-based approach, each vehicle is specified with a desired distance from the neighboring vehicles. This approach is flexible as the vehicle can change their position while maintaining the desired behavior with the neighboring vehicles.

Some of the challenges in cooperative missions are reliability to maintain a stable communication among all agents, collision avoidance, robustness to disturbances such as wind effect, and feasibility, that is, the vehicle should be able to follow the commanded control signals etc. In [66], the bounds on control gains were derived for maintaining a stable network connectivity among the robots while maintaining a desired formation. Rajnikant et al.[67] proposed an algorithm using graph theory for target capturing problems when there were uncertainties in the target information. Some applications may demand to change the formation shape to another as time progresses. For example, if a swarm of vehicles has to pass through a narrow hilly way, several re-configurations of the formation are required. In [68] and [69], authors proposed guidance algorithms for such recon-

1.2 Literature Review

figuration problems. The node failure problem in a connected network, that is, one vehicle among the group fails to exchange the information, was solved in [70] using Laplacian graph theory. The other challenge is the feasibility of the developed guidance law that controls the input demand with required bounds. Consolini et al. [59] proposed a feasible guidance law for two-vehicle systems based on leader-follower strategy. An extended collision cone approach was proposed in [71] for obstacle avoidance problems in formation flight.

The typical guidance of a swarm of vehicles is broadly divided in two ways; in the first case, all the vehicles form around a point on the scheduled trajectory and in the second case, all the vehicles form around a point that is treated as a virtual leader following the desired path. In the second case, any path following algorithm can be applied to the virtual leader. In most applications, the guidance problem has to be solved to generate a path for vehicles to follow a set of waypoints, considering various constraints such as travel time, total distance to be traveled, and path continuity. The capacity of the vehicles to plan and track trajectories is the key to expand their utility. The features of a path e.g. curvature, directly impact the control commands sent to vehicles, and a rapid change in curvature along the path may cause harm to the vehicles. The most straightforward path for waypoint following is generated by connecting the waypoints with straight lines. This would result in a C^0 curve, making the tracking of the path [72] extremely difficult for nonholonomic vehicles. In the literature, numerous algorithms generate C^1 smooth trajectories. In [8], [7], and [73], time-optimal paths based on the Dubins paths were investigated. However, transitions between arcs and straight lines create a discontinuous profile of the curvature. Multiple efforts have been made to generate C^2 smooth paths using geometric curves such as clothoids [74], hypocycloids [75], Bezier curves [76], B-splines [77], NURBS curves [78], and Fermat Spirals [79].

Desired formation with a desired heading is of typical interest for a motion constrained vehicle. Zhao *et al.* [80] presented a generalized way to produce coordination control law for nonholonomic vehicles from well-known motion unconstrained coordination control laws. Fathian *et al.* [81] proposed a distributed formation control law using Laplacian graph theory for stable formation with a desired heading. In the work, an integral model for a UAV was used and the dynamic constraints were imposed using orthogonal projection concept. Wang and

Xin [82] introduced an unified optimal control framework to integrate formation control, trajectory tracking, and obstacle avoidance objectives into a single cost functional for optimization. Peng *et al.* [83] proposed a control algorithm for simultaneous target tracking with the help of a circular formation. In the work, a Lyapunov phase function was generated as a potential function and the formation control law was developed by ensuring Lyapunov stability. These approaches offer valuable insights into achieving desired formations and headings in challenging scenarios.

1.3 Objectives of the Thesis

The guidance algorithms for unmanned vehicles are developed to achieve the following objectives:

- To follow a set of waypoints in 2D by generating smooth paths of C^2 continuity with curvature bounds using a vector field guidance algorithm.
- To extend the waypoint following problem for 3D space fulfilling the same objectives as those of 2D case.
- To develop a guidance algorithm for standoff target tracking by a fixed-wing UAV in the presence of wind.
- To develop a singular-free guidance law for a swarm of UAVs in formation to follow waypoints avoiding collision with static and dynamic obstacles in a cluttered environment.
- To validate the developed algorithms using numerical simulations as well as real-time experiments using UAVs/UGVs wherever applicable.

1.4 Thesis Organization

There are six chapters in this thesis. A summary of each chapter is given below.

1.4 Thesis Organization

- Chapter 1 is about the introduction of the thesis. Problem statement, motivation, literature survey and contributions made in the thesis have been highlighted here.
- Chapter 2 presents Lyapunov Guidance Vector Field (LGVF) based C^2 continuous near-optimal paths generation to guide unmanned vehicles to follow a sequence of waypoints in a 2D plane. The waypoints are connected by joining them sequentially with straight lines and then straight line convergence technique is applied to solve the problem. The proposed vector field guidance algorithm can generate C^2 smooth paths with curvature bounds. In this study, the convergence and circulation components of the LVGF are carefully selected to ensure that the integral curves of the vector field produce C^2 smooth paths. The control parameters of the designed vector field are determined through curvature analysis. The proposed model is then validated both numerically and experimentally. The proposed algorithm has been successfully implemented on a quadrotor in an indoor environment, demonstrating its effectiveness in real-world applications.
- Chapter 3 presents the vector field guidance algorithm for UAVs to follow waypoints in 3D space. In order to adapt the 2D guidance algorithm to 3D scenarios, we define a reference frame at the current waypoint in such a way that both the UAV and the current waypoint segment to be followed lie in the XY-plane of the introduced reference frame. Then the vector field guidance algorithm developed in Chapter 2 is applied on this XY-plane. The proposed algorithm has been successfully implemented on a quadrotor in an indoor environment, demonstrating its effectiveness in real-world applications.
- Chapter 4 describes the development of a guidance algorithm for UAVs to track standoff targets using the LVFG framework. The convergence and circulation terms of the LVFG have been modified to ensure faster and smoother convergence to the standoff circle. Additionally, the algorithm takes into account the presence of wind to ensure its applicability in real-world scenarios. To demonstrate the effectiveness of the proposed algorithm, a comparison with relevant literature is conducted, which shows faster convergence

to the standoff circle. The curvature of the vector field is also analyzed to accommodate different minimum turn radius capabilities of different vehicles. Although originally developed for fixed-wing UAVs, this work explores the practical realizability on microcontrollers by testing the algorithm on a wheeled mobile robot operating at a constant speed.

- Chapter 5 presents a novel approach for guiding a group of UAVs with non-holonomic constraints to follow a sequence of waypoints located in an environment filled with static and dynamic obstacles. The proposed method utilizes a vector field guidance technique to provide heading and speed commands to each UAV, to ensure that the desired formation of the UAV is maintained. The stability of the proposed vector field is rigorously analyzed and proved to be asymptotically stable. To showcase the effectiveness of the algorithm, a comparison is made with a recent literature using a three-vehicle system. One significant advantage of our formation approach is its reliance on relative position measurements in local coordinate frames, eliminating the need for inter-vehicle communication. Furthermore, a proximity-based obstacle avoidance technique that generates safe trajectories for each UAV is introduced. Unlike obstacle avoidance methods based on artificial force fields, the presented approach is free from singularities which occur when the obstacle field cancels out the main vector field and leads to unreliable guidance. In the proposed technique the guidance vector field is modulated by solely adjusting the vector direction, effectively avoiding singularities. Finally, the proposed methods are validated through extensive numerical simulations conducted in the MATLAB environment.
- Chapter 6 presents the summary of the presented work and future scopes of research in this domain.

Waypoint Following in 2D Plane

2.1 Introduction

This chapter presents generation of continuous curvature smooth paths for waypoint following in a 2D plane using Lyapunov Guidance Vector Field (LGVF) framework. First, the waypoints are connected sequentially with straight lines, and then the straight-line convergence technique is applied to solve the problem. Different techniques are available in the literature for straight-line convergence, but none of them can generate C^2 smooth paths with curvature bounds. In this study, the convergence and circulation components of the LGVF are carefully selected to ensure that the integral curves of the vector field produce C^2 smooth paths. The control parameters of the designed vector field are determined through curvature analysis. The proposed model is then validated both numerically and experimentally. For the experimental validation, an indoor quadrotor lab setup is used, and the results show the efficacy of the proposed algorithms.

This chapter is organized as follows: The problem is formally defined in Section 2.2. In Section 2.3, we analyze the Dubins path and design a guidance vector field for straight-line following. The stability analysis of this vector field is presented in Section 2.4, while the curvature analysis is conducted in Section 2.5. The ex-

perimental and numerical results are presented in Section 2.6, and finally, Section 2.7 provides the concluding remarks of this study.

2.2 Problem Definition

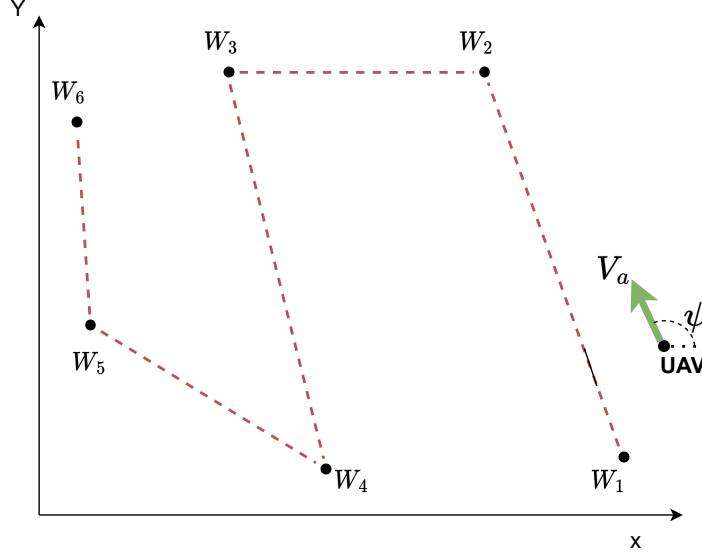


Figure 2.1: Waypoint Following Problem

A typical architecture of control problems to navigate an autonomously driven vehicle is a two-layer hierarchical process. The outer control layer generates the linear and the angular speed commands and the inner layer generates corresponding forces and moments. This work explores the outer control layer for UAVs, aiming to guide them along a series of straight lines connecting consecutive waypoints using C^2 smooth paths with minimum length.

Let us consider the UAV be in a fixed inertial frame of reference in the Cartesian coordinate system. The following kinematic equations govern these vehicles.

$$\dot{x} = V_a \cos \psi \quad (2.1)$$

$$\dot{y} = V_a \sin \psi \quad (2.2)$$

$$\dot{\psi} = u_0 \quad (2.3)$$

2.2 Problem Definition

where, $\mathbf{p} = \begin{bmatrix} x & y \end{bmatrix}^T$, $\mathbf{v} = \begin{bmatrix} \dot{x} & \dot{y} \end{bmatrix}^T$ are the position and velocity vectors of the UAV in the inertial reference frame, respectively. V_a is the airspeed of the UAV, which is set as constant using an autopilot for fuel efficiency. $\psi \in (-\pi, \pi]$ is the heading angle of the UAV. Consider N waypoints, denoted as W_i , where $i \in [1, N]$, in the inertial frame of reference, as illustrated in Fig. 2.1.

Consider a vector field, $\mathbf{F} = \begin{bmatrix} \dot{x}_d & \dot{y}_d \end{bmatrix}^T$, as a function of position, x and y , which can guide the UAV towards the desired waypoint on the prescribed straight line. From the vector field, the desired heading angle of the UAV, ψ_d , can be written as,

$$\psi_d = \arctan\left(\frac{\dot{y}_d}{\dot{x}_d}\right) \quad (2.4)$$

Moreover, to evaluate the proposed algorithm for the waypoint following problem, we assume that the UAV is equipped with a simple first-order heading hold autopilot as follows

$$\dot{\psi} = \alpha(\psi_d - \psi) \quad (2.5)$$

where, α is a positive proportional gain. And, from (2.3) and (2.5),

$$u_0 = \alpha(\psi_d - \psi) \quad (2.6)$$

u_0 is the commanded heading rate that acts as a guidance command and is constrained by

$$|u_0| \leq \dot{\psi}_{max} \quad (2.7)$$

As the heading rate is constrained by its upper limit $\dot{\psi}_{max}$, it imposes a constraint on the the curvature of the path of the UAV. The maximum curvature (κ_{max}) of the path is represented by,

$$\kappa_{max} = \frac{\dot{\psi}_{max}}{V_a} \quad (2.8)$$

and the minimum radius of turn is related to it as,

$$r_{min} = \frac{1}{\kappa_{max}} \quad (2.9)$$

To achieve successful operation, the vector field $\mathbf{F}$ must fulfill the following objectives:

1. The straight line to be followed should be the stable attractor of the vector field.
2. The curvature constraint of $\mathbf{F}$ must satisfy the condition $\frac{\dot{\psi}d}{V_a} \leq \kappa_{max}$ to guarantee that the UAV can feasibly navigate the generated path.
3. For constant speed applications, all vectors in $\mathbf{F}$ should remain constant and have a magnitude equal to V_a .

To address this problem, a novel LVGF based guidance law is proposed to achieve faster convergence to the desired path with a smooth curvature profile. In this method the vectors represent the desired heading angles of the UAV, which direct the UAV towards the path to which it will converge.

2.3 Proposed Lyapunov Vector Field

In this section, a LVGF algorithm for faster and smoother convergence to a straight line is proposed. Firstly, a short review of the LVGF for unmanned vehicles is presented. Lawrence et al. [18] constructed the LVGF with the desired closed attractor as a globally asymptotically stable limit cycle. This was achieved by considering a Lyapunov function, $V(x, y)$, such that it has zero energy on the desired attractor, positive energy elsewhere, and is radially unbounded. The vector field, $\mathbf{F}$, is defined as the sum of the convergence term f_1 and the circulation term f_2 , ensuring that when $\mathbf{F}$ is $\begin{bmatrix} \dot{x} & \dot{y} \end{bmatrix}^T$, the total time derivative of the Lyapunov function is negative definite. Thus, the designed vector field can provide reference speed commands to the UAV to track the attractor.

2.3.1 Straight Line Following

The vector field is constructed by considering the straight-line path to be the globally asymptotically stable attractor, similar to [21] .

2.3 Proposed Lyapunov Vector Field

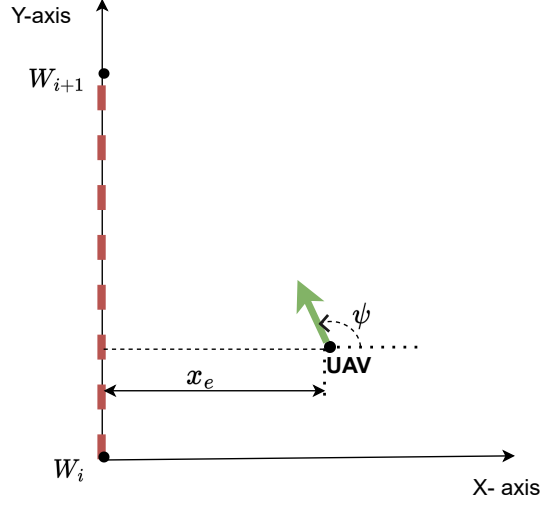


Figure 2.2: Straight line following

Assuming the straight line connecting the two consecutive waypoints $W_i(X_i, Y_i)$ and $W_{i+1}(X_{i+1}, Y_{i+1})$ to be followed is on the Y-axis, as shown in Fig. 2.2. The lateral distance error, denoted as x_e , from the UAV to the straight line will be the X-component of the position of the UAV. A positive scalar-valued Lyapunov function V is defined by treating the desired straight line path as a zero level set, given by:

$$V = \frac{1}{2}x_e^2 \quad (2.10)$$

The total time derivative of the Lyapunov function is,

$$\dot{V} = x_e \dot{x}_e \quad (2.11)$$

$\dot{x}_e$ along with $\dot{y}_e$ gives the velocity vector of the UAV. Here, we define a LVGF, $\mathbf{F}$, as follows

$$\begin{bmatrix} \dot{x}_e \\ \dot{y}_e \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}. \quad (2.12)$$

where, f_1 , the convergence term, and f_2 , the circulation term, of the vector field are defined as:

$$f_1 = -V_a \tanh(cx_e) \quad (2.13)$$

$$f_2 = V_a \operatorname{sech}(cx_e) \quad (2.14)$$

Here, c is a positive and real valued control parameter. From (2.11) and (2.12), $\dot{V}$ can be calculated as

$$\dot{V} = x_e \dot{x}_e \quad (2.15)$$

$$= x_e f_1 \quad (2.16)$$

$$= -V_a x_e \tanh(cx_e) \quad (2.17)$$

From (2.17), as the sign of the hyperbolic tangent function of x_e aligns with the sign of x_e , it can be inferred that $\dot{V}$ in (2.11) exhibits negative definiteness. Consequently, the equilibrium set $x_e = 0$ demonstrates asymptotic stability. Additionally, it is noteworthy that the magnitude of the proposed vector field, denoted by $\mathbf{F}$, remains constant, always equating to V_a .

The above concept of straight line convergence technique is used to converge to the first waypoint segments made by joining W_1 and W_2 . Once the UAV is on the first waypoint segment it will track the segment and at a particular distance (which depends on the minimum turn radius value of the UAV) from W_2 it will transit to the next waypoint segment made by joining W_2 and W_3 . The process will continue till the last waypoint segment. For transition from one waypoint segment to the next, another LVGF algorithm is adopted to get smooth curvature paths as presented next.

2.3.2 Switching Waypoint Segments

Let us consider any three consecutive waypoints, (W_{i-1}, W_i, W_{i+1}) , as shown in Fig. 2.3 which generate two waypoint segments in an acute angle at W_i . Assume the UAV is currently located on the waypoint segment connecting W_{i-1} and W_i , and it needs to switch to the waypoint segment connecting W_i and W_{i+1} . The design of convergence and circulation terms (f_1, f_2) for such case plays an important role in convergence time and smoothness of path. The construction of these terms are guided by Dubins paths for optimal convergence ([57],[84]). To

2.3 Proposed Lyapunov Vector Field

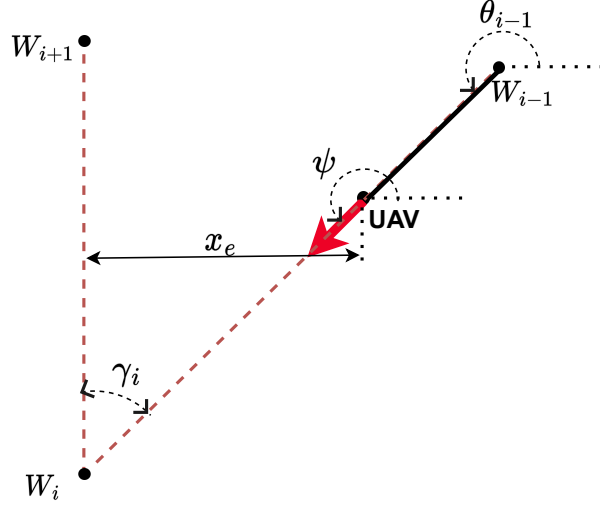


Figure 2.3: Straight line following

visualize this, let us consider an example of a waypoint following where there are two waypoint segments: (i) connecting waypoints (100, 100) and (0, -45) and (ii) connecting (0, -45) and (0, 100). The optimal Dubins path for transiting from one waypoint segment to another is shown in Fig. 2.4.

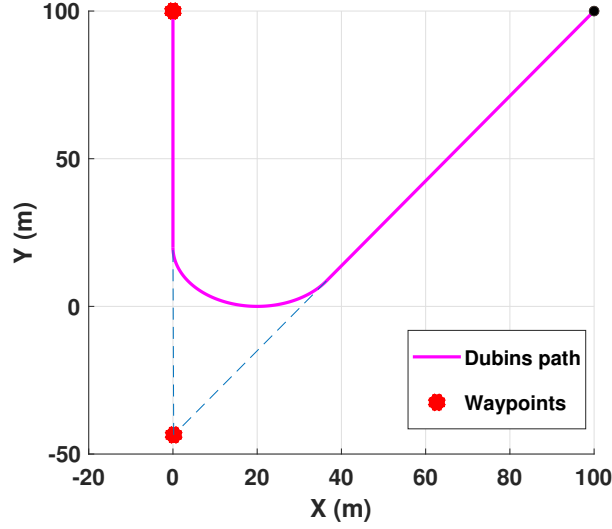


Figure 2.4: Dubins Path for transition

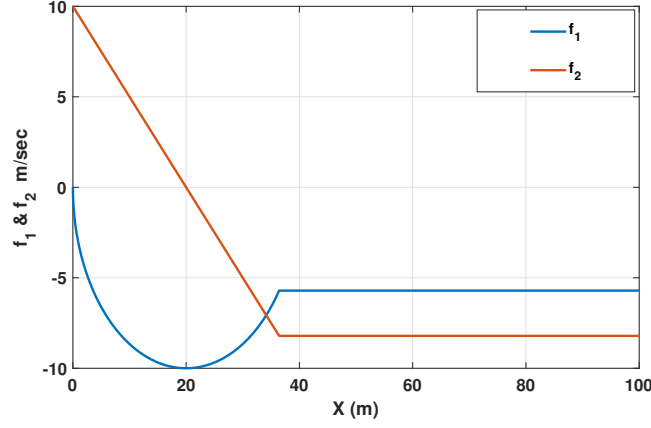


Figure 2.5: Convergence (f_1) and circulation (f_2) terms for Dubins path

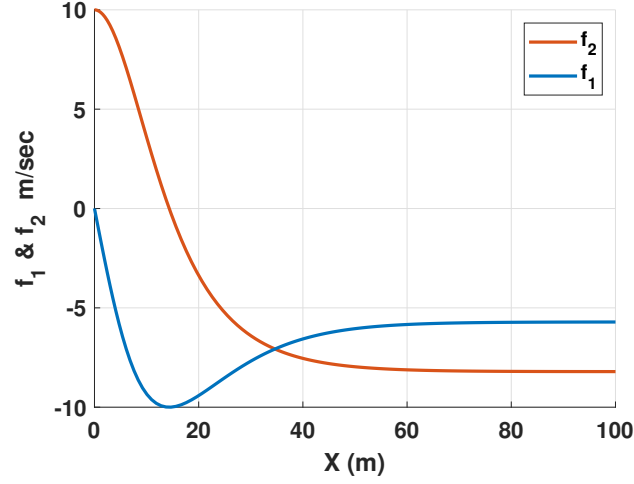


Figure 2.6: Convergence (f_1) and circulation (f_2) terms of the proposed approach

The vehicle pose data are collected for this Dubins path, and the data are mapped to the desired vector field model (2.12). The profiles of the convergence term (f_1) and the circulation term (f_2) are shown in Fig. 2.5. Analytical equations for these profiles are difficult to obtain. Also, following these profiles only leads to C^1 continuous path.

To generate a smoother path which is close to the optimal one (Dubins path)

2.3 Proposed Lyapunov Vector Field

the convergence term (f_1) and the circulation term (f_2) are proposed as,

$$f_1 = -\text{sign}(x_e)\sqrt{V_a^2 - f_2^2} \quad (2.18)$$

$$f_2 = (V_a - V_r) \text{sech}(cx_e) + V_r \quad (2.19)$$

where,

$$\text{sign}(x_e) = \begin{cases} -1, & x_e < 0 \\ 0, & x_e = 0 \\ 1, & x_e > 0 \end{cases} \quad (2.20)$$

where, $V_r = V_a \sin(\theta_{i-1})$, where, θ_{i-1} can be expressed as

$$\theta_{i-1} = \arctan\left(\frac{Y_i - Y_{i-1}}{X_i - X_{i-1}}\right) \quad (2.21)$$

Using f_1 (2.18) and f_2 (2.19), from (2.12), the vector field that provides smooth and feasible trajectories for the UAV to switch between the waypoint segments can be obtained.

Figure 2.6 shows the profiles of the convergence and circulation terms of the proposed approach. In the previous analysis, the waypoints (W_i , W_{i+1}) are as-

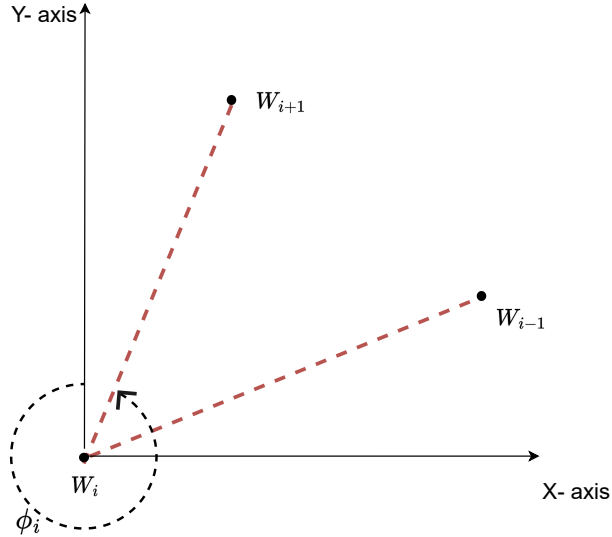


Figure 2.7: Three waypoints geometry

sumed on the Y -axis. However, if these waypoints are not on the Y -axis as shown in Fig. 2.7, a coordinate transformation can be applied to place them on the Y -axis of the transformed system to find the desired vector field as stated above. The generated vector field can then be transformed back to its original coordinate system with the help of the tilt angle, ϕ_i , which is measured with respect to the global Y -axis and can be written as

$$\phi_i = \arctan \left(\frac{-(X_{i+1} - X_i)}{(Y_{i+1} - Y_i)} \right) \quad (2.22)$$

In this case, the lateral distance error x_e can be calculated as:

$$x_e = (x - X_i) \cos \phi_i + (y - Y_i) \sin \phi_i \quad (2.23)$$

The proposed vector field for this generalized case is:

$$\begin{bmatrix} \dot{x}_d \\ \dot{y}_d \end{bmatrix} = \begin{bmatrix} \cos \phi_i & -\sin \phi_i \\ \sin \phi_i & \cos \phi_i \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \quad (2.24)$$

Integration of the tangent curves of the vector field gives the smooth trajectories to follow the path as shown in Fig. 2.8.

2.4 Stability Analysis

Consider the Lyapunov function $W = \frac{1}{2}x_e^2$ which satisfies the following conditions,

1. $W = 0$, for $x_e = 0$
2. $W > 0$, for $x_e > 0$
3. $\dot{W} \leq 0, \forall x_e \geq 0$

2.4 Stability Analysis

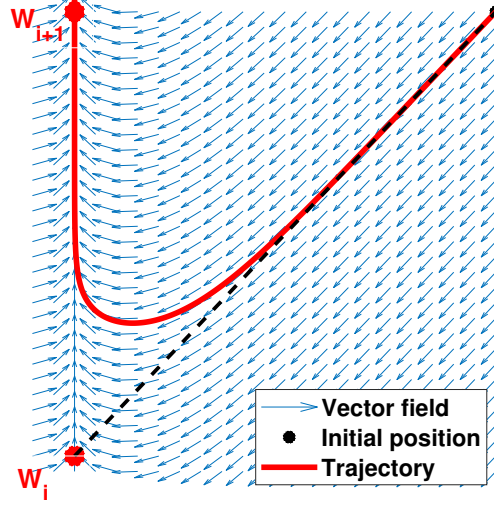


Figure 2.8: Guidance vector field for transition from one waypoint segment to the next

The total time derivative of the Lyapunov function is

$$\begin{aligned}
 \dot{W} &= x_e \dot{x}_e \\
 &= x_e \frac{d}{dt} ((x - X_i) \cos \phi_i + (y - Y_i) \sin \phi_i) \\
 &= x_e (\dot{x} \cos \phi_i + \dot{y} \sin \phi_i) \\
 &= x_e (f_1 \cos \phi_i - f_2 \sin \phi_i) \cos \phi_i \\
 &\quad + x_e (f_1 \sin \phi_i + f_2 \cos \phi_i) \sin \phi_i \\
 &= x_e (f_1 \cos^2 \phi_i + f_1 \sin^2 \phi_i) \\
 &= -x_e \operatorname{sign}(x_e) \sqrt{V_a^2 - f_2^2}
 \end{aligned} \tag{2.25}$$

From the above analysis using Lyapunov Direct Method, it can be deduced that $x_e = 0$ to be a globally asymptotically stable attractor set. This ensures that the trajectories generated by the vector field will always asymptotically converge to the desired path.

2.5 Curvature Analysis

For proposed vector field (2.24), the time derivative of desired heading angle (2.4) is

$$\dot{\psi}_d = \frac{\dot{x}_d \ddot{y}_d - \dot{y}_d \ddot{x}_d}{\dot{x}_d^2 + \dot{y}_d^2} \quad (2.26)$$

The curvature of the integral curves of a vector field can be expressed as

$$\kappa = \frac{\dot{\psi}_d}{V_a} \quad (2.27)$$

Using this equation (2.26) and (2.27), the curvature can be expressed as,

$$\begin{aligned} \kappa = \frac{1}{V_a^3} M_1 \left(\cos \phi_i \sin \phi_i (\dot{y}_d^2 - \dot{x}_d^2) + \dot{x}_d \dot{y}_d (\cos^2 \phi_i - \sin^2 \phi_i) \right) \\ + \frac{1}{V_a^3} M_2 (\dot{x}_d \cos \phi_i + \dot{y}_d \sin \phi_i)^2 \end{aligned} \quad (2.28)$$

where, M_1 and M_2 are defined as follows:

$$M_1 = -\text{sign}(x_e) \frac{f_2}{\sqrt{V_a^2 - f_2^2}} c (V_a - a) \text{sech}(cx_e) \tanh(cx_e) \quad (2.29)$$

$$M_2 = -c(V_a - a) \text{sech}(cx_e) \tanh(cx_e) \quad (2.30)$$

The path generated by the vector field (2.24) is feasible if $\kappa \leq \kappa_{max}$. Since it is observed that κ is the function of a single parameter, c , it becomes a necessity to choose c appropriately to ensure the feasibility of the path generated by the vector field.

Selection of Control Parameter (c) for Optimality

In this section the curvature of the designed path will be optimized to utilize the maximum available control effort of the UAV to reach the closest distance from

2.6 Simulation Results

the waypoint. The optimization problem can be formulated as,

$$\begin{aligned} & \max \kappa(x_e, c), \quad \text{or} \quad \min -\kappa(x_e, c) \\ & \text{such that} \quad \begin{cases} \kappa(x_e, c) - \kappa_{max} \leq 0 \\ x_{lb} \leq x_e \leq x_{ub} \\ c_{lb} \leq c \leq c_{ub} \end{cases} \end{aligned} \quad (2.31)$$

where, x_{lb} and c_{lb} are the lower bounds of x_e and c , respectively. And x_{ub} and c_{ub} are the upper bounds of x_e and c , respectively. $\kappa_{max} = \frac{1}{r_{min}}$, is the maximum curvature bound of the UAV.

Note that the minimum turn radius of a quadcopter for constant speed depends on various factors such as the size and weight of the quadcopter, the thrust generated by the motors, the speed of the quadcopter, and the angle of the bank during the turn. The minimum turn radius for a quadcopter is typically determined experimentally by analyzing the performed trajectories in given conditions. In this work an estimated value of the minimum turn radius is obtained by considering the point mass model of the quadcopter.

$$r_{min} = \frac{V_a^2}{g \sin(\phi_p)} \quad (2.32)$$

where, ϕ_p is the pitch angle made by the quadcopter with respect to the horizontal plane. g is the acceleration due to gravity.

The nonlinear programming solver ‘fmincon’ in MATLAB is used to solve the optimization problem to find out the optimal value of c . For a trajectory in Fig. 2.8 the optimal c value which is obtained using this optimization is 0.1 and the curvature profile of the trajectory is depicted in the Fig. 4.7. The trajectories obtained using this algorithm will have continuous curvature as shown in Fig. 4.7 and take the benefit of the maximum possible curvature to reach the closest distance to the waypoint.

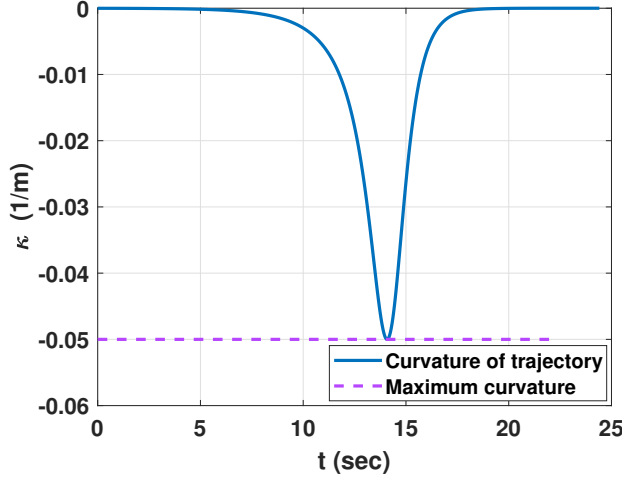


Figure 2.9: Curvature profile

2.6 Simulation Results

In this section, the experimental and numerical simulation results are presented for waypoint following using the proposed guidance algorithm. All the numerical simulations were carried out in MATLAB 2020b on a machine with an Intel Core i7-9750H Processor and 16 GB RAM. The practical experiments conducted on a quadrotor in an indoor lab setup. The lab setup consists of a quadrotor, 6 motion capture cameras, a desktop ground station, and a Wi-Fi router, with the connections between each component as illustrated in Fig. 2.10. The proposed guidance algorithm operates on the ground station, utilizing sensor data from the motion capture cameras and user-defined waypoints to generate the desired position commands. These reference command inputs are transmitted via Wi-Fi wireless communication to the onboard computer of the quadrotor. The embedded computer on the quadrotor processes the desired poses into the four control parameters (Roll, Pitch, Yaw angles, and total Thrust), which are then mapped to the four angular velocity inputs of the four rotors of the quadrotor. The embedded computer executes these control signals in real-time to drive the motors. The objective of the experimental study is to accurately follow the desired waypoint segments to the maximum possible length without compromising the turn radius constraints. In this section, the performance of the vector field for two different

2.6 Simulation Results

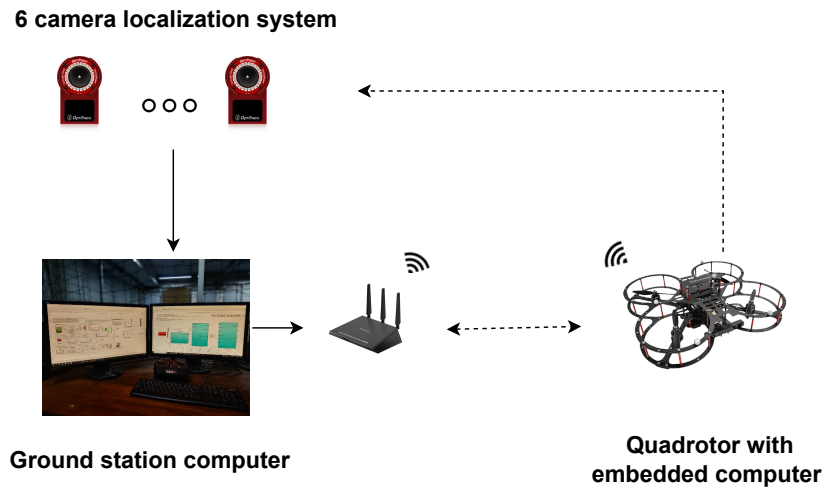


Figure 2.10: Quadrotor indoor lab setup

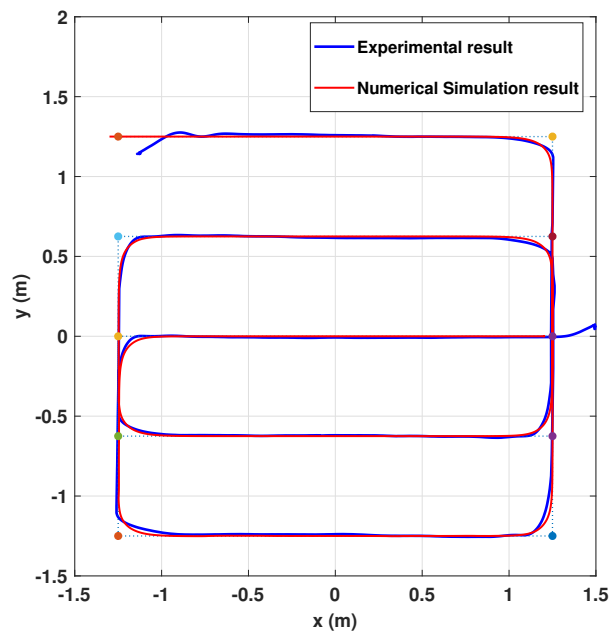


Figure 2.11: Experimental and numerical results of waypoint following

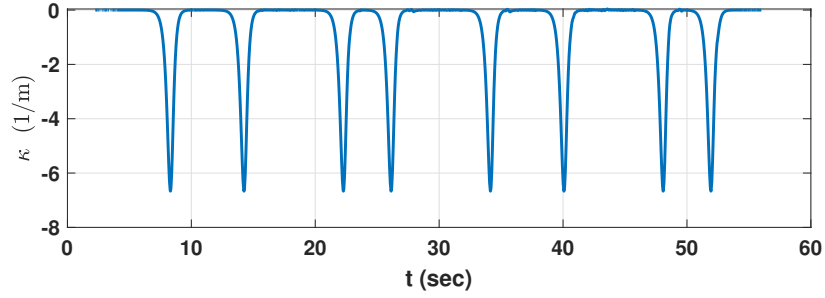


Figure 2.12: curvature profile

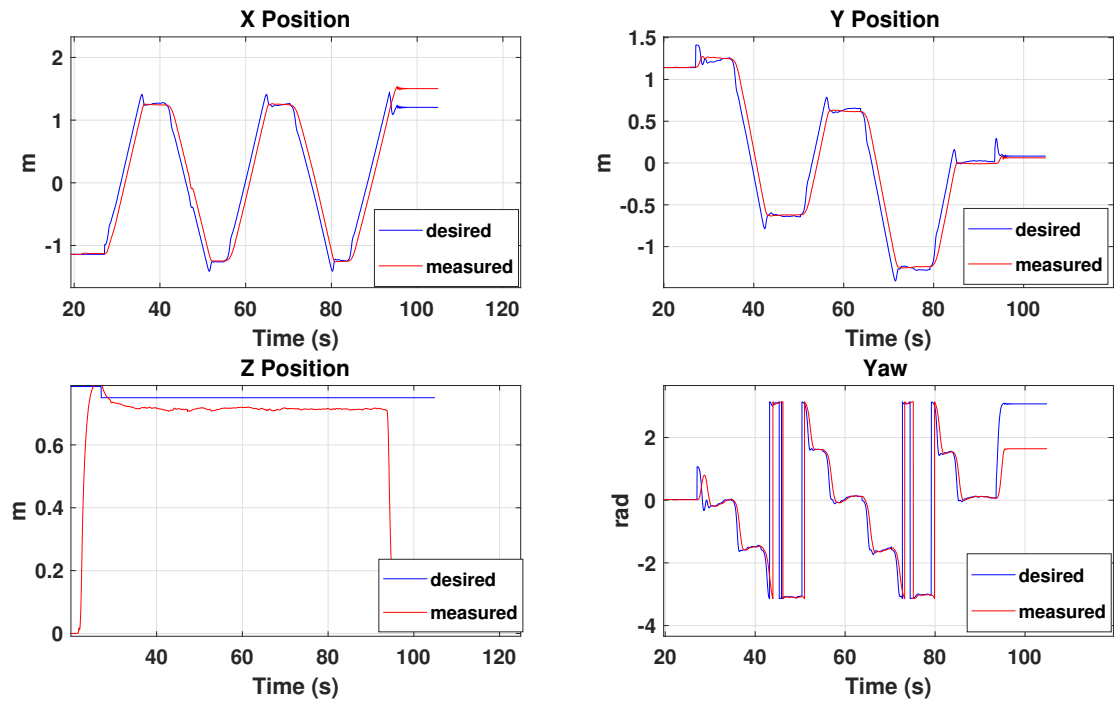


Figure 2.13: X, Y, Z positions and Yaw angle profiles

2.6 Simulation Results

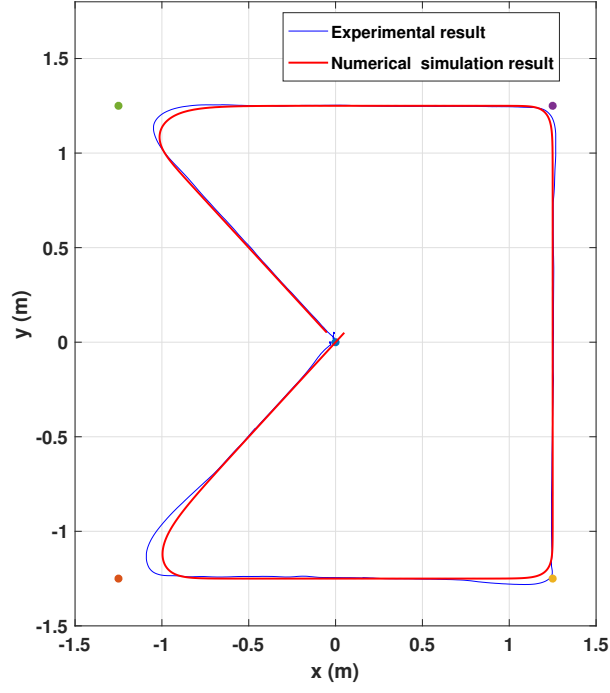


Figure 2.14: Experimental and numerical results of waypoint following

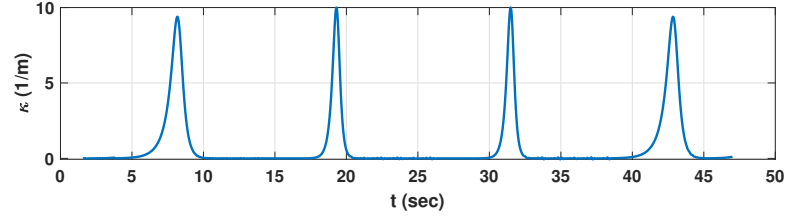


Figure 2.15: curvature profile

scenarios are demonstrated.

In the first example, a set of waypoints with following X and Y coordinates are considered,

$$X_w = 1.25 \begin{bmatrix} -1 & 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \end{bmatrix}^T$$

$$Y_w = 1.25 \begin{bmatrix} 1 & 1 & -0.5 & -0.5 & 0.5 & 0.5 & -1 & -1 & 0 & 0 \end{bmatrix}^T$$

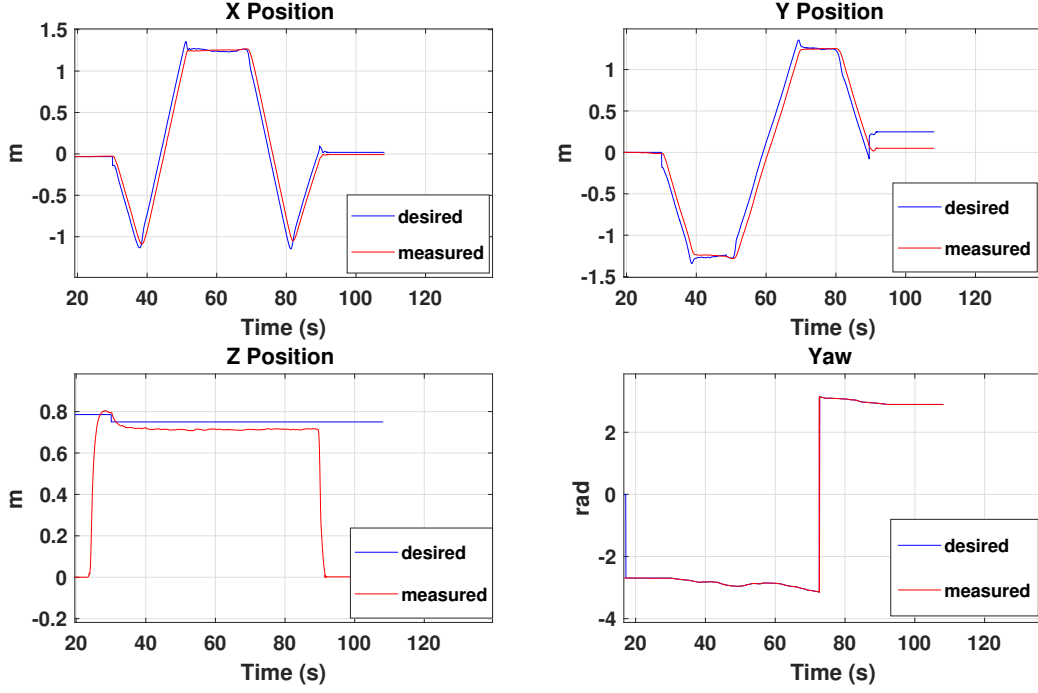


Figure 2.16: X, Y, Z positions and Yaw angle profiles

The results of the experimental and numerical waypoint following are shown in Fig. 2.11, indicating accurate tracking of the generated paths by the quadrotor. Furthermore, the curvature profile of the numerically simulated path is depicted in Fig. 2.12, demonstrating that the curvature remains smooth throughout the trajectory. The desired and achieved X and Y positions of the quadrotor along with the yaw angle are depicted in Fig. 2.13. In the second example, another set of 6 waypoints with the following x and y coordinates are considered:

$$X_w = 1.25 \begin{bmatrix} 0 & -1 & 1 & 1 & -1 & 0 \end{bmatrix}^T$$

$$Y_w = 1.25 \begin{bmatrix} 0 & -1 & -1 & 1 & 1 & 0 \end{bmatrix}^T$$

The experimental and numerical results show that the quadrotor can accurately track the generated path, as depicted in Fig. 2.14. Furthermore, the curvature profile of the numerically simulated path remains smooth throughout the trajectory, as illustrated in Fig. 2.15. The desired and achieved X and Y positions of

2.7 Summary

the quadrotor along with the yaw angle, are presented in Fig. [2.16](#).

2.7 Summary

In this chapter, a novel Lyapunov Guidance Vector Field (LGVF) algorithm has been proposed for guiding UAVs towards waypoints, while taking into account the minimum turn radius constraint. The LGVF algorithm utilizes normalized even and odd functions to achieve path convergence, while maintaining constant velocity and altitude, and it is computationally simple. The algorithm produces a continuous curvature and near-optimal trajectory with respect to time as the path generation is guided by Dubins optimal path. The LGVF-based algorithm is applicable in several waypoint following applications in real world missions. To demonstrate it the proposed algorithm has been successfully implemented and tested on a quadrotor platform. The proposed LGVF algorithm has significant potential for improving the performance and efficiency of autonomous navigation systems. Its simplicity makes it a promising candidate for use in a wide range of surveillance and target tracking applications.

Waypoint Following in 3D Space

3.1 Introduction

The proposed Lyapunov Guidance Vector Field (LGVF) approach for 2D waypoint following is now extended to three-dimensional (3D) space taking into account turn rate constraints. The method minimizes tracking errors and provides practical solutions for 3D surveillance applications suitable for nonholonomic autonomous vehicles. Numerical simulations have been conducted to evaluate the performance of the proposed algorithm, demonstrating its effectiveness in waypoint navigation in 3D environments.

The problem is detailed in Section 3.2. Proposed solution algorithm i.e. the Lyapunov Vector Field approach, is discussed in Section 3.3. Simulation results are presented in Section 3.4 and experimental findings are shown in Section 3.5. Finally, Section 3.6 summarizes the contribution.

3.2 Problem Definition

This work explores the guidance problem for UAVs to guide them along straight lines connecting consecutive waypoints in three-dimensional space, using C^2 smooth paths.

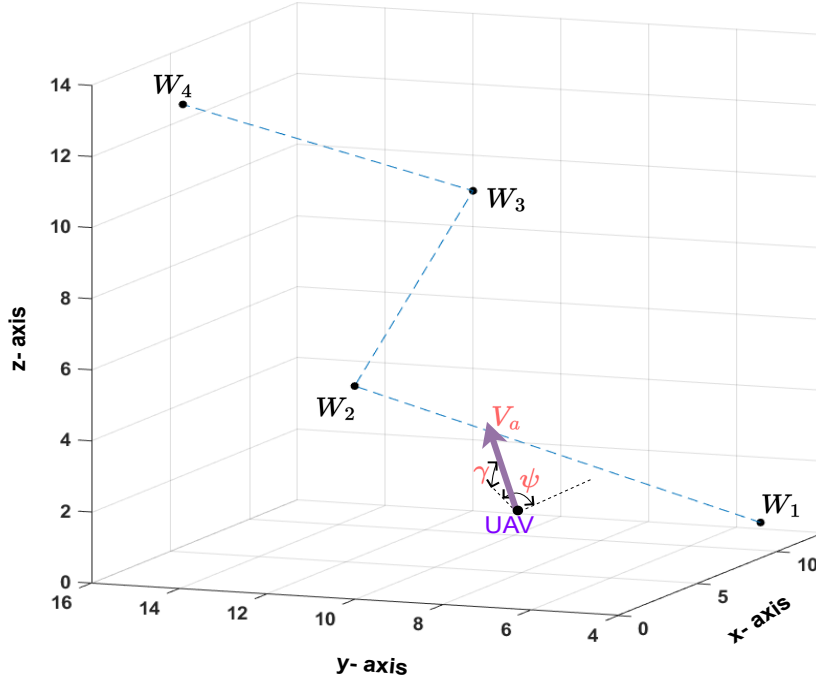


Figure 3.1: Illustration of waypoint following in 3D scenario

Consider a UAV is positioned in a fixed inertial frame of reference in a Cartesian coordinate system. The following kinematic equations govern the UAV.

$$\begin{aligned}
 \dot{x} &= V_a \cos \gamma \cos \psi \\
 \dot{y} &= V_a \cos \gamma \sin \psi \\
 \dot{z} &= V_a \sin \gamma \\
 \dot{\psi} &= u_\psi \\
 \dot{\gamma} &= u_\gamma
 \end{aligned} \tag{3.1}$$

Here, $p_v = [x \ y \ z]^T$ and $\mathbf{v} = [\dot{x} \ \dot{y} \ \dot{z}]^T$ represent the position and velocity vectors of the UAV in the inertial reference frame, respectively. V_a denotes the constant airspeed of the UAV. $\psi \in (-\pi, \pi]$ and $\gamma \in (-\pi, \pi]$ represent the heading and flight path angles, respectively.

3.3 Proposed Lyapunov Vector Field

Consider a set of N waypoints denoted as W_i , where $i \in [1, N]$, in the inertial frame of reference, as shown in Fig. 3.1. The control inputs u_ψ and u_γ are used to guide the UAV along the waypoint segments connecting consecutive waypoints. In this work, the objective is to determine the appropriate control variables, u_ψ and u_γ , to achieve this guidance.

To address this problem, the proposed LVGF based guidance law presented in Chapter 2 is extended in the following section to achieve faster convergence by generating C^2 smooth paths for waypoint following in 3D space.

3.3 Proposed Lyapunov Vector Field

Here, a right handed reference coordinate frame $\mathcal{F}_w$, centered at the waypoint W_i is assumed. Let $p_{ww} = W_{i+1} - W_i$ and $p_{wv} = p_v - p_w$ be the positional vectors of the next waypoint and the UAV from the current waypoint, respectively as shown in Fig. 3.2. U_{ww} and U_{wv} are the unit vectors in the direction of p_{ww} and p_{wv} , respectively. The position vectors of the UAV and the next waypoint from the current waypoint lie on the XY -plane of the reference frame ($\mathcal{F}_w$). In this XY -plane, Y -axis points to the direction of position vector, p_{ww} . Let U_x , U_y and U_z be the unit vectors along the X , Y and Z axis of the reference frame $\mathcal{F}_w$.

The Z -axis to be perpendicular to the plane formed by U_{ww} and U_{wv} , so,

$$U_z = \frac{U_{wv} \times U_{ww}}{|U_{wv} \times U_{ww}|} \quad (3.2)$$

And the unit vector in the direction of X - axis is,

$$U_x = \frac{U_y \times U_z}{|U_y \times U_z|} \quad (3.3)$$

Let θ_x , θ_y , and θ_z represent three ZYX sequence Euler angles to define the current attitude of the frame $\mathcal{F}_w$ with respect to the global reference frame. Then the rotational matrix, R_{gw} , which transforms the coordinates from the global frame ($\mathcal{F}_g$) to $\mathcal{F}_w$, is given by:

$$R_{gw} = R_z R_y R_x \quad (3.4)$$

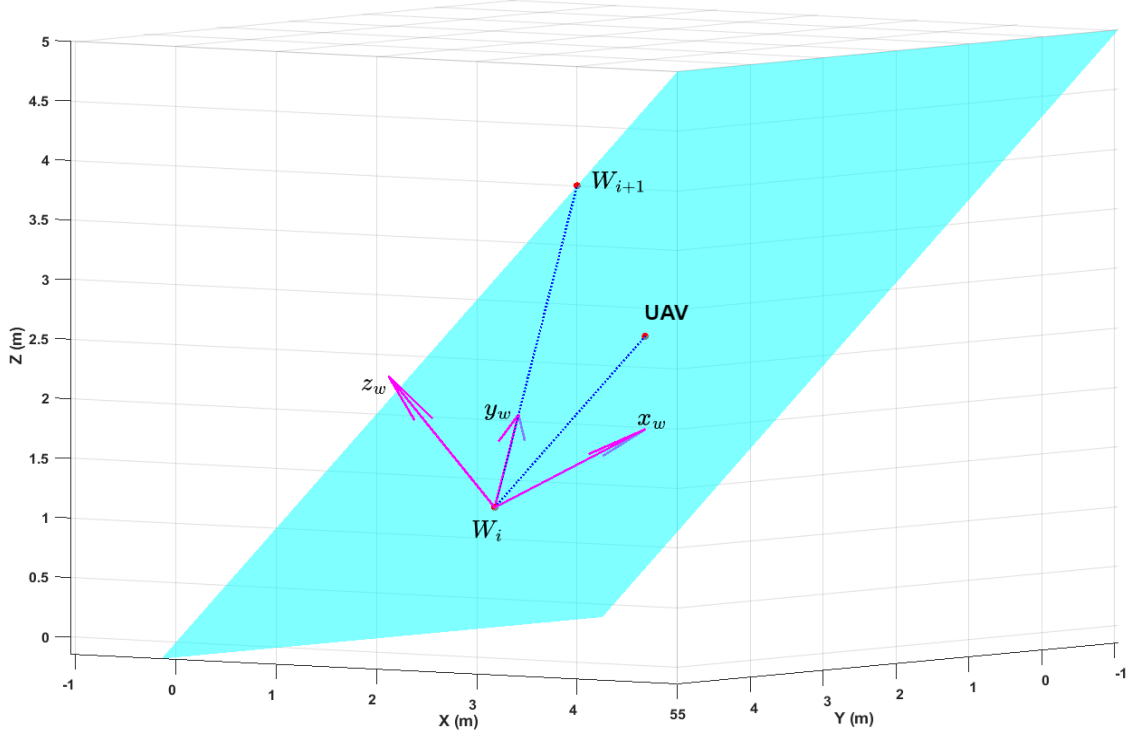


Figure 3.2: Illustration of waypoint following in 3D scenario

where,

$$R_z = \begin{bmatrix} \cos \theta_z & -\sin \theta_z & 0 \\ \sin \theta_z & \cos \theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3.5)$$

$$R_y = \begin{bmatrix} \cos \theta_y & 0 & \sin \theta_y \\ 0 & 1 & 0 \\ -\sin \theta_y & 0 & \cos \theta_y \end{bmatrix} \quad (3.6)$$

$$R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta_x & -\sin \theta_x \\ 0 & \sin \theta_x & \cos \theta_x \end{bmatrix} \quad (3.7)$$

The approach involves aligning the XY -plane of the Waypoint fixed reference frame with the global XY -plane. Next, the lateral distance error x_e is determined in this $2D$ plane. A guidance vector field is then developed based on this error,

3.4 Simulation Results

as discussed in Chapter 2. Finally, the $2D$ field is transformed back into $3D$ to obtain the reference commands as follows

$$\begin{bmatrix} W_x \\ W_y \\ W_z \end{bmatrix} = R_z R_y R_x \begin{bmatrix} \dot{x}_d \\ \dot{y}_d \\ 0 \end{bmatrix} \quad (3.8)$$

From (3.8), the commanded heading and flight path angles can be obtained as follows:

$$\begin{aligned} \psi_c &= \arctan(W_y/W_x) \\ \gamma_c &= \arctan(W_z/\sqrt{W_x^2 + W_y^2}) \end{aligned} \quad (3.9)$$

Using ψ_c and γ_c , the control inputs can be generated using any convergence guaranteed steering algorithms. Here, the control inputs u_ψ and u_γ are obtained using first order hold loops as follows:

$$\begin{aligned} u_\psi &= \alpha_1(\psi_c - \psi) \\ u_\gamma &= \alpha_2(\gamma_c - \gamma) \end{aligned} \quad (3.10)$$

where, α_1 and α_2 are two positive proportional gains.

3.4 Simulation Results

This section presents numerical simulation results for waypoint following in 3D space to validate the proposed vector field guidance algorithm. All the numerical simulations have been carried out in MATLAB 2020b on a machine with an Intel Core i7-9750H Processor and 16 GB RAM. Following data are considered for the simulation.

Set of 9 waypoints with following X , Y and Z coordinates are considered:

$$X_w = \begin{bmatrix} -1 & 1 & 1 & -1 & 1 & 1 & -1 & 1 & -1 \end{bmatrix} \times 1.25$$

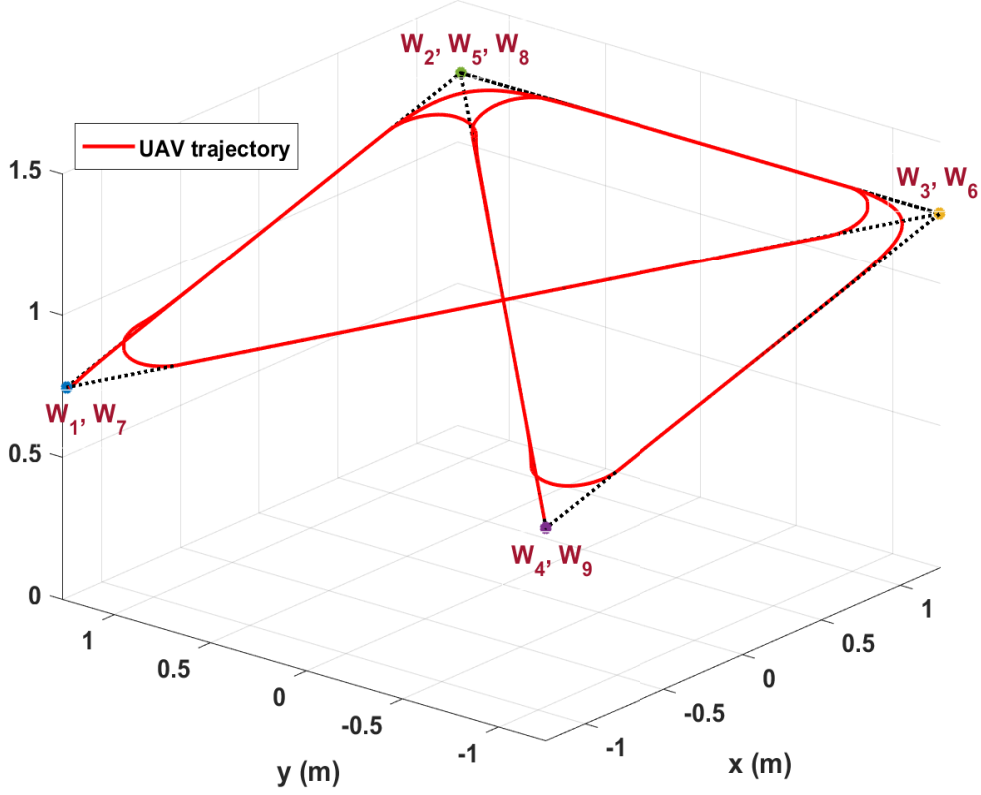


Figure 3.3: Waypoint following in 3D space

$$Y_w = \begin{bmatrix} 1 & 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix} \times 1.25$$

$$Z_w = \begin{bmatrix} 0.75 & 1.25 & 1.25 & 0.75 & 1.25 & 1.25 & 0.75 & 1.25 & 0.75 \end{bmatrix}$$

The UAV is moving with a constant linear speed of 0.4 m/sec to track the desired waypoints from its initial position, (1.25, 1.25, 0.75) m, with initial heading angle 0° and flight path angle of 0° . The maximum and the minimum angular speeds of the UAV are 2 rad/sec and -2 rad/sec, respectively. Fig. 3.3 shows the trajectory of the UAV. It can be seen from the figure that the path is very smooth and the tracking accuracy to follow waypoint segments connected between the waypoints

3.5 Experimental Results

is very good.

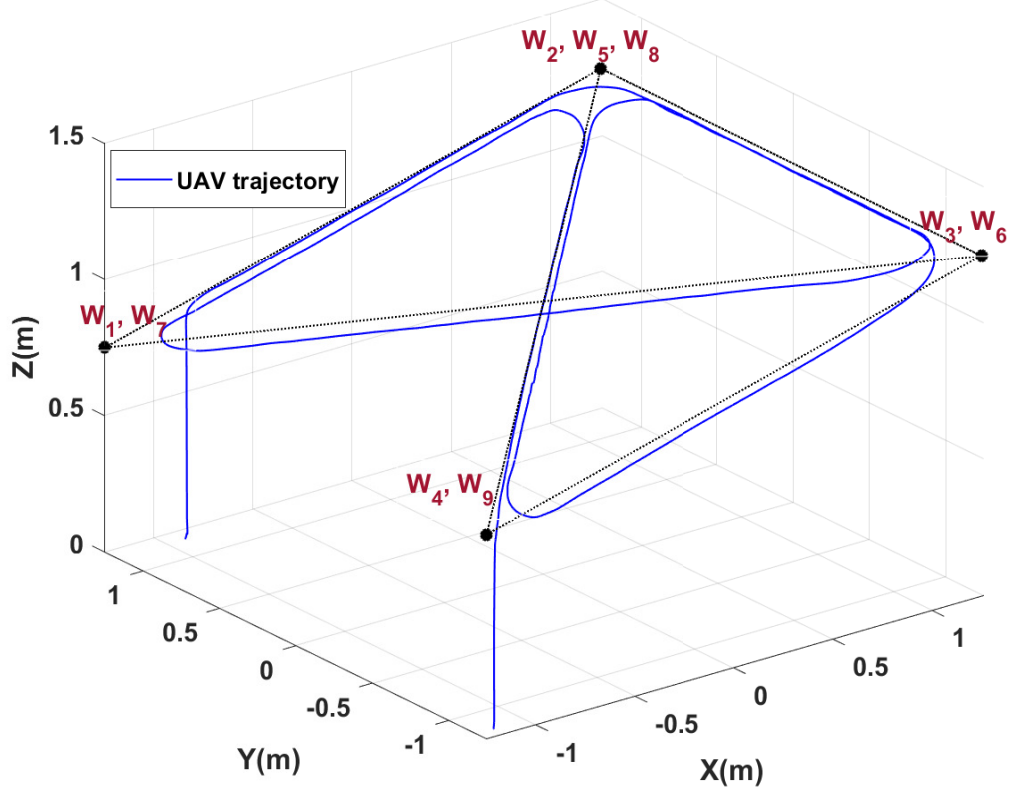


Figure 3.4: Experimental result of waypoint following in 3D

3.5 Experimental Results

In this section, the performance of the proposed guidance algorithm is demonstrated practically using a quadrotor in an indoor lab. A brief introduction for this lab setup has been given Chapter 2, Section 2.6. Fig. 3.4 shows the trajectory of the quadrotor. The desired and achieved X , Y and Z positions of the quadrotor are depicted Fig. 3.5. It shows the applicability of the proposed approach in the real-world.

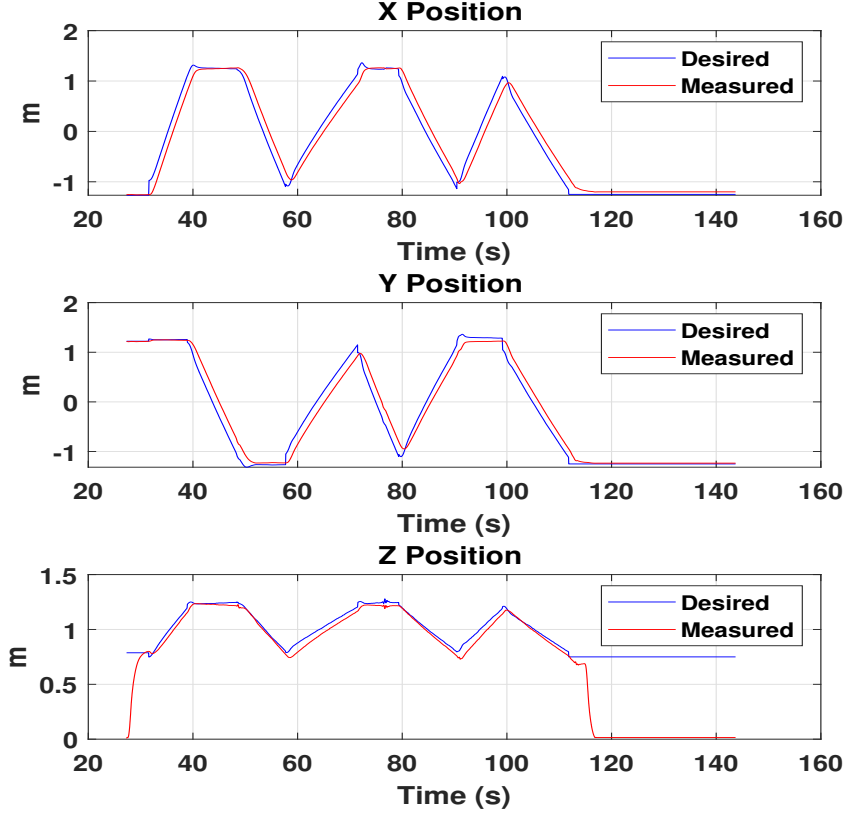


Figure 3.5: X, Y and Z position profiles

3.6 Summary

This chapter introduces a new method for guiding UAVs in 3D space along a sequence of waypoints. To achieve this, the existing 2D guidance algorithm is modified to work in 3D scenarios. A new reference frame is created at each waypoint, aligning the path of the UAV and the waypoint segment to the XY-plane of this frame. Then the vector field guidance algorithm from Chapter 2 is applied to this XY-plane to guide the UAV towards the waypoints smoothly. To test the proposed approach, experiments are conducted using a quadrotor in an indoor lab setup. The results demonstrate that the algorithm successfully guides the UAV for waypoint following in 3D while adhering to the curvature constraints. This research opens up promising possibilities for autonomous UAV navigation in

3.6 Summary

complex environments, benefiting applications such as surveillance, mapping, and search-and-rescue missions, contributing to the advancement of UAV technology in real-world.

Standoff Target Tracking

4.1 Introduction

Target tracking while maintaining a constant standoff distance from the target is called as standoff target tracking. It can be accomplished by a fixed-wing UAV by making it loitered around a target. The speed of the UAV should be more than the speed of the target as the fixed-wing UAV needs to maintain a forward motion to avoid stalling. In two dimensional (2D) plane it can be accomplished by a robot moving on the same plane or by a fixed-wing UAV loitering around the target in a circular path while keeping its altitude constant. There are different approaches available for target tracking in the existing literature.

A wide variety of vector fields can be generated by the Lyapunov vector field framework for path following, target tracking and obstacle avoidance applications. However, constructing a simple vector field to guide vehicles for faster convergence to a desired standoff circle considering maneuverability constraints of a vehicle is still a challenging area as in many applications UAVs are initially deployed from sufficiently large distance away from the desired circle but the the life of the battery of UAVs is limited. Hota and Ghose [57] and Alok and Hota [58] solved the optimal path planning problem for converging to a straight line and a circle by using geometrical arguments but the curvature profile of the path is not

continuous. This paper uses the findings of Ref. [58] to generate a path with the novelty that it has a smooth curvature profile and is very close to the optimal path. The guidance law is also explicitly a function of the minimum turn radius of the vehicle, and hence it is applicable to different maneuverable vehicles. The other advantage of the algorithm is that the computation time is very fast, which makes it feasible for implementation in real-time as demonstrated through simulations.

The chapter is organized as follows: In Section 4.1, the concept of standoff target tracking is introduced. The problem is formally defined in Section 4.2. The background of the problem is discussed in 4.3. Our proposed solution, the Lyapunov Vector Field approach, is detailed in Section 4.4, demonstrating its effectiveness in efficiently tracking targets from a standoff distance. Section 4.6 showcases the results of extensive simulation experiments, evaluating the performance of our approach in various tracking scenarios. Experimental results are presented in Section 4.7. Finally, in Section 4.8, our findings in standoff target tracking are summarized.

4.2 Problem Definition

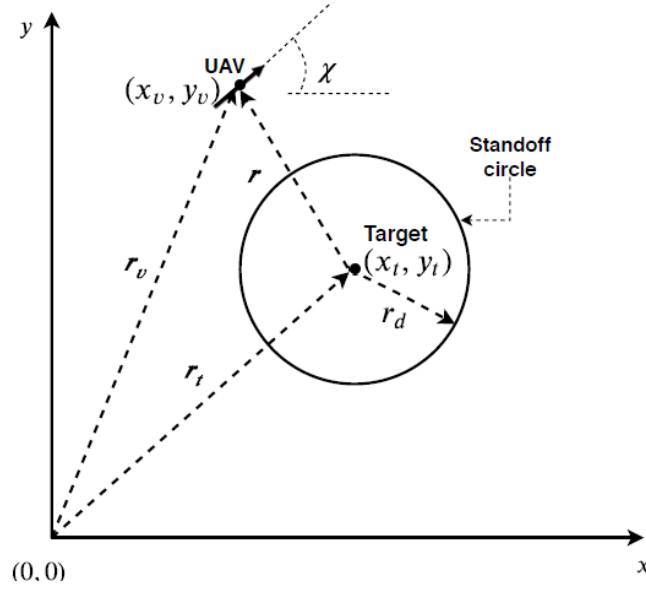


Figure 4.1: Scenario of target tracking

4.2 Problem Definition

In the fixed reference frame as shown in Fig. 4.1, the target and the UAV are located at $(x_t \ y_t)$ and $(x_v \ y_v)$, respectively. Here, $\mathbf{r}_v$ is the position vector of the UAV and $\mathbf{r}_t$ is the position vector of the target. The relative position vector of the UAV with respect to the target is denoted by $\mathbf{r} (= \mathbf{r}_v - \mathbf{r}_t)$. To converge to the circular path of radius r_d as shown in the figure, a feasible Lyapunov guidance vector field ($\dot{\mathbf{r}}$) with an objective of faster convergence has to be constructed. This field can produce the reference course angle (χ_d) commands to the higher-level control (guidance) loop. The guidance loop generates angular velocity commands, which should be followed by the vehicle for circular path convergence.

To study the performance of the circular path convergence the following kinematic equations of an unmanned vehicle are considered.

$$\begin{aligned}\dot{x} &= V_a \cos \psi + W_x \\ \dot{y} &= V_a \sin \psi + W_y\end{aligned}\tag{4.1}$$

where, V_a is the airspeed and ψ is the heading angle of the UAV. W_x and W_y are the wind velocity components along the X and Y directions, respectively as shown in Fig. 4.2. It is assumed that altitude and airspeed are held constant by the control of longitudinal dynamics in line with the work presented in Ref. [26, 28] and [51].

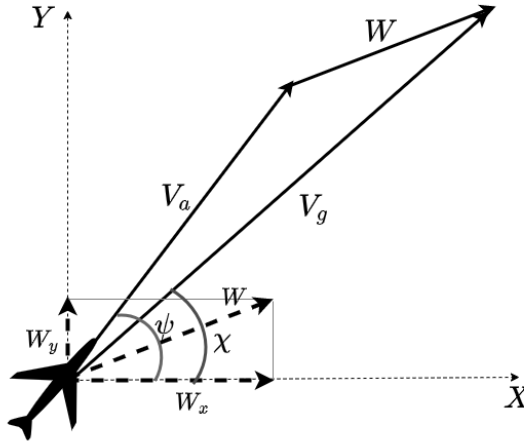


Figure 4.2: Velocity triangle

The kinematic equations (4.1) can be rewritten as

$$\begin{aligned}\dot{x} &= V_g \cos \chi \\ \dot{y} &= V_g \sin \chi\end{aligned}\tag{4.2}$$

where, χ is the course angle of the UAV. V_g is the ground speed of the UAV, which can be expressed as,

$$V_g = \sqrt{V_a^2 + W^2 + 2V_a(W_x \cos \psi + W_y \sin \psi)}\tag{4.3}$$

where,

$$W = \sqrt{W_x^2 + W_y^2}$$

To generate the guidance law, here we consider the first-order course angle dynamics as

$$\dot{\chi} = \alpha(\chi_d - \chi) \quad (\chi_d - \chi) \in (-\pi, \pi]\tag{4.4}$$

where, α is the proportional gain. The relation between the course angle rate and the heading angle rate can be described as [51]

$$\dot{\chi} = F(\psi)\dot{\psi}\tag{4.5}$$

where $F(\psi)$ can be calculated as follows:

$$\dot{\chi} = \frac{d}{dt} \tan^{-1} \left(\frac{V_a \sin \psi + W_y}{V_a \cos \psi + W_x} \right)\tag{4.6}$$

$$= \dot{\psi} \frac{V_a^2 + V_a W_x \cos \psi + V_a W_y \sin \psi}{(V_a \cos \psi + W_x)^2 + (V_a \sin \psi + W_y)^2}\tag{4.7}$$

$$= \dot{\psi} \frac{V_a^2 + V_a(W_x \cos \psi + W_y \sin \psi)}{V_g^2}\tag{4.8}$$

Therefore, the function $F(\psi)$ is given by:

$$F(\psi) = \frac{V_a^2 + V_a(W_x \cos \psi + W_y \sin \psi)}{V_g^2}\tag{4.9}$$

4.3 Background

The heading rate constraints on the UAV can be imposed indirectly by constraining the course angle rate using (4.5). For feasible operation $W < V_a$ and hence, from (4.9), it can be observed that $F(\psi) \in (0.5, 1]$. The course angle rate constraints can be set as $\dot{\chi} = 0.5\dot{\psi}$ for safe operation.

4.3 Background

In this section a brief review on the Lyapunov vector field based guidance for unmanned vehicles is presented.

Lawrence et al.[21] constructed the Lyapunov vector fields with the desired closed attractor as a globally asymptotically stable limit cycle. It is done by considering the Lyapunov function, $V(\mathbf{r})$ such that on the desired attractor it has zero energy, elsewhere it has positive energy and also $V(\mathbf{r})$ is radially unbounded, that is, as $\mathbf{r} \rightarrow \infty, V(\mathbf{r}) \rightarrow \infty$.

The total time derivative of Lyapunov function is given by

$$\dot{V}(\mathbf{r}) = \frac{\partial V}{\partial \mathbf{r}} \dot{\mathbf{r}} \quad (4.10)$$

For simpler analysis, without the loss of generality let us assume that the target is located at the origin so, $(x_t, y_t) = (0, 0)$ and hence $\mathbf{r}$ is $\begin{bmatrix} x_v & y_v \end{bmatrix}$.

To ensure that the total time derivative of the Lyapunov function is negative except on the attractor, the vector field ($\dot{\mathbf{r}}$) is defined as the sum of convergence ($A(\mathbf{r})$) and circulation terms ($S(\mathbf{r})$) as given below:

$$\dot{\mathbf{r}} = A(\mathbf{r}) + S(\mathbf{r}) \quad (4.11)$$

such that

$$\frac{\partial V}{\partial \mathbf{r}} A(\mathbf{r}) < 0, \quad (4.12)$$

$$\frac{\partial V}{\partial \mathbf{r}} S(\mathbf{r}) = 0 \quad (4.13)$$

and for a constant vector field

$$\sqrt{A(\mathbf{r})^2 + S(\mathbf{r})^2} = V_a \quad (4.14)$$

$S(\mathbf{r})$ signifies the required modulation of the tangent curves of the vector field while converging to the desired attractor.

Satisfying the equations, (4.12) (4.13) and (4.14), Lawrence et al. [18] proposed a Lyapunov guidance vector field for a standoff target tracking, which can be described as

$$\begin{bmatrix} \dot{x}_d \\ \dot{y}_d \end{bmatrix} = \begin{bmatrix} -\frac{r^2-r_d^2}{r^2+r_d^2}x_v - \frac{2rr_d}{r^2+r_d^2}y_v \\ -\frac{r^2-r_d^2}{r^2+r_d^2}y_v + \frac{2rr_d}{r^2+r_d^2}x_v \end{bmatrix} \quad (4.15)$$

where, $r (= \mathbf{r}^T \hat{\mathbf{r}})$ is the horizontal range between the unmanned vehicle and the target on the loitering plane. $\dot{x}_d$ and $\dot{y}_d$ are the desired inertial velocity components along X -axis and Y -axis, respectively. The unit vector in the direction of $\mathbf{r}$ is represented by $\hat{\mathbf{r}}$.

Similarly, by modifying the circulation term for faster convergence Pothen and Ratnoo [20] proposed a curvature constrained Lyapunov guidance vector field, which can be described as

$$\begin{bmatrix} \dot{x}_d \\ \dot{y}_d \end{bmatrix} = \frac{-V_a}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \begin{bmatrix} \frac{r^2-r_d^2}{r}x_v + cr_dy_v \\ \frac{r^2-r_d^2}{r}y_v - cr_dx_v \end{bmatrix} \quad (4.16)$$

where,

$$c = \begin{cases} \frac{r}{r_d} & \text{if } r < r_d \\ \frac{r_d}{r} & \text{if } r \geq r_d \end{cases}$$

4.4 Proposed Lyapunov Vector Field

In this section, a new guidance vector field is proposed using the Lyapunov vector field guidance approach for faster convergence to the standoff circle. The objective is to generate the path which is of smooth curvature profile satisfying the maximum curvature limit of a fixed-wing vehicle and the length of the path should be close to the optimal one. For this, a positive definite scalar valued function $V(\mathbf{r})$ is considered by treating the desired standoff circle with respect to the target as a zero level set, which is given below:

$$V(\mathbf{r}) = (r - r_d)^2 \quad (4.17)$$

4.4 Proposed Lyapunov Vector Field

The total time derivative of the Lyapunov function is given by

$$\dot{V}(\mathbf{r}) = \frac{\partial V}{\partial \mathbf{r}} \frac{d\mathbf{r}}{dt} \quad (4.18)$$

where,

$$\begin{aligned} \frac{\partial V}{\partial \mathbf{r}} &= \begin{bmatrix} \frac{\partial V}{\partial x_v} & \frac{\partial V}{\partial y_v} \end{bmatrix} \\ &= \frac{2(r - r_d)}{r} \begin{bmatrix} x_v & y_v \end{bmatrix} \\ &= \frac{2(r - r_d)}{r} \mathbf{r} \\ &= 2(r - r_d) \hat{\mathbf{r}} \end{aligned} \quad (4.19)$$

The desired vector field, $\dot{\mathbf{r}}$, can be defined as the sum of the convergence (f_1) and circulation (f_2) components given below and they should be selected in such a manner that the total time derivative of the Lyapunov function, $\dot{V}(\mathbf{r})$, is negative definite.

$$\dot{\mathbf{r}} = -f_1 \hat{\mathbf{r}}^T \pm f_2 \hat{\mathbf{n}} \times \hat{\mathbf{r}}^T \quad (4.20)$$

where, $\hat{\mathbf{n}}$ is the unit normal vector to the loitering plane. There are two possible normal vectors: i) the positive convention is to indicate the normal vector which points up-side of the loitering plane, ii) the negative convention is to indicate the normal vector which points downside of the loitering plane.

Here, the convergence (f_1) and circulation (f_2) terms are normalized odd and even functions of $(r - r_d)$. They are defined by decomposing an exponential function, $a^{c(r-r_d)}$, into odd and even functions, and are normalized with the desired speed of the vehicle to generate a constant speed vector field as expressed below,

$$f_1 = v_g \frac{a^{c(r-r_d)} - a^{-c(r-r_d)}}{a^{c(r-r_d)} + a^{-c(r-r_d)}} \quad (4.21)$$

$$f_2 = v_g \frac{2}{a^{c(r-r_d)} + a^{-c(r-r_d)}} \quad (4.22)$$

where, the base of the exponential function, a , can be any positive constant greater than 1 and c is the control parameter which will be designed later to satisfy the

curvature bounds. Using (4.21) and (4.22), the equation (4.20) can be written as,

$$\dot{\mathbf{r}} = -v_g \frac{a^{c(r-r_d)} - a^{-c(r-r_d)}}{a^{c(r-r_d)} + a^{-c(r-r_d)}} \hat{\mathbf{r}}^T \pm v_g \frac{2}{a^{c(r-r_d)} + a^{-c(r-r_d)}} \hat{\mathbf{n}} \times \hat{\mathbf{r}}^T \quad (4.23)$$

Here,

$$\hat{\mathbf{n}} \times \hat{\mathbf{r}}^T = \begin{bmatrix} \cos(\frac{\pi}{2}) & -\sin(\frac{\pi}{2}) \\ \sin(\frac{\pi}{2}) & \cos(\frac{\pi}{2}) \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}}_v \\ \hat{\mathbf{y}}_v \end{bmatrix} \quad (4.24)$$

So, $\dot{\mathbf{r}}$ in (4.23) will become

$$\dot{\mathbf{r}} = -v_g \frac{a^{c(r-r_d)} - a^{-c(r-r_d)}}{a^{c(r-r_d)} + a^{-c(r-r_d)}} \begin{bmatrix} \hat{\mathbf{x}}_v \\ \hat{\mathbf{y}}_v \end{bmatrix} \pm v_g \frac{2}{a^{c(r-r_d)} + a^{-c(r-r_d)}} \begin{bmatrix} -\hat{\mathbf{y}}_v \\ \hat{\mathbf{x}}_v \end{bmatrix} \quad (4.25)$$

For clockwise loitering, the desired vector field can be represented in the Cartesian coordinate as

$$\begin{aligned} \dot{x}_d &= -\frac{v_g}{r} \frac{a^{c(r-r_d)} - a^{-c(r-r_d)}}{a^{c(r-r_d)} + a^{-c(r-r_d)}} x_v + \frac{v_g}{r} \frac{2}{a^{c(r-r_d)} + a^{-c(r-r_d)}} y_v \\ \dot{y}_d &= -\frac{v_g}{r} \frac{a^{c(r-r_d)} - a^{-c(r-r_d)}}{a^{c(r-r_d)} + a^{-c(r-r_d)}} y_v - \frac{v_g}{r} \frac{2}{a^{c(r-r_d)} + a^{-c(r-r_d)}} x_v \end{aligned} \quad (4.26)$$

The vector field in (4.26) forms the desired circle as an asymptotically stable limit cycle. The vectors of this field give the desired course angles as a function of position to follow the desired limit cycle. The expression for the desired course angle is,

$$\chi_d = \arctan \left(\frac{\dot{y}_d}{\dot{x}_d} \right) \quad (4.27)$$

Using (4.26) the above equation becomes,

$$\chi_d = \arctan \left(\frac{(a^{c(r-r_d)} - a^{-c(r-r_d)}) y_v - 2x_v}{(a^{c(r-r_d)} - a^{-c(r-r_d)}) x_v + 2y_v} \right) \quad (4.28)$$

The curvature ($\kappa = \frac{\dot{\chi}_d}{v_g}$) of the path developed by these vector fields is,

$$\kappa = \frac{2}{a^{c(r-r_d)} + a^{-c(r-r_d)}} \left[c \ln(a) \frac{a^{c(r-r_d)} - a^{-c(r-r_d)}}{a^{c(r-r_d)} + a^{-c(r-r_d)}} - \frac{1}{r} \right] \quad (4.29)$$

4.4 Proposed Lyapunov Vector Field

4.4.1 Target Tracking

Alok and Hota [58] generated optimal paths for convergence to the desired circle from different initial positions and orientations. It was shown there that if the initial position was sufficiently far from the circular path to be followed, the straight-line segment of the optimal CSC (Circular path of minimum turn radius- Straight line- Circular path of minimum turn radius) path should pass through the center of the circle, which is shown in Fig. 4.3. In this figure the UAV with a minimum turn radius of $r_{\min}$ converges to the circular path of radius r_d starting from the location (x_0, y_0) . Here, d and θ are defined as,

$$d = (r_{\min} + r_d) \cos \theta \quad (4.30)$$

$$\theta = \arcsin \left(\frac{r_{\min}}{r_{\min} + r_d} \right) \quad (4.31)$$

For optimal time convergence to the desired circular path the following properties can be observed:

Region 1. if $r > d$, the guidance law for optimal convergence is composed of the convergence term only.

Region 2. if $r_d < r \leq d$, the guidance law for optimal convergence is composed of both the convergence and the circulation terms.

Region 3. if $r = r_d$, the guidance law for optimal convergence is composed of the circulation term only.

The profiles of convergence and circulation terms in case of optimal time convergence are analyzed here by rewriting (4.25) as given below:

$$\begin{bmatrix} \dot{x}_v \\ \dot{y}_v \end{bmatrix} = \begin{bmatrix} -\hat{x}_v & \hat{y}_v \\ -\hat{y}_v & -\hat{x}_v \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix} \quad (4.32)$$

This gives,

$$\begin{bmatrix} f_1 \\ f_2 \end{bmatrix} = \begin{bmatrix} -\frac{x_v}{r} & -\frac{y_v}{r} \\ \frac{y_v}{r} & -\frac{x_v}{r} \end{bmatrix} \begin{bmatrix} \dot{x}_v \\ \dot{y}_v \end{bmatrix} \quad (4.33)$$

To visualize the variations of convergence and circulation terms with respect to r , an example is considered here with the following data: $x = -1.5 \text{ m}$, $y = 0 \text{ m}$,

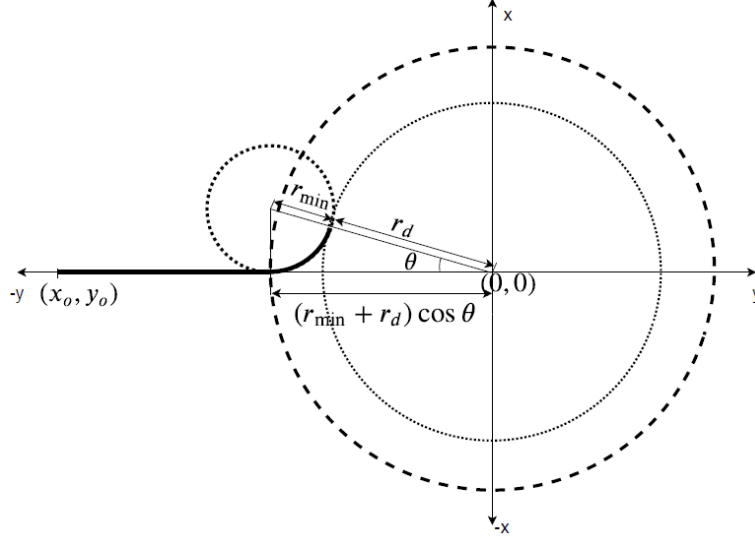


Figure 4.3: Dubins path with zero initial orientation angle

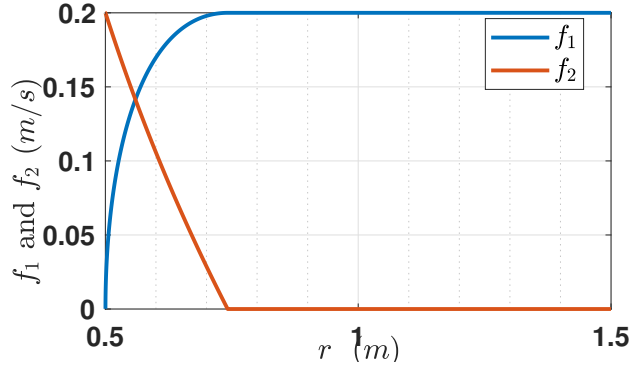


Figure 4.4: Variation of the convergence and circulation terms for optimal convergence

$r_d = 0.5 \text{ m}$, $r_{min} = 0.3 \text{ m}$ and $v_a = 0.2 \text{ m/s}$. The profiles of these two terms for optimal convergence are plotted in Fig. 4.4.

But the main problem to adopt the optimal CSC path for implementation is that the curvature profile of the path is not continuous, which makes it difficult for tracking by fixed-wing UAVs. This motivates us to choose the control parameter, c , of our guidance algorithm in such a way that it generates a smooth path and the length of which is very close to the optimal one.

4.4 Proposed Lyapunov Vector Field

4.4.2 Proposed Control Parameter, c

The convergence term, f_1 of (4.21) and the circulation term, f_2 of (4.22) can also be written as

$$f_1 = v_a \tanh(c(r - r_d) \ln(a)) \quad (4.34)$$

$$f_2 = v_a \operatorname{sech}(c(r - r_d) \ln(a)) \quad (4.35)$$

Let us define, $m = c(r - r_d) \ln(a)$. The variations of the convergence and the circulation functions with the variable m of the proposed guidance law is depicted in Fig. 4.6.

As stated above for optimal convergence, switching from Region 1 to Region 2, should occur at $r = d$. So the value of m at the transition, defined as k , will be expressed as

$$c(d - r_d) \ln a = k \quad (4.36)$$

which gives the control parameter, c as,

$$c = \frac{k}{(d - r_d) \ln a} \quad (4.37)$$

A trade-off between the feasibility and the optimality is made to choose the constant, k which can be of any value in the range $(0, 4)$ as shown in Fig. 4.6. The exact value of this can be selected using an optimization algorithm for the nonlinear multivariable function (5.29).

Let us define, $k_1 = (d - r_d) \ln a$. Note that for given r_d and r_{min} , k_1 is a constant. The equation, (5.29) can be re-written as

$$\kappa(r, k) = \frac{2}{a^{\frac{k}{k_1}(r-r_d)} + a^{-\frac{k}{k_1}(r-r_d)}} \left[\frac{k}{k_1} \ln(a) \frac{a^{\frac{k}{k_1}(r-r_d)} - a^{-\frac{k}{k_1}(r-r_d)}}{a^{\frac{k}{k_1}(r-r_d)} + a^{-\frac{k}{k_1}(r-r_d)}} - \frac{1}{r} \right] \quad (4.38)$$

The optimization problem can be formulated as

$$\max \kappa(r, k) \quad \text{such that} \quad \begin{cases} \kappa(r, k) - \kappa_{max} = 0 \\ r_{lb} \leq r \leq r_{up} \\ k_{lb} \leq k \leq k_{up} \end{cases} \quad (4.39)$$

where, r_{lb} and k_{lb} are the lower bounds of r and k , respectively. And r_{ub} and k_{ub} are the upper bounds of r and k , respectively. $\kappa_{max} = \frac{1}{r_{min}}$, is the maximum curvature bound of the UAV. The nonlinear programming solver ‘fmincon’ in MATLAB is used to solve the optimization problem to find out the optimal value of k . The above analysis holds good for convergence when the initial position is outside the standoff circle.

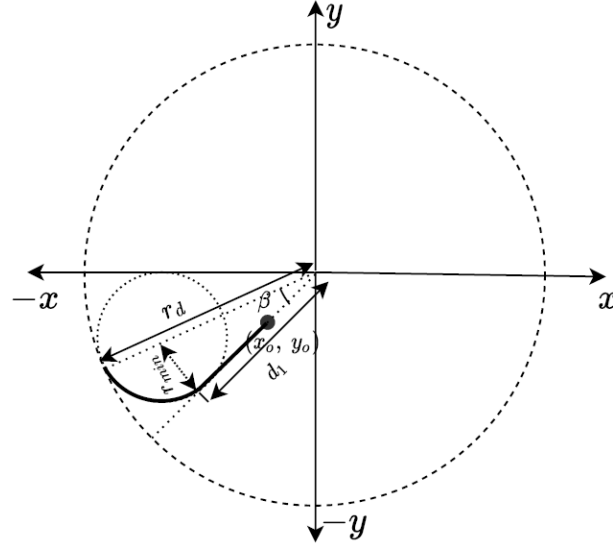


Figure 4.5: Optimal path of converging from inside standoff circle

Similar analysis can be done if the initial position is inside the standoff circle as shown in Fig. 4.5. In that case, the parameter, c , should be selected as

$$c = \frac{l}{d_1 - r_d} \quad (4.40)$$

here, l is a positive constant similar to k and in the range of $(0, 4)$. d_1 can be defined as

$$d_1 = (r_d - r_{min}) \cos \beta \quad (4.41)$$

4.5 Dynamic Model of a UAV for Path Following

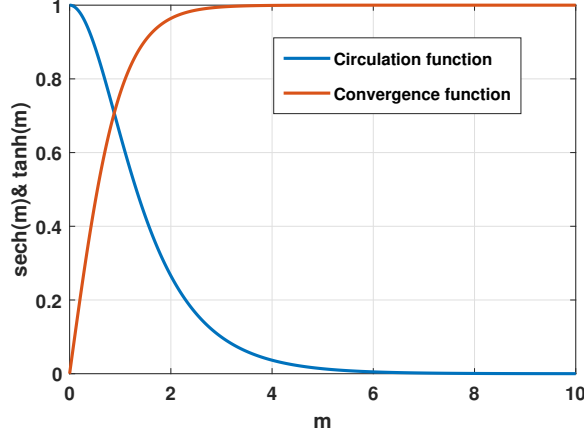


Figure 4.6: Circulation and convergence functions in the proposed guidance law

where,

$$\beta = \arcsin \left(\frac{r_{\min}}{r_d - r_{\min}} \right) \quad (4.42)$$

4.5 Dynamic Model of a UAV for Path Following

In the preceding section, we discussed the standoff target tracking algorithm utilizing the kinematic model of the UAV, treating the vehicle as a point mass in space. This approach was chosen to simplify the computational complexity of optimal path generation during trajectory planning. However, for algorithms designed to operate on real UAVs, accounting for the dynamics of the UAV becomes necessary. In this section, we utilize the dynamics of the vehicle to follow the pre-generated path. We employ a six-degree-of-freedom (6-DOF) model of a UAV [85] to execute the guidance commands required by proposed target tracking algorithm.

A rigid body in three-dimensional space is characterized by twelve state variables, representing three translational and three rotational degrees of freedom. The translational motion is defined by three position states: (x, y, z) , and three velocities along the body axis: (u, v, w) . Rotational motion is described by three angular rates about the body axis: (p, q, r) , and the Euler angles of the UAV: (ϕ, θ, ψ) .

The position of UAV can be determined using the following equations[86]:

$$\begin{aligned}\dot{x} = & \cos(\theta) \cos(\psi)u + (\sin(\phi) \sin(\theta) \cos(\psi) - \cos(\phi) \sin(\psi))v \\ & + (\cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi))w\end{aligned}\quad (4.43)$$

$$\begin{aligned}\dot{y} = & \cos(\theta) \sin(\psi)u + (\sin(\phi) \sin(\theta) \sin(\psi) + \cos(\phi) \cos(\psi))v \\ & + (\cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi))w\end{aligned}\quad (4.44)$$

$$\dot{z} = \sin(\theta)u - \sin(\phi) \cos(\theta)v - \cos(\phi) \cos(\theta)w \quad (4.45)$$

The rotational kinematics are described by the following equations,

$$\dot{\phi} = p + q \sin(\phi) \tan(\theta) + r \cos(\phi) \tan(\theta) \quad (4.46)$$

$$\dot{\theta} = q \cos(\phi) - r \sin(\phi) \quad (4.47)$$

$$\dot{\psi} = q \sin(\phi) \sin(\theta) + r \cos(\phi) \sec(\theta) \quad (4.48)$$

The translational dynamic equations are expressed as

$$\dot{u} = rv - qw + \frac{F_x}{m} \quad (4.49)$$

$$\dot{v} = pw - ru + \frac{F_y}{m} \quad (4.50)$$

$$\dot{w} = qu - pv + \frac{F_z}{m} \quad (4.51)$$

where, m is the mass of the UAV and the forces F_x , F_y and F_z along X, Y and Z directions, respectively, are

$$F_x = -mg \sin(\theta) + \frac{1}{2}\rho V^2 S(C_{X0} + C_{X\alpha}\alpha + C_{Xq}\frac{cq}{V} + C_{X\delta_e}\delta_e) \quad (4.52)$$

$$+ \frac{1}{2}\rho S_{\text{prop}} \left(k_{\text{motor}}^2 C_{2X} \frac{\delta_T}{\delta_T^2} - V^2 \right) \quad (4.53)$$

$$F_y = mg \cos(\theta) \sin(\phi) + \frac{1}{2}\rho V^2 S(C_{Y0} + C_{Y\beta}\beta + C_{Yp}\frac{bp}{2V} + C_{Yr}\frac{br}{2V} + C_{Y\delta_a}\delta_a + C_{Y\delta_r}\delta_r) \quad (4.54)$$

$$F_z = mg \cos(\theta) \cos(\phi) + \frac{1}{2}\rho V^2 S(C_{Z0} + C_{Z\alpha} + C_{Zq}\frac{cq}{V} + C_{Z\delta_e}\delta_e) \quad (4.55)$$

4.5 Dynamic Model of a UAV for Path Following

Where, all the aerodynamic coefficients are defined as

$$C_{X0} = -C_{D0} \cos(\alpha) + C_{L0} \sin(\alpha) \quad (4.56)$$

$$C_{X\alpha} = -C_{D\alpha} \cos(\alpha) + C_{L\alpha} \sin(\alpha) \quad (4.57)$$

$$C_{Xq} = -C_{Dq} \cos(\alpha) + C_{Lq} \sin(\alpha) \quad (4.58)$$

$$C_{X\delta_e} = -C_{D\delta_e} \cos(\alpha) + C_{L\delta_e} \sin(\alpha) \quad (4.59)$$

$$C_{Z0} = -C_{D0} \sin(\alpha) - C_{L0} \cos(\alpha) \quad (4.60)$$

$$C_{Z\alpha} = -C_{D\alpha} \sin(\alpha) - C_{L\alpha} \cos(\alpha) \quad (4.61)$$

$$C_{Zq} = -C_{Dq} \sin(\alpha) - C_{Lq} \cos(\alpha) \quad (4.62)$$

$$C_{Z\delta_e} = -C_{D\delta_e} \sin(\alpha) - C_{L\delta_e} \cos(\alpha) \quad (4.63)$$

The moment equations corresponding to the three rotational degrees of freedom are,

$$\dot{p} = \Gamma_1 pq - \Gamma_2 qr + \Gamma_3 l + \Gamma_4 n \quad (4.64)$$

$$\dot{q} = \Gamma_5 pr - \Gamma_6 (p^2 - r^2) + \frac{m}{J_y} \quad (4.65)$$

$$\dot{r} = \Gamma_7 pq - \Gamma_1 qr + \Gamma_4 l + \Gamma_8 n \quad (4.66)$$

where,

$$\Gamma_1 = \frac{J_{xz}(J_x - J_y + J_z)}{\Gamma} \quad (4.67)$$

$$\Gamma_2 = \frac{J_z(J_z - J_y) + J_{xz}^2}{\Gamma} \quad (4.68)$$

$$\Gamma_3 = \frac{J_z}{\Gamma} \quad (4.69)$$

$$\Gamma_4 = \frac{J_{xz}}{\Gamma} \quad (4.70)$$

$$\Gamma_5 = \frac{J_z - J_x}{J_y} \quad (4.71)$$

$$\Gamma_6 = \frac{J_{xz}}{J_y} \quad (4.72)$$

$$\Gamma_7 = \frac{(J_x - J_y)J_x + J_{xz}^2}{\Gamma} \quad (4.73)$$

$$\Gamma_8 = \frac{J_x}{\Gamma} \quad (4.74)$$

where, $\Gamma = J_x J_z - J_{xz}^2$. The moments about X, Y, and Z axis are,

$$l = \frac{1}{2}\rho V_a^2 \frac{sb}{2} \left(C_{l0} + C_{l\beta}\beta + C_{lp}\frac{b_p}{2V_a} + C_{lr}\frac{b_r}{2V_a} + C_{l\delta_a}\delta_a + C_{l\delta_r}\delta_r \right) \quad (4.75)$$

$$m = \frac{1}{2}\rho V_a^2 S_c \left(C_{m0} + C_{m\alpha}\alpha + C_{mq}\frac{c_q}{V_a} + C_{m\delta_e}\delta_e \right) \quad (4.76)$$

$$n = \frac{1}{2}\rho V_a^2 \frac{sb}{2} \left(C_{n0} + C_{n\beta}\beta + C_{np}\frac{b_p}{2V_a} + C_{nr}\frac{b_r}{2V_a} + C_{n\delta_a}\delta_a + C_{n\delta_r}\delta_r \right) \quad (4.77)$$

Parameter	Value
m	1.56 kg
J	$\begin{pmatrix} 0.1147 & 0 & 0.0015 \\ 0 & 0.0576 & 0 \\ 0.0015 & 0 & 0.1712 \end{pmatrix} \text{ kg m}^2$
S	0.2589 m ²
b	1.4224 m
c	0.3302 m
S_{prop}	0.0314 m ²
ρ	1.2682 kg/m ³
$C_{X\delta_T}$	1
k_{motor}	20

Table 4.1: Parameter Values

The parameters of the UAV are listed in Table 4.1, while the stability and control coefficients are provided in Table 4.2. The permissible range for deflection of the elevator and aileron control surfaces is ± 40 degrees. It becomes evident that if the aforementioned twelve nonlinear equations are utilized to formulate the optimal trajectory for this specific issue, it will result in a two-point boundary value problem, which proves to be exceedingly challenging to solve in real-time. Consequently, the decision is made to employ a point mass model for generating the optimal trajectory. Subsequently, in the following section, an autopilot will be

4.5 Dynamic Model of a UAV for Path Following

Longitudinal Coefficients		Lateral Coefficients	
Coefficient	Value	Coefficient	Value
C_{L_0}	0.28	C_{Y_0}	0
C_{D_0}	0.03	C_{l_0}	0
C_{m_0}	0	C_{n_0}	0
C_{L_α}	3.45	C_{Y_β}	-0.98
C_{D_α}	0.30	C_{l_β}	-0.12
C_{M_α}	-0.38	C_{n_β}	0.25
C_{L_q}	0	C_{Y_p}	0
C_{D_q}	0	C_{l_p}	-0.26
C_{M_q}	-3.6	C_{n_p}	0.022
$C_{L_{\delta e}}$	-0.36	C_{Y_r}	0
$C_{D_{\delta e}}$	0	C_{l_r}	0.14
$C_{M_{\delta e}}$	0.5	C_{n_r}	-0.35
$C_{Y_{\delta a}}$	0		
$C_{l_{\delta a}}$	0.08		
$C_{n_{\delta a}}$	0.06		

Table 4.2: UAV Coefficients

developed to follow the trajectory computed by the guidance algorithms discussed in the preceding section.

4.5.1 Autopilot Design

In this section, the autopilot design will be discussed for the UAV [85] to track the generated paths. The assumption is that longitudinal and lateral dynamics of the UAV are decoupled[87].

4.5.1.1 Longitudinal Autopilot

One of the inputs to the longitudinal autopilot remains constant during the simulation, which is the commanded velocity V^c . Similarly, another input, the commanded altitude z^c , is also maintained constant for two-dimensional path following. The longitudinal autopilot produces two outputs: the deflection of the elevator δ_e and the throttle command δ_t as shown in Fig. 4.7. The altitude hold autopilot translates altitude error into a commanded pitch angle θ^c . Subsequently, the pitch attitude hold autopilot transforms pitch attitude error into

a commanded pitch rate q^c . This pitch rate hold autopilot then converts pitch rate error into elevator command δ_e . Finally, the velocity hold autopilot translates velocity error into throttle command δ_t . Figure 4.8a depicts the altitude hold controller within the longitudinal autopilot, while Figure 4.8b illustrates the velocity hold controller. The coordinated action of the attitude and velocity hold loops aids the UAV in sustaining level flight. Both controllers are configured with proportional-integral-derivative (PID) parameters.

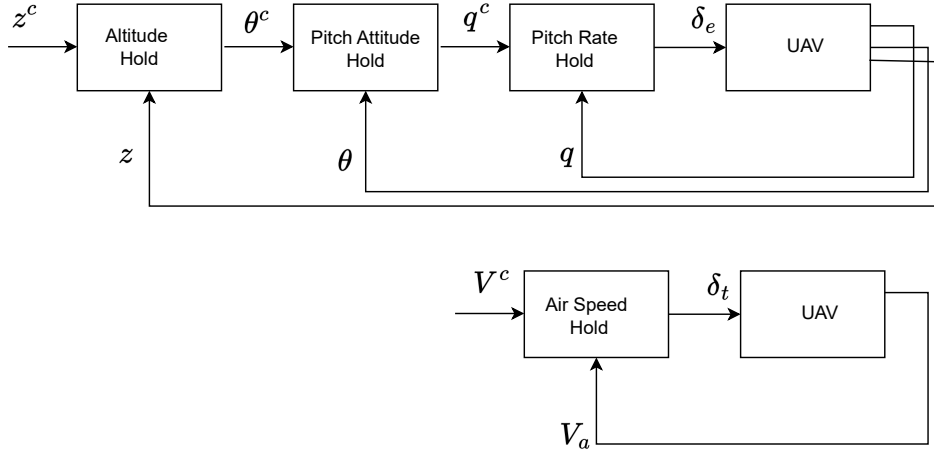
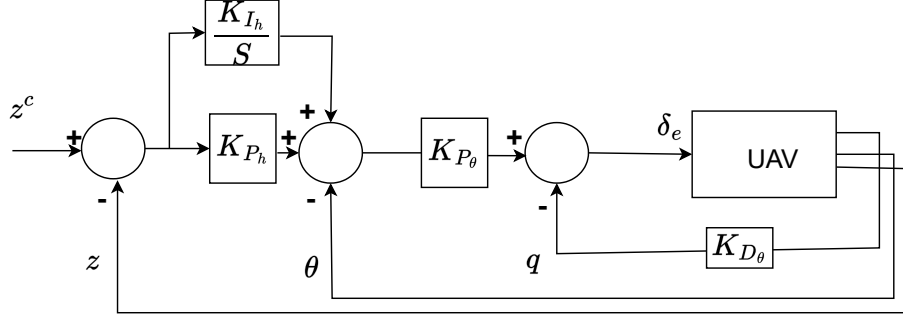


Figure 4.7: Longitudinal autopilot

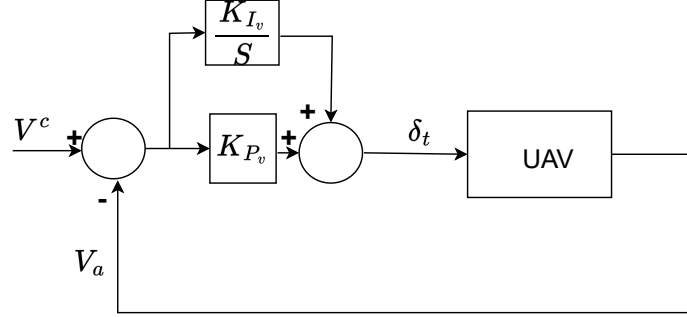
4.5.1.2 Lateral Autopilot

The commanded heading angle ψ^c is a time-varying input to the lateral autopilot and is obtained from the heading angle profile of the designed path obtained using kinematic model of the UAV. The output of the lateral autopilot is the aileron command, δ_a , as shown in Figure 4.9. The heading hold autopilot converts the heading error to roll attitude command, ϕ^c . The roll attitude hold autopilot converts the roll angle error to roll rate command, p^c . The roll rate hold autopilot converts the roll rate error to aileron command δ_a . A realization of lateral autopilot for attitude control using PID is shown in Figure 4.10.

4.6 Simulation Results



(a) Altitude hold Controller



(b) Airspeed hold Controller

Figure 4.8: Longitudinal autopilot

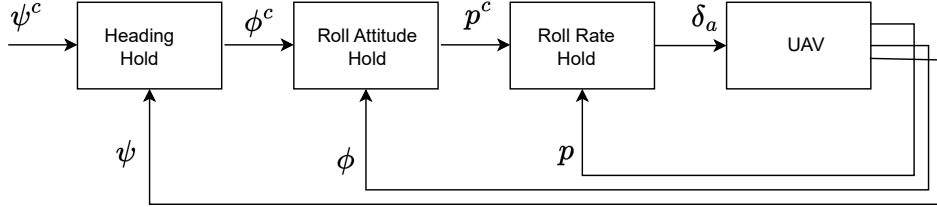


Figure 4.9: Lateral Autopilot

4.6 Simulation Results

4.6.1 Stationary Target Tracking

In this section, simulation results are presented to validate the proposed Lyapunov vector field guidance law, using the kinematic model of the vehicle (4.1) with the following simulation data:

The maximum curvature limit of the vehicle is $\kappa = \frac{1}{r_{min}} = 0.0143 \text{ m}^{-1}$. The

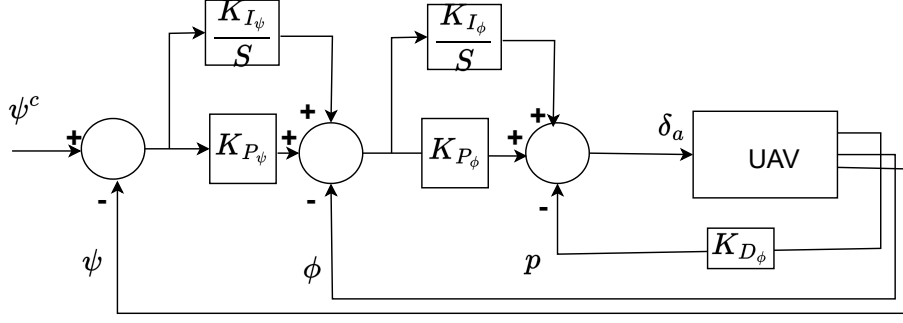


Figure 4.10: Attitude hold controller

airspeed is $v_a = 15 \text{ m/s}$ [88]-[89], and a wind of velocity $W = (5 \ 5) \text{ m/s}$ is considered for each case. The vehicle has a guidance loop (4.4) with control gain $\alpha = 10$. The target is located at $(0 \ 0) \text{ m}$ and the initial position of the UAV is $(-300 \ 0) \text{ m}$. The desired standoff distance is considered as $r_d = 100 \text{ m}$.

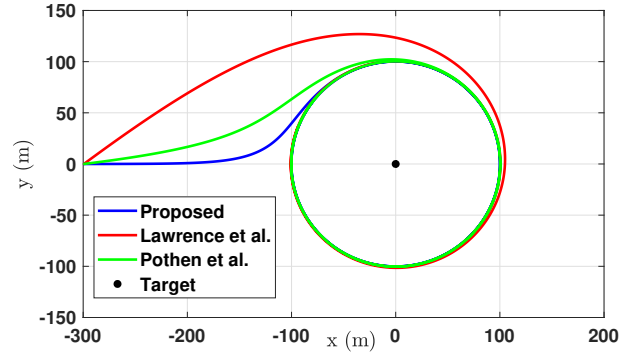
The same data are also considered for generating paths using guidance laws in the existing literature (Ref. [18] and [20]) to compare and to establish the superiority of the proposed guidance algorithm.

Figure. 4.11a shows UAV trajectories generated by the proposed guidance field along with the algorithms presented in Ref. [18] and [20]. It shows all the three trajectories converge and follow the standoff circle whose center is the location of the stationary target. The curvature variation of guidance vector fields while converging from outside of the standoff circle is shown in Fig. 4.11b, which shows that the curvature limit is also satisfied. Fig. 4.11c shows the time required to converge to the circular path. It is clearly seen that the time taken by the proposed guidance law is lower in comparison to the work presented in Ref. [18] and Ref. [20]. With the proposed guidance law, the convergence time of the UAV is 17 sec, whereas, it is 23 sec with the guidance algorithm in Ref. [20] and 40 sec with the guidance algorithm in Ref. [18].

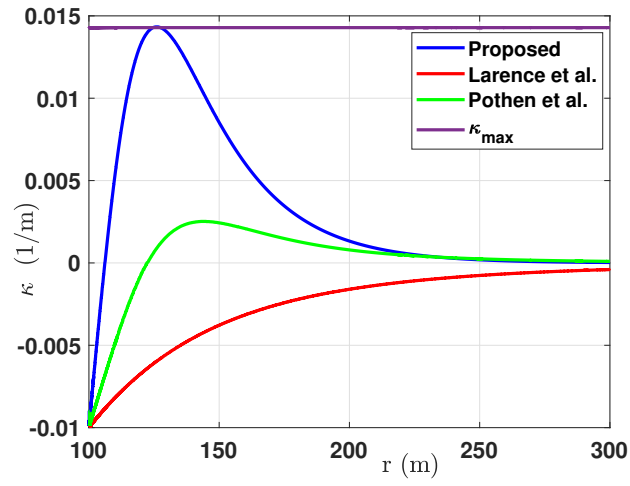
It is to mention that the proposed guidance vector field (4.26) has a singular point at the center of the standoff circle as at $r = 0$, the curvature is not defined but it has no practical implication.

As the algorithm [20] is constrained by $r_d \geq \frac{3.0126}{\kappa_{max}}$, for case studies when the UAV starts from the inside of the standoff circle, the radius of the circular path is

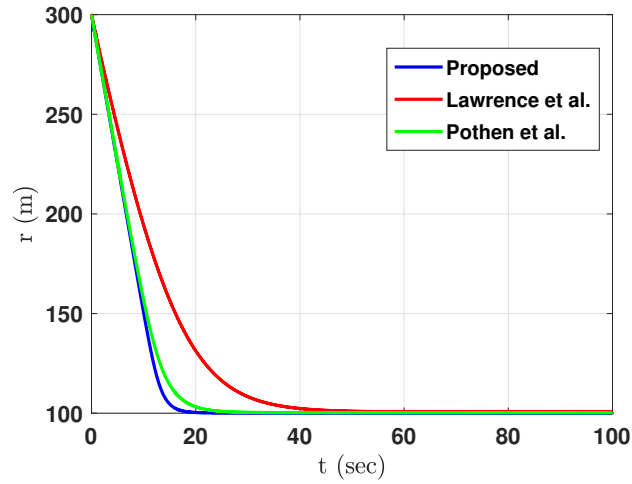
4.6 Simulation Results



(a) Trajectories for stationary target tracking

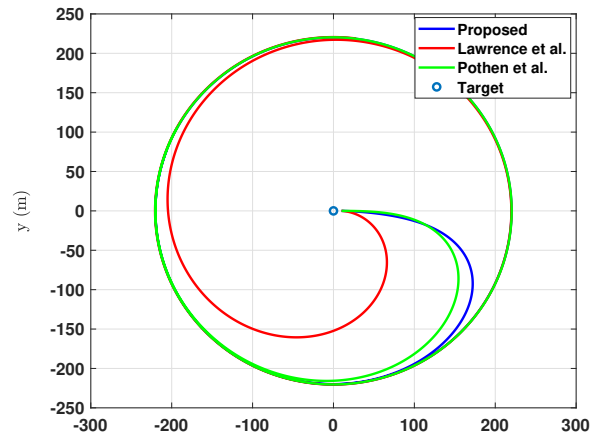


(b) Curvature variation

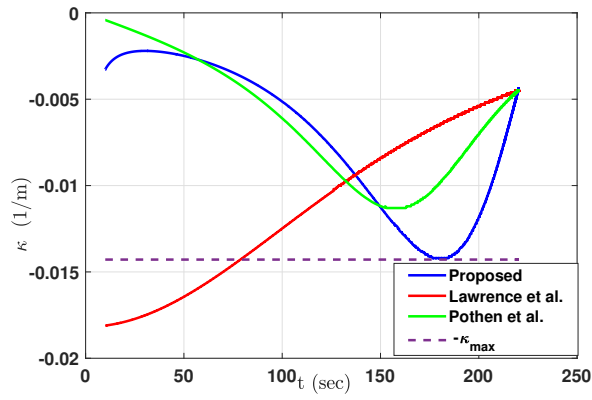


(c) Convergence time

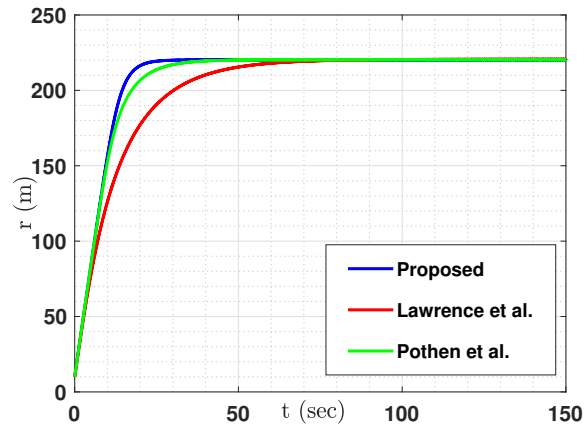
Figure 4.11: Stationary target tracking when the UAV starts from outside of the standoff circle



(a) Stationary target tracking from inside standoff circle



(b) Curvature variation



(c) Converging time profile

Figure 4.12: Stationary target tracking

4.6 Simulation Results

considered as $r_d = 220 \text{ m}$. The initial position is considered as $(5 \ 0) \text{ m}$. Fig. 4.12a, Fig. 4.12b and Fig. 4.12c show trajectories, the curvature variation and the time required to converge to the circular path for all three algorithms, respectively. It is seen that the performance of the proposed algorithm is far better.

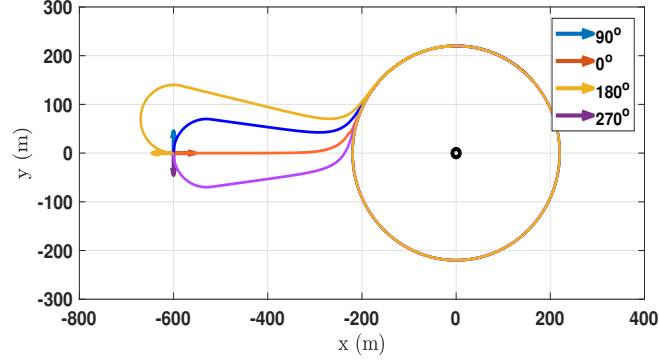


Figure 4.13: Trajectories for different initial orientation angles

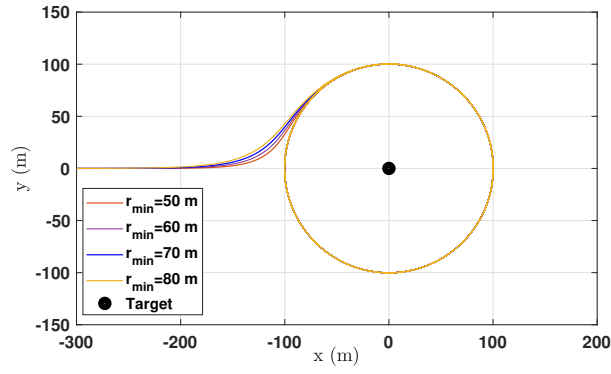
In real-time scenarios, the control input gets saturated when it crosses its maximum bound. Such a situation is also considered here by setting the initial orientation angles as shown in Fig. 4.13. The initial circular arcs indicate the saturation of the control signal.

Fig. 4.14a shows trajectories for four different vehicles with different values of minimum radius of curvature (50 m , 60 m , 70 m and 80 m , respectively). The corresponding scaling parameters (k) are obtained by using ‘fmincon’ of MATLAB and the values are, 2.1346, 2.15, 2.1634, and 2.1751, respectively. Corresponding curvature plots are shown in Fig. 4.14b. From the figure it can be clearly seen that all the trajectories are within their respective curvature bounds. This implies that the proposed guidance algorithm is equally applicable for different vehicles with different maneuvering capabilities.

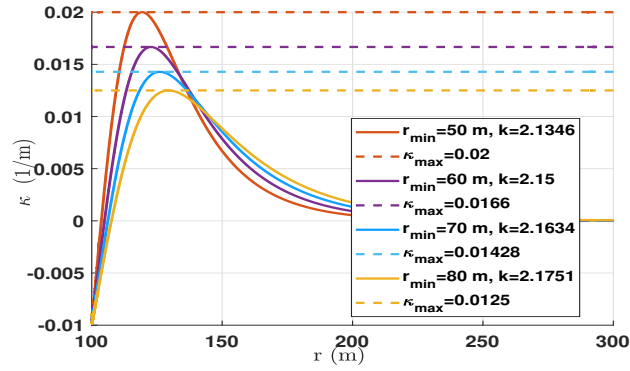
If the wind is time varying, the algorithm will still be useful to generate the paths for standoff target tracking. Let us consider an example of variable wind case where the wind model is similar to the model used in Ref. [90], which is given below:

$$W = a + b \sin \frac{\pi t}{T} \quad (4.78)$$

where, the constants are $a = 5$, $b = 1$, $T = 10$, from which W_x and W_y can be



(a) Trajectories for three different vehicles



(b) Curvature analysis of three different vehicles

Figure 4.14: Stationary target tracking

4.6 Simulation Results

expressed as

$$W_x(t) = W_y(t) = \frac{a}{\sqrt{2}} + \frac{b}{\sqrt{2}} \sin \frac{\pi t}{T} \quad (4.79)$$

With the above said wind condition, the path has been generated as shown in Fig. 4.15 for different initial poses. In all the cases it can be seen that the paths converge to the desired circular path.

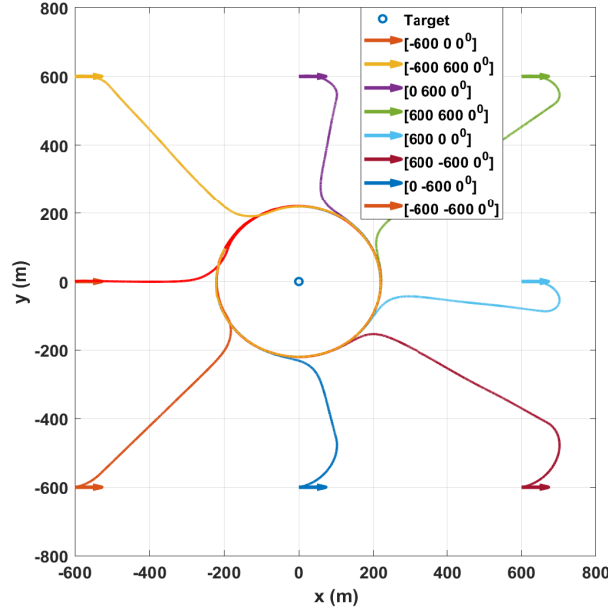


Figure 4.15: Stationary target tracking in variable wind condition for different initial poses

4.6.2 Moving Target Tracking

Let us consider the case when the target is moving with a velocity of $\begin{pmatrix} 7.5 & 7.5 \end{pmatrix} m/s$ from the origin as shown in Fig. 4.16a. Let us assume that the UAV with the air-speed of $15 m/s$ from the starting position $\begin{pmatrix} -350 & 0 \end{pmatrix} m$ has to track the moving target while maintaining the standoff distance of $220 m$. Fig. 4.16a shows trajectories of the UAV while tracking the target. Corresponding range plot between the UAV and the target is shown in Fig. 4.16b, which shows that the deviation from the standoff distance is about $30 m$ at steady state, which is due to the con-

stant linear speed input and nonholonomic constraints of the UAV. The curvature plot of the trajectory of the UAV is depicted in Fig. 4.16c to illustrate that the curvature of the designed trajectory is within its maximum curvature bounds.

4.6.3 Path following using dynamic model

The trajectory of standoff target tracking is carried out using the 6 Degree of Freedom (DOF) UAV simulation model along with the autopilot system as explained earlier. The simulation employs the following parameters: airspeed $V_a = 15$ m/s, initial heading $\chi_0 = 0$ rad, and turn radius $r = 70$ m. The altitude is maintained at a constant 15 m throughout the simulation.

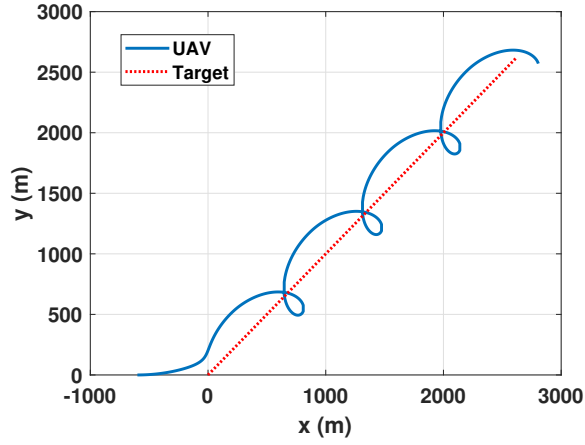
4.6.4 Wind Absent Case

In wind absent condition, the path generated by the dynamic model of the UAV is shown in Fig. 4.17, indicating accurate tracking. Figures 4.18a, 4.18b and 4.18c show the responses the heading angle, altitude and airspeed, obtained from the 6DOF model, respectively. Additionally, Figure 4.18d displays the variation of the bank angle. The control inputs—throttle command, elevator deflection, rudder deflection, and aileron deflection—are depicted in Figures 4.18e-4.18h respectively.

4.6.5 Wind Case

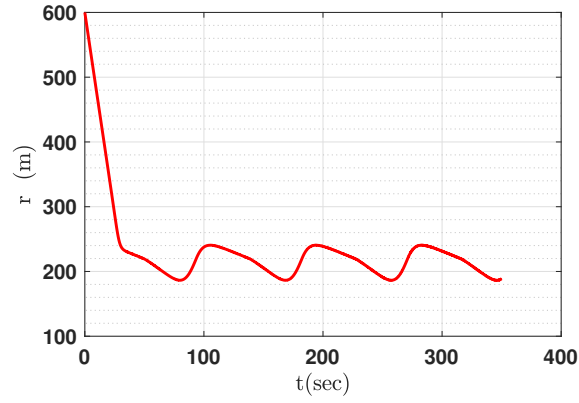
In the presence of windy conditions, a wind vector of $[5, 5]$ m/s is considered. The trajectory of the UAV under these conditions is illustrated in Fig. 4.19, demonstrating precise tracking of the target. The responses of the heading angle, altitude, and airspeed, obtained from the 6-degree-of-freedom (6DOF) model, are presented in Figures 4.20a, 4.20b, and 4.20c, respectively. Furthermore, Figure 4.20d depicts the variation of the bank angle. The control inputs, including throttle command, elevator deflection, rudder deflection, and aileron deflection, are illustrated in Figures 4.20e to 4.20h, respectively.

4.6 Simulation Results

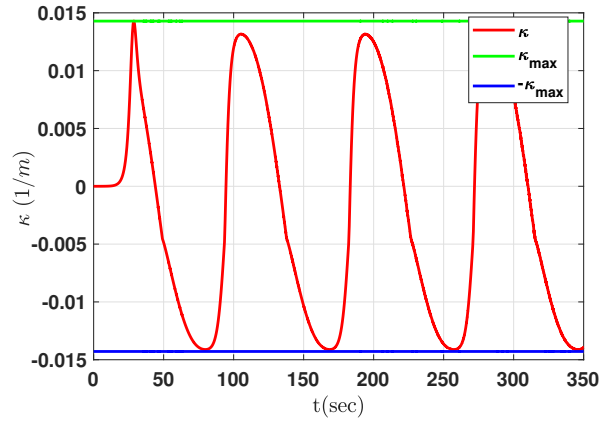


(a) Moving target tracking

s



(b) Moving target tracking time profile



(c) Curvature analysis

Figure 4.16: Moving target tracking

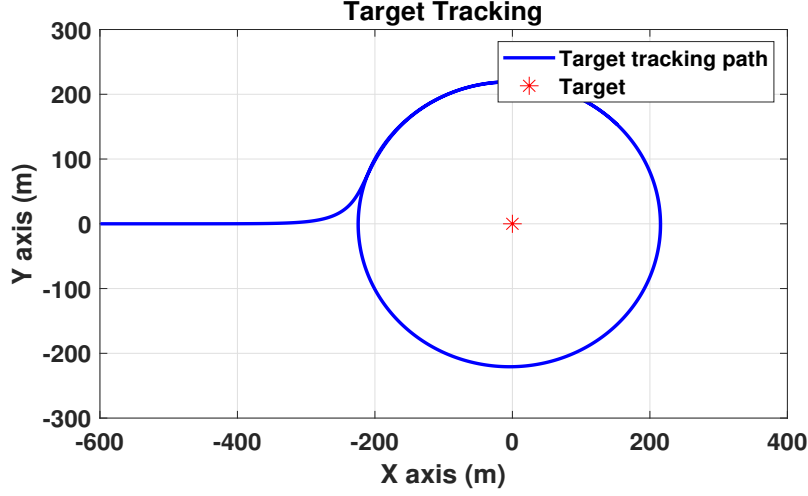


Figure 4.17: Stationary target tracking using dynamic model of the UAV

4.7 Experimental Results

To demonstrate real-time applications, the proposed guidance algorithm is implemented on Quanser's QBot2[91]. QBot 2 is a 2-Wheel mobile ground robot with differential drive mechanism and it is equipped with an on-board computer. Specifications of QBot2 are listed in Table 4.3. Fig. 4.21 shows control flow diagram for autonomous operation of the QBot2. Localization of the robot is done by six synchronized infrared camera setup along with the motive software. This is responsible for 6-DOF pose measurements of the robot in a defined fixed coordinate system. For sending and receiving data from the host computer, it is in communication with the robot via WI-FI using TCP/IP protocol. Quarc software was installed in the host computer, which is responsible for generating C code from Simulink control models. And, Quarc is also used to interface SIMULINK control models with the physical robot through Quanser's data acquisition card.

In this work, experiments are conducted using the proposed guidance algorithm in the high-level control loop and with no low-level controller. The guidance loop generates angular speed commands as a control signal, but as discussed earlier the linear speed is kept constant. The inverse kinematics block generates left and right wheel speed commands from linear and angular speeds. The following parameters are considered for experiment: linear speed (v_a) = 0.2 m/sec, the

4.7 Experimental Results

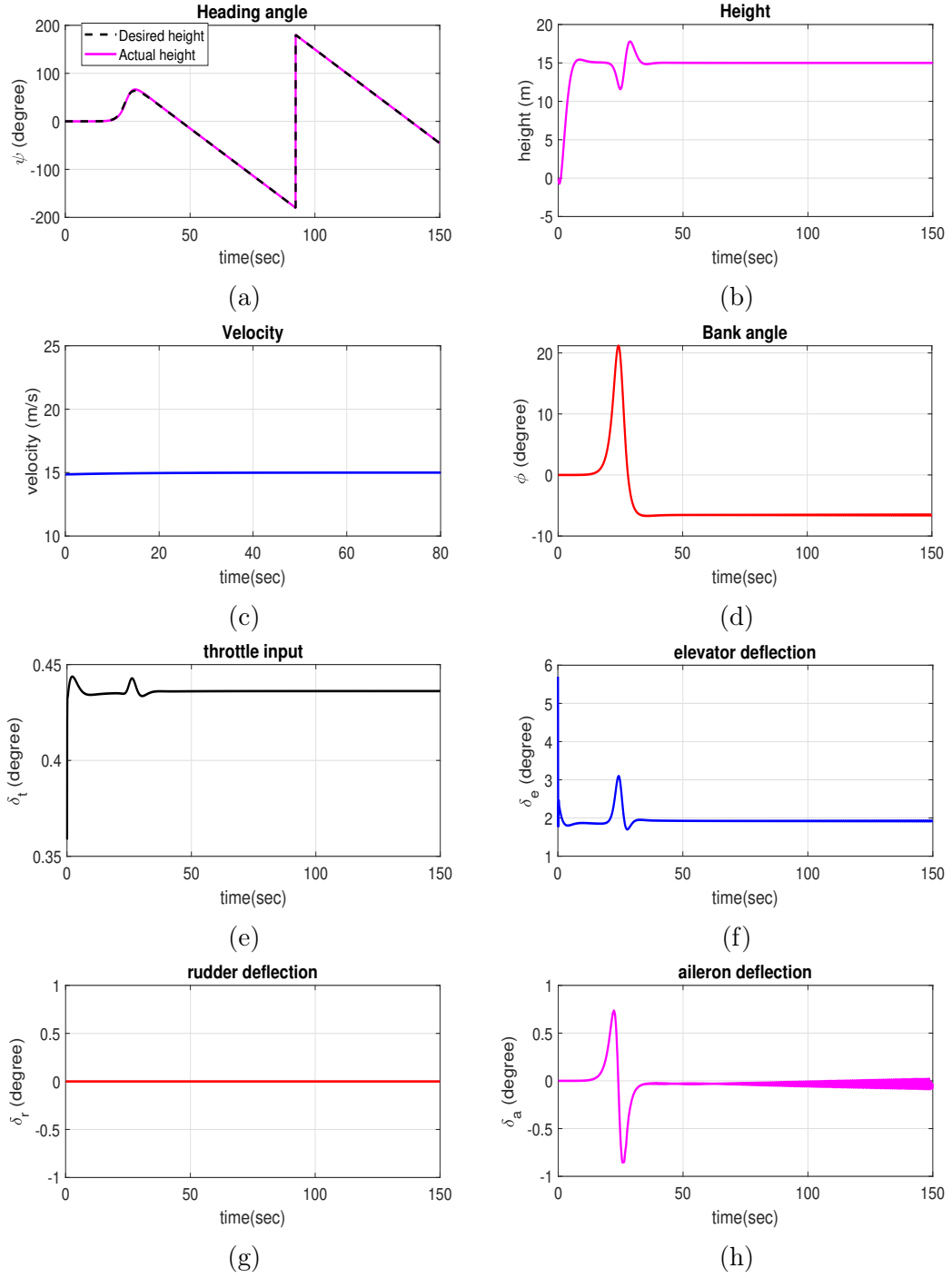


Figure 4.18: State and control variables for standoff target tracking (a) heading angle, (b) altitude profile, (c) airspeed profile, (d) bank angle, (e) throttle input, (f) elevator input, (g) rudder input, and (h) aileron input.

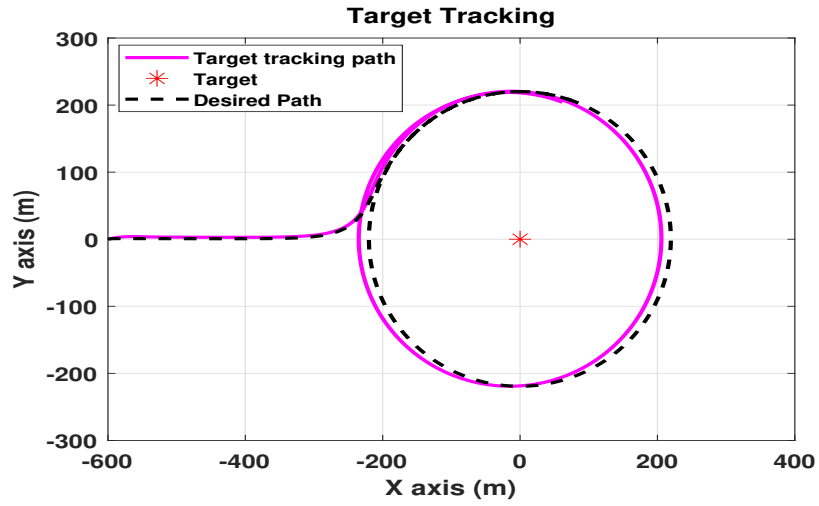


Figure 4.19: Stationary target tracking in windy condition using dynamic model of the UAV

Table 4.3: Specifications of QBot 2.

QBot2 diameter	0.35 m
Maximum linear velocity	0.7 m/s
On board processor	1 GHz ARM cortex A9 processor
Quarc maximum sample rate	1,000 Hz

4.7 Experimental Results

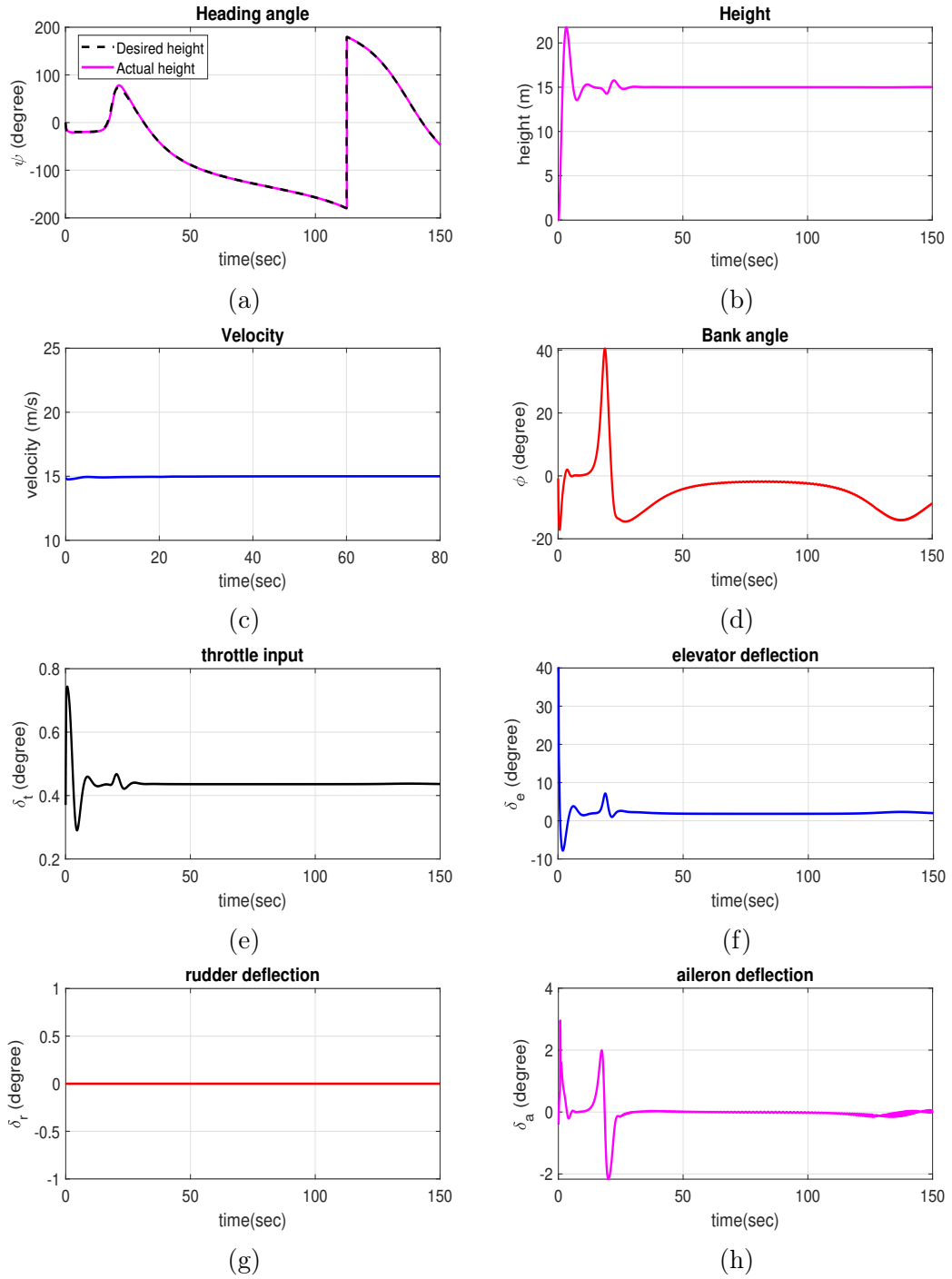


Figure 4.20: State and control variables for standoff target tracking (a) heading angle, (b) altitude profile, (c) airspeed profile, (d) bank angle, (e) throttle input, (f) elevator input, (g) rudder input, and (h) aileron input.

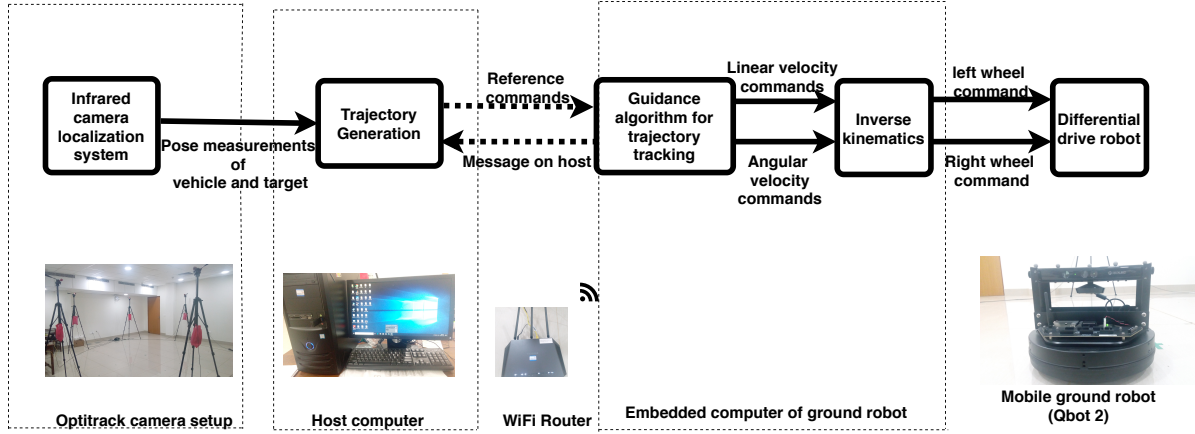
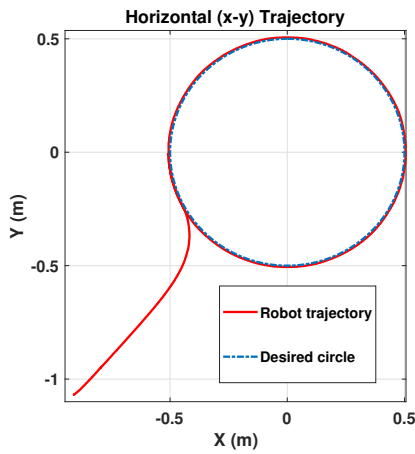
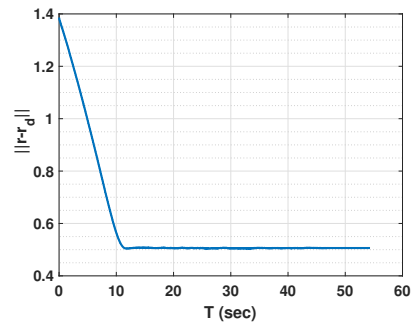


Figure 4.21: Architecture of control flow for autonomous operation by the ground robot



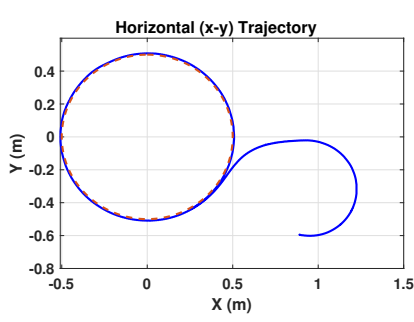
(a) Ground robot trajectory



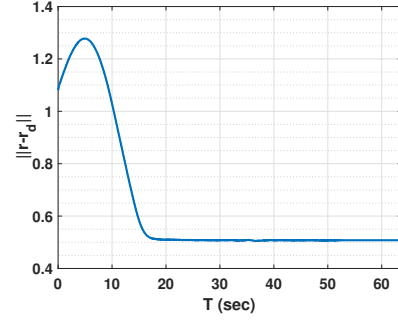
(b) Range profile from the center of the circle

Figure 4.22: Circle following by the ground robot

4.8 Summary



(a) Ground robot trajectory



(b) Range profile from the center of the circle

Figure 4.23: Circle following by the ground robot

maximum curvature of vehicle $\kappa = \frac{1}{r_{min}} = 1/0.3 \text{ m}^{-1}$, the radius of desired circle $r_d = 0.5 \text{ m}$.

Experiments were conducted in a $1.5\text{m} \times 1.5\text{m}$ workspace area, with different initial poses around the desired circle. Fig. 4.22 shows experimental results with an initial orientation angle 42.3° and position of $(-0.9 \ -1) \text{ m}$. Fig. 4.22a shows corresponding ground robot trajectory and Fig. 4.22b shows the error profile which shows that the robot converges to the desired circle as time progresses. The ground robot is also tested with different initial orientation angles when the robot is close to the circular path to be followed. Fig. 4.23a shows trajectory of the ground robot with an initial orientation angle -1.33° and position of $(0.8916 \ 0.3003) \text{ m}$. Fig. 4.23b shows the error profile while converging to the desired circle as time progresses.

4.8 Summary

A fast convergence guidance algorithm based on the Lyapunov guidance vector field theory for standoff target tracking by UAVs has been presented in this work for both stationary and moving targets in the presence of wind. The proposed guidance law is explicitly a function of the minimum turn radius of the vehicle, and hence it is applicable to different maneuverable vehicles. The proposed law has been compared with other LVGF based algorithms in literature to show its faster convergence performance. The other advantage of the algorithm is that it

generates a smooth near optimal path in lower time in comparison to the similar work in the literature, which makes it feasible for real-time applications. The work can be extended to perform real-time target tracking by UAVs in 3-dimensional environments.

Coordinated Waypoint Following by Group of UAVs

5.1 Introduction

A novel vector field guidance to a group of unmanned vehicles with nonholonomic constraints is proposed in this chapter for waypoint following problems in an environment strewn with static and dynamic obstacles. Unlike the existing position and distance-based topologies, this work proposes a novel behavioral-based balance formation topology where all the vehicles are distributed evenly in the desired formation circle by using a guidance vector field method to provide the desired heading and speed commands to each vehicle. The asymptotic stability of the proposed field is established. The proposed algorithm is also compared with the Barycentric coordinate-based control approach (BCB) for a three-vehicle system to illustrate the faster convergence of the proposed work especially when the initial positions are far from the desired formation distances. The important advantage of the proposed formation approach is that it can also be developed based on the relative position measurements in local coordinate frames of the respective vehicles and the communication among the vehicles is also not required. Then, a proximity-based obstacle avoidance technique is developed to generate

safe trajectories for each vehicle, which is singularity free, unlike other force field-based obstacle avoidance methods existing in the literature. Finally, the work is validated numerically through simulations using the MATLAB environment.

The chapter is structured as follows: In Section 5.1, an introduction is given to explain coordinated waypoint following problems. The formal problem definition is presented in Section 5.2. Balanced formation topology is explained in Section 5.3. Our proposed solution, the Guiding Vector Field approach, is detailed in Section 5.4, explaining its theoretical basis and advantages in achieving coordinated waypoint following. Additionally, Section 5.5 addresses obstacle avoidance technique adopted by the UAV group during navigation. The waypoint following by the virtual vehicle is discussed in Section 5.6. Comparative analysis is presented in 5.7. To evaluate the performance of our approach, simulation results are presented in Section 5.8, demonstrating the effectiveness of the coordinated navigation strategy.

5.2 Problem Definition

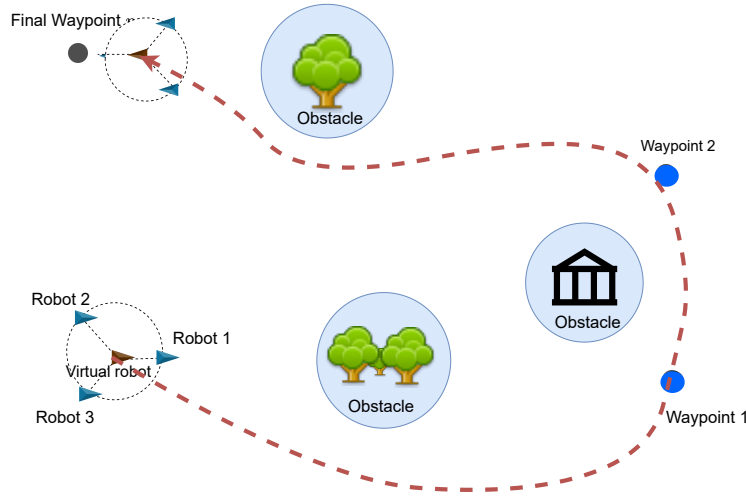


Figure 5.1: Waypoint following by formation of three agents

A typical architecture of control problems to navigate an autonomously driven vehicle is a two-layer hierarchical process. The outer control layer generates the

5.2 Problem Definition

linear and the angular speed commands and the inner layer generates corresponding forces and moments. This work deals with the outer layer of the control to generate speed commands as control inputs to autonomous vehicles for waypoint following problems. Moreover, in this work, autonomously driven non-holonomic vehicles are regarded as robots.

Let us consider a group of three autonomously driven non-holonomic robots in a fixed inertial frame of reference in the Cartesian coordinate system. The following kinematic equations govern these vehicles.

$$\begin{aligned}\dot{x}_i &= v_i \cos \chi_i \\ \dot{y}_i &= v_i \sin \chi_i \\ \dot{\chi}_i &= u_i\end{aligned}\tag{5.1}$$

where, $i = 1, 2, 3$. v_i and u_i are the linear and the angular speeds, respectively. $p_i = (x_i, y_i) \in \mathbb{R}^2$ is the position and χ_i is the heading angle of the i^{th} vehicle. v_i and u_i follow the constraints given below:

$$v_{min} \leq v_i \leq v_{max}\tag{5.2}$$

and

$$\omega_{min} \leq u_i \leq \omega_{max}\tag{5.3}$$

where, v_{max} and ω_{max} are the maximum bounds on the linear and the angular speeds of all the vehicles. v_{min} and ω_{min} are the lower bounds on the linear and the angular speeds of all the vehicles. The objective is to track the desired waypoints in an obstacle-filled environment by the formation of three robots as depicted in Fig.5.1 and at the same time, the heading direction of each vehicle should also be the same.

To generate the guidance laws for the formation control, consider the speed hold autopilot and the heading hold autopilot for each vehicle as follows:

$$\dot{v}_i = \beta_1(v_{ic} - v_i)\tag{5.4}$$

$$\dot{\chi}_i = \beta_2(\chi_{ic} - \chi_i)\tag{5.5}$$

where, v_{ic} and χ_{ic} are the commanded speed and the heading signals, respectively. β_1 and β_2 are two positive constants.

5.3 Balanced Formation Topology

Let us assume initially the virtual robot is at the centroid of all the actual robots. The centroid, $p_o = (x_o, y_o)$ is,

$$p_o = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right) \quad (5.6)$$

Consider the following kinematic equations that govern the virtual robot

$$\begin{aligned} \dot{x}_o &= v_o \cos \chi_o \\ \dot{y}_o &= v_o \sin \chi_o \\ \dot{\chi}_o &= u_o \end{aligned} \quad (5.7)$$

where, v_o is the speed of the virtual robot, which is constant, and its value can be approximately close to the value of the cruise speed of the actual robot, but it should be less than its maximum bound. χ_o is the heading angle of the virtual robot. u_o is the angular speed of the virtual robot, and its bounds are the same as those of the actual robots, that is,

$$-\omega_{max} \leq u_o \leq \omega_{max} \quad (5.8)$$

In this work, a balanced formation topology is adopted, which can be defined as, at any instant, the sum of the position vectors from all the robots to the virtual robot is zero. Hence, the formation flight control problem can be solved by finding the control inputs (v_i and u_i) of each vehicle to achieve the above objective.

The balance formation topology is quite analogous to the balanced 3- phase voltage or current where at any instant the vector sum of all phase voltages/currents is zero. The geometry of the balanced formation topology is like three position vectors of equal magnitudes and having angle difference with the adjacent vehicle as $360/3$ as shown in Fig. 5.2.

5.3 Balanced Formation Topology

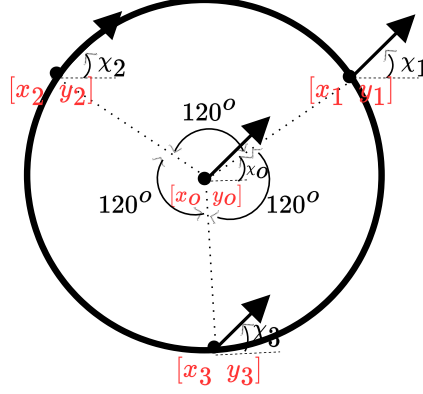


Figure 5.2: Balanced formation topology

Theorem : Three vehicles with positions, $(x_i \ y_i)$, $(i = 1, 2, 3)$ and which satisfy the equations (5.9), (5.10) and (5.11) are evenly distributed on a circle of radius r_f , centre at $(x_o \ y_o)$ to make a balanced formation.

$$\sum_{i=1}^3 x_{oi} = 0, \quad (5.9)$$

$$\sum_{i=1}^3 y_{oi} = 0 \quad (5.10)$$

and

$$\sqrt{x_{oi}^2 + y_{oi}^2} = r_f \quad (5.11)$$

where, $\mathbf{r}_{oi} = [x_{oi} \ y_{oi}]^T = [x_i - x_o \ y_i - y_o]^T$ is the position vector of the i^{th} robot from the virtual robot and r_{oi} is the magnitude of $\mathbf{r}_{oi}$.

Proof: In polar form the position vectors of three robots can be expressed as $r_{o1}\angle\theta_{o1}$, $r_{o2}\angle\theta_{o2}$ and $r_{o3}\angle\theta_{o3}$, respectively. Considering $r_{o1}\angle\theta_{o1}$ as the reference vector, these three vectors can be rewritten as $r_{o1}\angle 0$, $r_{o2}\angle(\theta_{o2} - \theta_{o1})$ and $r_{o3}\angle(\theta_{o3} - \theta_{o1})$, respectively.

Now, (5.9), (5.10) can be written as

$$r_{o1} \cos 0 + r_{o2} \cos(\theta_{o2} - \theta_{o1}) + r_{o3} \cos(\theta_{o3} - \theta_{o1}) = 0 \quad (5.12)$$

$$r_{o1} \sin 0 + r_{o2} \sin(\theta_{o2} - \theta_{o1}) + r_{o3} \sin(\theta_{o3} - \theta_{o1}) = 0 \quad (5.13)$$

(5.12) and (5.13) can be simplified as follows using (5.11):

$$\cos(\theta_{o2} - \theta_{o1}) + \cos(\theta_{o3} - \theta_{o1}) = -1 \quad (5.14)$$

$$\sin(\theta_{o2} - \theta_{o1}) + \sin(\theta_{o3} - \theta_{o1}) = 0 \quad (5.15)$$

From (5.15), we can get,

$$(\theta_{o3} - \theta_{o1}) = -(\theta_{o2} - \theta_{o1}) \quad (5.16)$$

Using (5.16), we can obtain the following results from (5.14)

$$(\theta_{o2} - \theta_{o1}) = 120^\circ$$

and

$$(\theta_{o3} - \theta_{o1}) = -120^\circ$$

So, three position vectors of the robots will evenly be distributed on the circle of radius r_f with angle difference between any two vectors as 120° , which is shown in Fig. 5.2.

5.4 Proposed Guiding Vector Field

This section presents a novel guidance algorithm for convergence of all the vehicles to the defined formation topology. In addition, the heading directions of all the robots should be the same as that of the virtual robot.

A positive definite scalar valued function (Lyapunov function) is considered as,

$$V = \frac{1}{2} \left(\sum_{i=1}^3 x_{oi} \right)^2 + \frac{1}{2} \left(\sum_{i=1}^3 y_{oi} \right)^2 + \frac{1}{2} \sum_{i=1}^3 (r_{oi} - r_f)^2 \quad (5.17)$$

5.4 Proposed Guiding Vector Field

The total time derivative of the Lyapunov function (V) is:

$$\dot{V} = \sum_{i=1}^3 \left(\frac{\partial V}{\partial x_{oi}} \dot{x}_{oi} + \frac{\partial V}{\partial y_{oi}} \dot{y}_{oi} \right) \quad (5.18)$$

where,

$$\begin{bmatrix} \frac{\partial V}{\partial x_{oi}} \\ \frac{\partial V}{\partial y_{oi}} \end{bmatrix} = \begin{bmatrix} \left(\sum_{i=1}^3 x_{oi} \right) + (r_{oi} - r_f) \frac{x_{oi}}{r_{oi}} \\ \left(\sum_{i=1}^3 y_{oi} \right) + (r_{oi} - r_f) \frac{y_{oi}}{r_{oi}} \end{bmatrix} \quad (5.19)$$

The negative gradient of (5.17) will be with an angle ϕ_i and which can be expressed as,

$$\phi_i = \arctan 2 \left(\frac{-\partial V}{\partial y_{oi}}, \frac{-\partial V}{\partial x_{oi}} \right) \quad (5.20)$$

Any trajectory which follows the direction of (5.20) will converge to the desired formation behavior, but may not be feasible considering the dynamics of the vehicle. So to generate feasible trajectories the guiding vector field (GVF) concept as stated below is adopted here.

The proposed GVF for each robot is

$$\begin{aligned} \mathbf{W}_i &= v_{ic} \left[f_{1i} \begin{bmatrix} \cos \phi_i \\ \sin \phi_i \end{bmatrix} + f_{2i} \begin{bmatrix} -\sin \phi_i \\ \cos \phi_i \end{bmatrix} \right] \\ &= v_{ic} \begin{bmatrix} \cos \phi_i & -\sin \phi_i \\ \sin \phi_i & \cos \phi_i \end{bmatrix} \begin{bmatrix} f_{1i} \\ f_{2i} \end{bmatrix} \end{aligned} \quad (5.21)$$

where,

$$f_{1i} = \text{sech}(k_1(\phi_i - \chi_o)),$$

$$f_{2i} = \tanh(k_1(\phi_i - \chi_o)),$$

k_1 is a positive gain. v_{ic} is the commanded speed signal and which is set from the stability analysis presented in the next section.

For path following using GVF approach, the vector field is described as the tangent and the normal components to the level set. But, for the formation flight problem two unit vectors $\begin{bmatrix} \cos \phi_i & \sin \phi_i \end{bmatrix}^T$ and $\begin{bmatrix} -\sin \phi_i & \cos \phi_i \end{bmatrix}^T$ are considered, which are in the normal and the tangential vector directions to the equilibrium

set, respectively, and these two vectors are scaled by f_{1i} and f_{2i} , respectively, to generate the convergence and the circulation components. The gain k_1 should be selected such that when the heading angle of the virtual robot is in the direction of the normal to the equilibrium set, the circulation term vanishes so the vector field consists of only the convergence term. Further, if the heading directions of the virtual robot and the normal to the equilibrium set are not the same, then the vector field consists of both the convergence and the circulation terms.

The commanded heading signal, χ_{ic} , using the guidance vector field (5.21) can be generated as follows:

$$\chi_{ic} = \arctan \left(\frac{\mathbf{W}_i(2, 1)}{\mathbf{W}_i(1, 1)} \right) \quad (5.22)$$

$$= \arctan \left(\frac{f_{1i} \sin \phi_i + f_{2i} \cos \phi_i}{f_{1i} \cos \phi_i - f_{2i} \sin \phi_i} \right) \quad (5.23)$$

This can further be simplified to the following:

$$\chi_{ic} = \phi_i + \arctan \left(\frac{f_{2i}}{f_{1i}} \right) \quad (5.24)$$

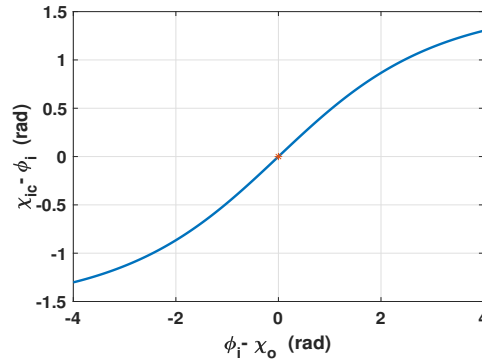


Figure 5.3: $\chi_{ic} - \phi_i$ vs $\phi_i - \chi_o$ profile

With the proposed vector field (χ_{ic}), asymptotic convergence to the balanced formation topology (which means χ_i converges to ϕ_i asymptotically) is established in the next section. Assume that with the heading hold loop (5.5) the heading angle of the i^{th} vehicle, χ_i , strictly follows commanded heading angle of the i^{th} vehicle, χ_{ic} . The relationship between $\chi_{ic} - \phi_i$ and $\phi_i - \chi_o$ can be directly seen by

5.4 Proposed Guiding Vector Field

rearranging (5.24) in the following manner:

$$\chi_{ic} - \phi_i = \arctan(\sinh(k_1(\phi_i - \chi_o))) \quad (5.25)$$

This relationship can be better visualised by plotting $\chi_{ic} - \phi_i$ vs $\phi_i - \chi_o$ and is shown in Fig. 5.3. From this figure, it can be noted that as the angle difference, $\chi_{ic} - \phi_i$, goes to zero, the angle difference, $\phi_i - \chi_o$, will also go to zero.

5.4.1 Stability

In this section, the stability analysis of the proposed guidance vector field is described with the help of the Lyapunov direct method. The Lyapunov function (5.17) is a positive definite function and is radially unbounded. The total time derivative of the Lyapunov function is

$$\dot{V} = \sum_{i=1}^3 \begin{bmatrix} \frac{\partial V}{\partial x_{oi}} & \frac{\partial V}{\partial y_{oi}} \end{bmatrix} \begin{bmatrix} \dot{x}_{oi} \\ \dot{y}_{oi} \end{bmatrix} \quad (5.26)$$

$$= \sum_{i=1}^3 \begin{bmatrix} \frac{\partial V}{\partial x_{oi}} & \frac{\partial V}{\partial y_{oi}} \end{bmatrix} \begin{bmatrix} \dot{x}_i - \dot{x}_o \\ \dot{y}_i - \dot{y}_o \end{bmatrix} \quad (5.27)$$

By substituting the proposed vector field (5.21) in (5.27)

$$\dot{V} = \sum_{i=1}^3 v_{ic} \begin{bmatrix} \frac{\partial V}{\partial x_{oi}} & \frac{\partial V}{\partial y_{oi}} \end{bmatrix} \begin{bmatrix} f_{1i} \cos \phi_i - f_{2i} \sin \phi_i \\ f_{1i} \sin \phi_i + f_{2i} \cos \phi_i \end{bmatrix} - \sum_{i=1}^3 \begin{bmatrix} \frac{\partial V}{\partial x_{oi}} & \frac{\partial V}{\partial y_{oi}} \end{bmatrix} \begin{bmatrix} \dot{x}_o \\ \dot{y}_o \end{bmatrix} \quad (5.28)$$

$\cos \phi_i$ and $\sin \phi_i$ can be expressed as

$$\cos \phi_i = \frac{-\frac{\partial V}{\partial x_{oi}}}{\sqrt{\frac{\partial V}{\partial x_{oi}}^2 + \frac{\partial V}{\partial y_{oi}}^2}},$$

$$\sin \phi_i = \frac{-\frac{\partial V}{\partial y_{oi}}}{\sqrt{\frac{\partial V}{\partial x_{oi}}^2 + \frac{\partial V}{\partial y_{oi}}^2}}$$

By replacing $\cos \phi_{oi}$ and $\sin \phi_{oi}$ in (5.28), $\dot{V}$ can be expressed as

$$\begin{aligned} \dot{V} = & \sum_{i=1}^3 \frac{v_{ic}}{\sqrt{\frac{\partial V}{\partial x_{oi}}^2 + \frac{\partial V}{\partial y_{oi}}^2}} \left(-f_{1i} \left(\frac{\partial V}{\partial x_{oi}} \right)^2 + f_{2i} \frac{\partial V}{\partial x_{oi}} \frac{\partial V}{\partial y_{oi}} - f_{2i} \frac{\partial V}{\partial x_{oi}} \frac{\partial V}{\partial y_{oi}} \right. \\ & \left. - f_{1i} \left(\frac{\partial V}{\partial y_{oi}} \right)^2 \right) + \sum_{i=1}^3 \left(-\dot{x}_o \frac{\partial V}{\partial x_{oi}} - \dot{y}_o \frac{\partial V}{\partial y_{oi}} \right) \\ \dot{V} = & - \sum_{i=1}^3 \frac{v_{ic}}{\sqrt{\frac{\partial V}{\partial x_{oi}}^2 + \frac{\partial V}{\partial y_{oi}}^2}} f_{1i} \left(\frac{\partial V}{\partial x_{oi}}^2 + \frac{\partial V}{\partial y_{oi}}^2 \right) \\ & - \sum_{i=1}^3 \left(v_o \cos \chi_o \frac{\partial V}{\partial x_{oi}} + v_o \sin \chi_o \frac{\partial V}{\partial y_{oi}} \right) \end{aligned} \quad (5.29)$$

$\dot{V}$ expression can be further simplified to

$$\dot{V} = - \sum_{i=1}^3 v_{ic} f_{1i} \sqrt{\frac{\partial V}{\partial x_{oi}}^2 + \frac{\partial V}{\partial y_{oi}}^2} + \sum_{i=1}^3 v_o \cos(\phi_i - \chi_o) \sqrt{\frac{\partial V}{\partial x_{oi}}^2 + \frac{\partial V}{\partial y_{oi}}^2} \quad (5.30)$$

From (5.30), for $k_1 \in [0, 1)$, and for $(\phi_i - \chi_o) \in [-\pi, \pi)$,

- if $|\phi_i - \chi_o| \geq \pi/2$, $\dot{V}$ is always negative definite because $\cos(\phi_i - \chi_o) \leq 0$, as v_{ic} , f_{1i} , and v_o are always positive.
- if $|\phi_i - \chi_o| \leq \pi/2$, $\dot{V}$ is negative definite for $v_{ic} \geq v_o$ as $f_{1i} \geq \cos(\phi_i - \chi_o)$ and v_o and v_{ic} are always positive.

So, by ensuring the condition $v_{ic} \geq v_o$ when $|\phi_i - \chi_o| \leq \pi/2$, in the following section 5.4.2, a speed profile for stable convergence to the desired formation is presented.

5.4.2 Speed Profile:

In this section, the commanded speed profile, v_{ic} , for actual vehicles will be set based on the stability conditions and the constraints on the linear speed. The velocity vector of i^{th} vehicle can be written as

5.4 Proposed Guiding Vector Field

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = \begin{bmatrix} \dot{x}_{oi} + \dot{x}_o \\ \dot{y}_{oi} + \dot{y}_o \end{bmatrix} \quad (5.31)$$

where, $\begin{bmatrix} \dot{x}_{oi} & \dot{y}_{oi} \end{bmatrix}^T$ is the velocity vector of the i^{th} vehicle with respect to the virtual vehicle and $\begin{bmatrix} \dot{x}_o & \dot{y}_o \end{bmatrix}^T$ is the velocity vector of the virtual vehicle. Consider the velocity vector of i^{th} vehicle as

$$\begin{bmatrix} \dot{x}_i \\ \dot{y}_i \end{bmatrix} = \begin{bmatrix} -\frac{\partial V}{\partial x_{oi}} + v_o \cos \chi_o \\ -\frac{\partial V}{\partial y_{oi}} + v_o \sin \chi_o \end{bmatrix} \quad (5.32)$$

Let us assume the magnitude of the above velocity vector be S . So,

$$S = \sqrt{\left(-\frac{\partial V}{\partial x_{oi}} + v_o \cos \chi_o\right)^2 + \left(-\frac{\partial V}{\partial y_{oi}} + v_o \sin \chi_o\right)^2}$$

The above expression can be simplified as

$$S = \sqrt{v_{oi}^2 + v_o^2 + 2v_{oi}v_o \cos(\phi_i - \chi_o)} \quad (5.33)$$

where,

$$v_{oi} = \sqrt{\left(\frac{\partial V}{\partial x_{oi}}\right)^2 + \left(\frac{\partial V}{\partial y_{oi}}\right)^2}$$

The same can be represented in a vector diagram form as shown in Fig. 5.4.

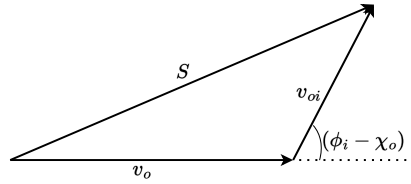


Figure 5.4: Vector diagram

From Fig. 5.4, the maximum bounds of the speed is

$$S_{max} = v_o + v_{oi} \quad \text{for} \quad \phi_i - \chi_o = 0$$

and the minimum bounds of the speed is,

$$S_{min} = |v_o - v_{oi}| \quad \text{for} \quad \phi_i - \chi_o = \pi$$

Note that $S = v_o$ when $v_{io} = 0$. The maximum and minimum bounds of S may cross the actual bounds of the speed of the vehicle, so to impose the bounds on speed, a commanded speed profile is given as follows:

$$v_{ic} = v_{min} + (v_{max} - v_{min}) \tanh(k_2 S) \quad (5.34)$$

where k_2 is a positive gain and it is set using the terminal condition that is when $\frac{\partial V}{\partial x_{oi}}$ and $\frac{\partial V}{\partial y_{oi}}$ are zero (which means the formation is tracking the virtual target), v_{ic} should be equal to v_o .

At the terminal condition

$$v_o = v_{min} + (v_{max} - v_{min}) \tanh(k_2 v_o) \quad (5.35)$$

From (5.35), k_2 can be calculated as follows

$$k_2 = \frac{1}{v_o} \tanh^{-1} \left(\frac{v_o - v_{min}}{v_{max} - v_{min}} \right) \quad (5.36)$$

From (5.35), the value of v_{ic} will be

$$v_{ic} \in [v_{min} \ v_o], \forall \|\phi_i - \chi_o\| \geq \pi/2 \quad (5.37)$$

$$v_{ic} \in (v_o \ v_{max}], \forall \|\phi_i - \chi_o\| < \pi/2 \quad (5.38)$$

$$v_{ic} = v_o, \quad v_{oi} = 0 \quad (5.39)$$

So, using these observations (5.37), (5.38) and (5.39), and from stability conditions (5.30), we can say that one can achieve the stable formation tracking condition with the commanded heading signal, v_{ic} .

5.5 Obstacle Avoidance

5.5.1 Static Obstacle Avoidance

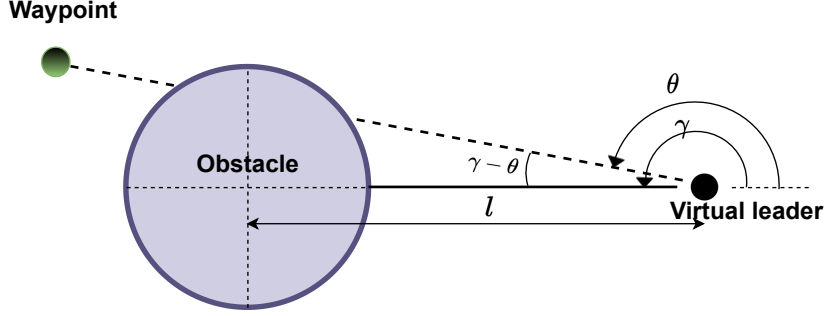


Figure 5.5: Geometry of obstacle avoidance

Let us consider a scenario where an obstacle is on the shortest path from the virtual robot to the waypoint as shown in Fig. 5.5. Here, θ denotes the line of sight (LOS) angle to the waypoint and γ represents the LOS angle to the obstacle. r_{ob} is the radius of the obstacle and l is the distance between the robot and the center of the obstacle as shown in the figure. Note that to avoid collision by the whole formation system, the effective radius of the obstacle, r_o , is considered as

$$r_o = \begin{cases} r_{ob} + r_f, & \text{if } r_{ob} \geq r_{min} \\ r_{min} + r_f, & \text{if } r_{ob} < r_{min} \end{cases}$$

where, r_{min} is the minimum radius of curvature of the vehicle and r_f is the radius of the desired formation circle.

The objective is to guide the virtual robot to the waypoint by avoiding the obstacle in an efficient way. The dynamics of the distance, l , is

$$\dot{l} = -v_o \cos(\theta - \gamma) \quad (5.40)$$

Any trajectory from the position of the virtual robot to the waypoint satisfying $\dot{l}$ as positive, will be a safe trajectory. If $\dot{l}$ is negative, and the virtual vehicle within the collision region then it should be deviated from its original LOS and a new LOS angle, θ' will be adopted by the virtual vehicle as per the following equation:

$$\theta' = \theta + \frac{\pi}{2} W \operatorname{sech}(k_3(\theta - \gamma)) \operatorname{sech}(k_4(l - r_o)) \quad (5.41)$$

where,

$$W = \begin{cases} \operatorname{sign}(\theta - \gamma), & \text{if } \theta - \gamma \neq 0 \\ 1, & \text{if } \theta - \gamma = 0 \end{cases}$$

Note that θ' will be adopted by the virtual vehicle only when the obstacle is in the range of the robot. The scaling term $\operatorname{sech}(k_3(\theta - \gamma))$ offers a high value when the obstacle is on the path of the robot and it offers a low value when the obstacle is not on the path. Similarly, the scaling term $\operatorname{sech}(k_4(l - r_o))$ offers a high value when the distance between the robot and the obstacle is low. The gains k_3 and k_4 are positive, and they should be selected suitably to generate an optimal and safe trajectory.

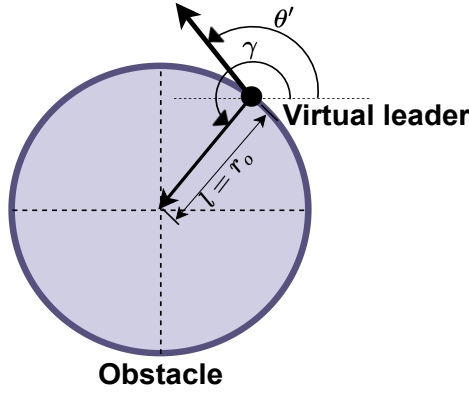


Figure 5.6: On the obstacle boundary

For setting k_3 , assume the vehicle on the boundary of the obstacle as shown in Fig. 5.6. In this case, $l = r_o$ which means, $\operatorname{sech}(k_4(l - r_o)) = 1$. When the vehicle is on the boundary of the obstacle for successful obstacle avoidance the condition to be met is,

$$\begin{bmatrix} \cos \gamma & \sin \gamma \end{bmatrix} \cdot \begin{bmatrix} \cos \theta' \\ \sin \theta' \end{bmatrix} \leq 0 \quad (5.42)$$

$$\cos(\theta' - \gamma) \leq 0 \quad (5.43)$$

5.5 Obstacle Avoidance

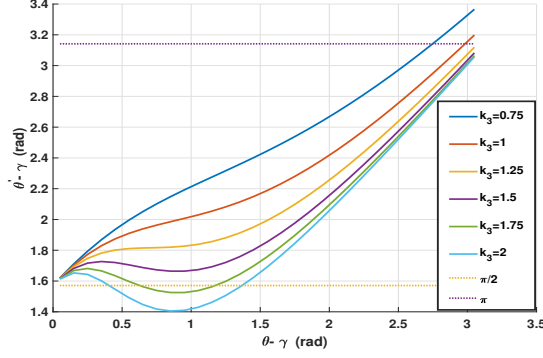


Figure 5.7: Angle difference between the proposed virtual LOS angle and LOS angle to the obstacle, $(\theta' - \gamma)$ vs. Angle difference between the virtual LOS angle and LOS angle to the obstacle, $(\theta - \gamma)$, for different values of k_3 .

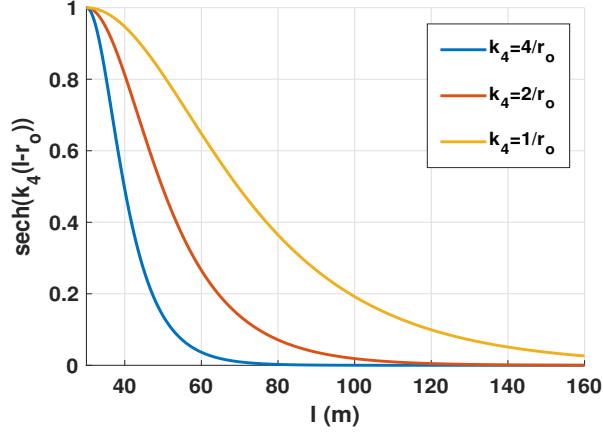


Figure 5.8: Profile of $\text{sech}(k_4(l - r_o))$ vs. l , for $k_4 = 4/r_o$, $2/r_o$ and $1/r_o$ ($r_o = 30$ m)

It implies,

$$\pi/2 \leq |(\theta' - \gamma)| \leq \pi \quad (5.44)$$

or

$$\pi/2 \leq |((\theta - \gamma) + \frac{\pi}{2} \text{sech}(k_3(\theta - \gamma)))| \leq \pi \quad (5.45)$$

For different values of k_3 , the plot of $((\theta - \gamma) + \frac{\pi}{2} \text{sech}(k_3(\theta - \gamma)))$ vs. $(\theta - \gamma)$ is shown in Fig. 5.7. Moreover, from this figure, we can observe that $k_3 = 1.5$ satisfies the boundary condition (5.45) and hence $k_3 = 1.5$ is the best possible value for successful obstacle avoidance.

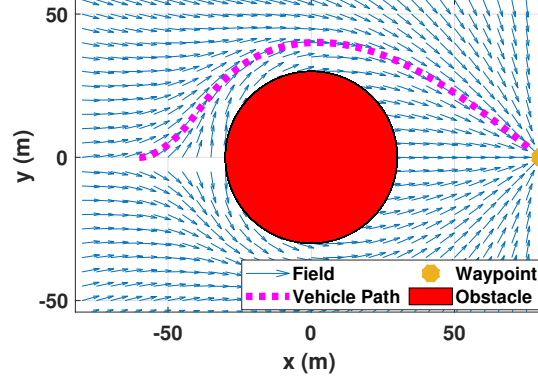
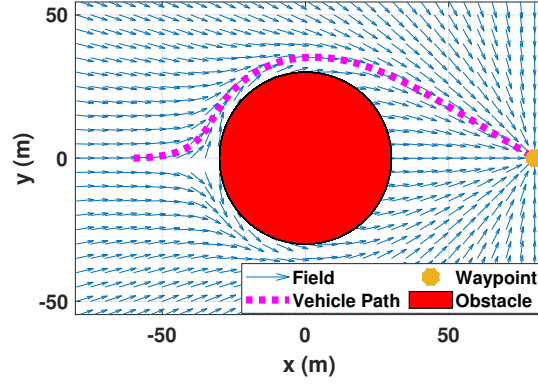

 (a) Vehicle trajectory for $k_4 = 2/r_o$

 (b) Vehicle trajectory for $k_4 = 4/r_o$

Figure 5.9: Obstacle avoidance

For setting the gain k_4 , consider the condition when $\theta - \gamma$ is zero, which implies that the virtual vehicle, the obstacle, and the waypoint are in the same line. So the vehicle must be diverted from its original path by following a smooth path to avoid the obstacle and the smoothness is dictated by the gain, k_4 . Fig. 5.8 shows the variation of the function, $\text{sech}(k_4(l - r_o))$ with respect to the distance between the virtual robot and the center of the obstacle (l). From this figure, it can be noted that for $k_4 = 4/r_o$, approximately 95% of change in the final value (final value is 1) occurs between r_o and $2r_o$. For $k_4 = 2/r_o$, approximately 95% change in the final value occurs between r_o and $3r_o$. Similarly, for $k_4 = 1/r_o$,

5.5 Obstacle Avoidance

approximately 95% change in the final value occurs between r_o and $5r_o$. As the value of k_4 decreases, the deviation from the trajectory increases. On the other hand, increasing the value of k_4 decreases the deviation from the trajectory, but the resultant trajectory may not be feasible. Therefore, selecting the appropriate value of k_4 requires a tradeoff between optimality and feasibility. The trajectory of the virtual vehicle for $k_4 = \frac{2}{r_o}$ and $k_3 = 1.5$ is shown in Fig. 5.9a, while the trajectory for $k_4 = \frac{4}{r_o}$ and $k_3 = 1.5$ is shown in Fig. 5.9b. In this example, the linear speed of the vehicle is 10 m/sec, the minimum radius of curvature of the vehicle 10 m and $r_o = 30$ m. From Fig. 5.9, one can say that change in the LOS angle has no effect once it surpasses the obstacle.

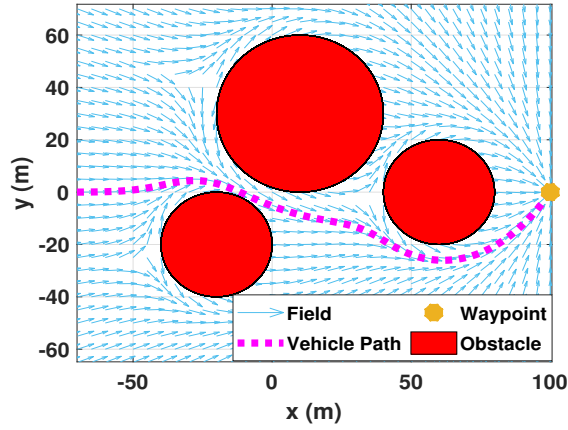


Figure 5.10: Obstacle avoidance

For the multiple obstacle case, the virtual LOS angle is

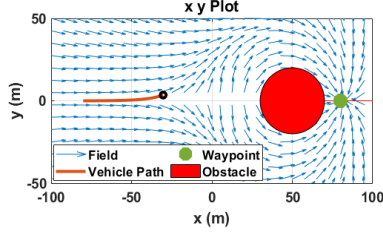
$$\theta' = \theta + \sum_{j=1}^m \frac{\pi}{2} W_j \text{sech}(k_3(\theta - \gamma_j)) \text{sech}(k_4(l_j - r_{oj})) \quad (5.46)$$

where, m is the number of obstacles. γ_j and l_j are the LOS angle and the distance of the virtual leader to the j^{th} obstacle, respectively.

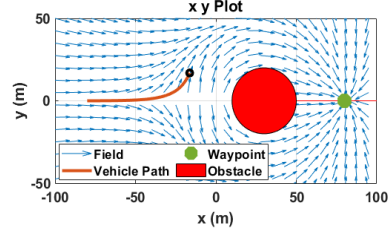
Fig. 5.10 illustrates the sample guiding vector field for the multiple obstacle avoidance case using the proposed virtual LOS angle (5.46). Furthermore, the figure includes the resulting trajectory of the vehicle, which has a minimum radius of curvature of 10 m. It is worth noting that in cases where two obstacles overlap,

they should be treated as a single obstacle.

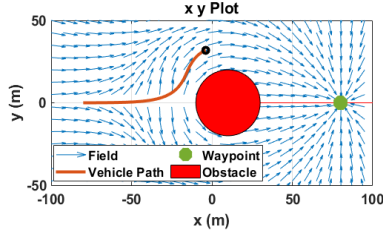
5.5.2 Moving Obstacle Avoidance



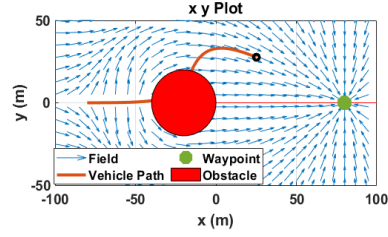
(a) Vehicle trajectory at $t = 5 \text{ sec}$



(b) Vehicle trajectory at $t = 7 \text{ sec}$



(c) Vehicle trajectory at $t = 9 \text{ sec}$



(d) Vehicle trajectory at $t = 12 \text{ sec}$

Figure 5.11: Vehicle trajectories for moving obstacle avoidance

The literature on moving obstacle avoidance algorithms suggests that a moving obstacle can be treated as a static obstacle by assuming that the vehicle moves with a relative velocity vector rather than its original velocity vector. The relative velocity vector is defined as follows:

$$v_{or} = \begin{bmatrix} v_o \cos \theta - v_{ob} \cos \psi_{ob} \\ v_o \sin \theta - v_{ob} \sin \psi_{ob} \end{bmatrix} \quad (5.47)$$

Here, v_{ob} represents the speed of the obstacle and ψ_{ob} denotes the angle made by the directional vector of the obstacle. For the moving obstacle case, we propose a virtual LOS angle given by the equation:

5.6 Waypoint Following

$$\theta' = \theta + \frac{\pi}{2} W \operatorname{sech}(k_3(\theta_{ov} - \gamma)) \operatorname{sech}(k_4(l - r_{om})) \quad (5.48)$$

where θ_{ov} is the angle made by the relative velocity vector and it is defined as

$$\theta_{ov} = \arctan \left(\frac{v_o \sin \theta - v_{ob} \sin \psi_{ob}}{v_o \cos \theta - v_{ob} \cos \psi_{ob}} \right)$$

To account for the speed of the obstacle on the distance range profile, an additional safety radius is defined as follows:

$$r_{om} = r_o \left(1 + \frac{v_{ob}}{v_o} \right)$$

Here, we examine the trajectory of a virtual vehicle in a scenario where the virtual vehicle is travelling at a speed of 10 m/sec while an obstacle with $r_o = 20$ m is moving in the opposite direction at the same speed. To visualize the effectiveness of the proposed approach, we present the trajectories of the virtual vehicle in Fig. 5.11. As demonstrated in the figure, the virtual vehicle follows the waypoint avoiding the obstacle.

The advantage of this proposed approach over artificial force field methods is that there is no singularity in the proposed method. In artificial force field methods, a robot experiences the repulsive force from the obstacles and the attractive force from the goal point. At the points when both forces cancel each other, singularities occur in the vector field whereas in the proposed approach the magnitude of the velocity vector is a constant. So, the singularity problem is eliminated.

5.6 Waypoint Following

Assuming a stable formation of all the robots with the virtual robot, the waypoint following problem by the swarm can be considered as the waypoint following problem by the single virtual robot. This section presents a steering algorithm for the virtual robot to track the desired waypoints in a given sequence. The desired heading angle is the LOS angle (θ) which is obtained by joining the current position of the virtual robot to the desired waypoint. The control input that is angular speed (u_o) of the virtual vehicle can be generated by considering the first-order heading

loop, that is,

$$u_o = \begin{cases} \alpha(\theta - \chi_o) & \text{if } \|\alpha(\theta - \chi_o)\| < \omega_{max} \\ \text{sign}(\theta - \chi_o) \cdot \omega_{max} & \text{if } \|\alpha(\theta - \chi_o)\| \geq \omega_{max} \end{cases} \quad (5.49)$$

In this work, sequential waypoint following is executed by making the next waypoint as the desired waypoint when the distance between the present waypoint and the virtual vehicle is less than the prefixed threshold value, δ .

Please make sure to include the necessary packages (algorithm and algorithmic) and the document class (article) at the beginning of your document. This code should compile without any errors and generate the desired algorithm layout.

5.7 Comparative Analysis

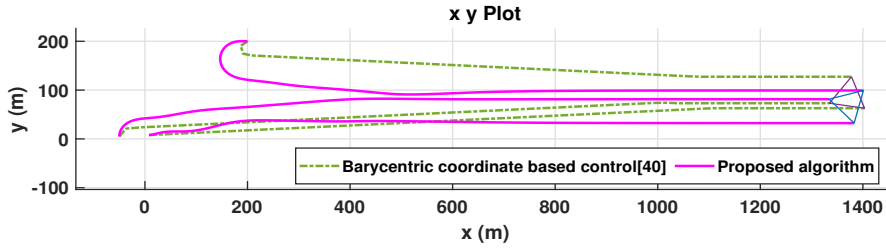


Figure 5.12: **Case 1:** Three vehicles in a triangular formation of $r_f = 40 \text{ m}$

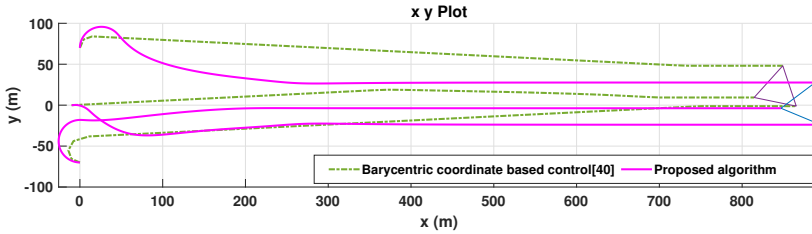


Figure 5.13: **Case 2:** Three vehicles in a triangular formation of $r_f = 30 \text{ m}$

5.7 Comparative Analysis

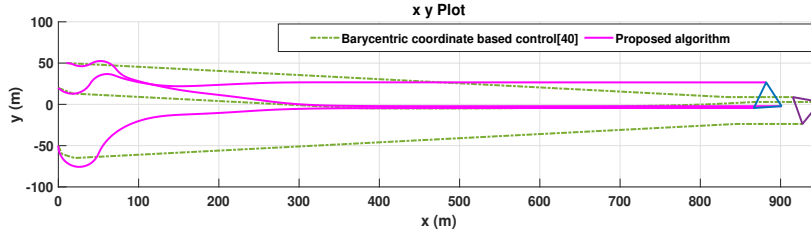


Figure 5.14: **Case 3:** Three vehicles in a triangular formation of $r_f = 20\text{ m}$

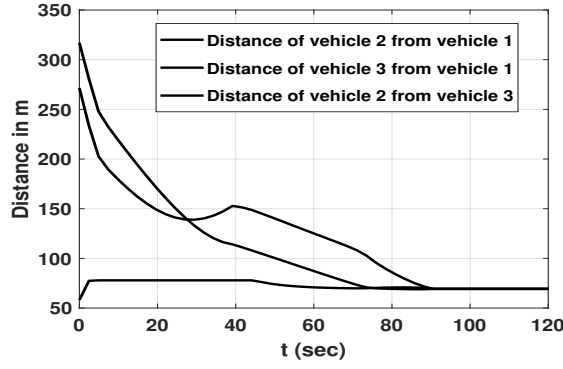


Figure 5.15: **Case 1:** Distance profiles of all the vehicles from their neighboring vehicles for Barycentric coordinate based control algorithm [40]

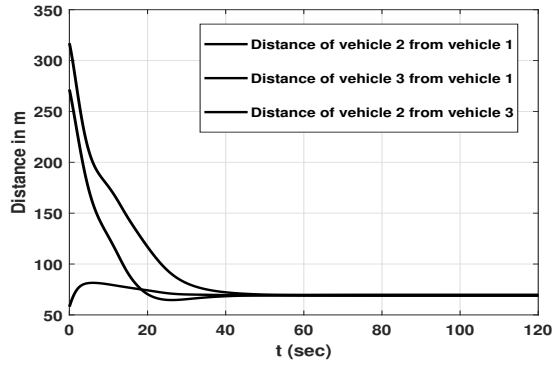


Figure 5.16: **Case 1:** Distance profiles of all the vehicles from their neighboring vehicles for the proposed algorithm

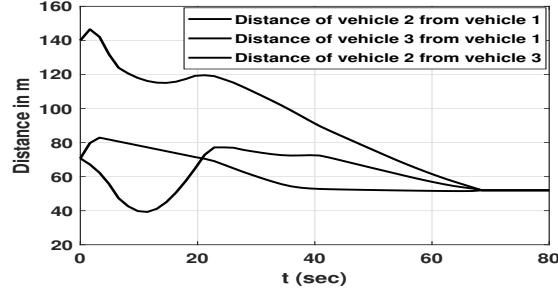


Figure 5.17: **Case 2:** Distance profiles of all the vehicles from their neighboring vehicles for Barycentric coordinate based control algorithm [40]

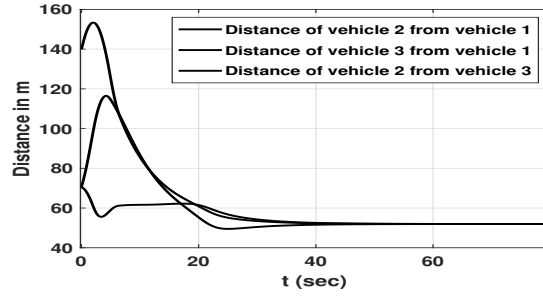


Figure 5.18: **Case 2:** Distance profiles of all the vehicles from their neighboring vehicles for the proposed algorithm

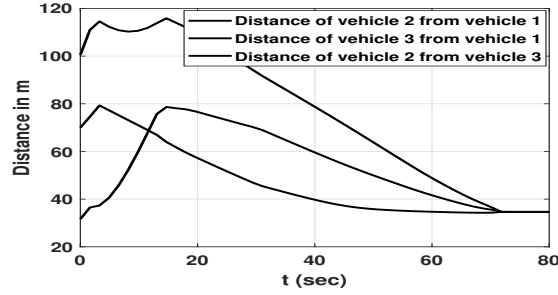


Figure 5.19: **Case 3:** Distance profiles of all the vehicles from their neighboring vehicles for Barycentric coordinate based control algorithm [40]

In this section the proposed algorithm is compared with the Barycentric coordinate based (BCB) control algorithm [92]. Here the objective is, all the vehicles should form an equilateral triangle and the formation should be heading towards $+x$ -direction.

In the BCB approach, the control vector for each of the vehicles is generated

5.7 Comparative Analysis

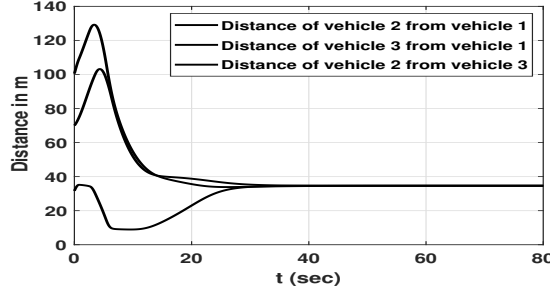


Figure 5.20: **Case 3:** Distance profiles of all the vehicles from their neighboring vehicles for the proposed algorithm

using the relative position vectors of the neighboring vehicles as given below:

$$\mathbf{U}_i = \sum_{j \in N_i} A_{ij} \begin{bmatrix} x_j - x_i \\ y_j - y_i \end{bmatrix} \quad (5.50)$$

where, N_i is the neighbour set of i^{th} vehicle. A_{ij} is the 2×2 control gain matrix. A_{ij} is designed by solving an optimization algorithm considering the formation problem as a semi definite programming problem. More detailed description on the BCB control algorithm can be found in [92].

From the control vector (5.50), the linear and angular speed commands are generated by projecting the control vector along the direction of heading and perpendicular to the direction of heading, respectively. That is,

$$v_i = \begin{bmatrix} \cos \chi_i & \sin \chi_i \end{bmatrix} (\mathbf{U}_i + c\mathbf{P}) \quad (5.51)$$

$$u_i = \begin{bmatrix} -\sin \chi_i & \cos \chi_i \end{bmatrix} (\mathbf{U}_i + c\mathbf{P}) \quad (5.52)$$

where, c is the positive gain value and $\mathbf{P}$ is the desired heading direction vector.

The proposed algorithms and the BCB algorithms have been simulated for three cases with the following input data: the linear speed and the angular speed bounds are $[7 \ 13]$ m/sec and $[-0.5 \ 0.5]$ rad/sec, respectively. The desired heading direction is $\mathbf{P} = [1 \ 0]^T$.

Case 1: Initial positions of three vehicles are $(200, 200)$, $(8, 8)$ and $(-50, 5)$ in meters with heading angles π , 0 and $\pi/2$ rad, respectively. The value of r_f is

considered as 40 m. From Fig. 5.12, it is evident that in both approaches, all the vehicles converge to the desired formation in finite time. Fig. 5.15 and Fig. 5.16 show convergence time profiles for both the approaches. In this case, the convergence time for the BCB approach is 84 sec, and in the proposed approach, it is only 43 sec.

Case 2: Initial positions of three vehicles are $(0, -70)$, $(-10, 0)$ and $(0, 70)$ in meters with heading angles π , 0 and $\pi/2$ rad, respectively. The value of r_f is considered as 30 m. In this case, the convergence time for the BCB approach is 68 sec, and in the proposed approach, it is only 45 sec.

Case 3: Initial positions of three vehicles are $(10, 50)$, $(0, 20)$ and $(0, -50)$ in meters with heading angles 0, $-\pi/4$ and $-\pi/2$ rad, respectively. The value of r_f is considered as 20 m. In this case, the convergence time for the BCB approach is 70 sec, and that of the proposed approach is only 40 sec.

From this comparative study, it can be observed that in both the approaches all the vehicles converge to the desired formation in finite time. The proposed approach shows faster convergence to the formation circle, especially when the initial positions of three vehicles are far from the desired formation distance.

5.8 Simulation Results

This section presents simulation results for waypoint following in a dynamic environment to validate the proposed vector field guidance algorithm. Following data are considered for the simulation. The desired sequence of waypoints are $(200, -400)$, $(-500, -100)$, $(-200, -200)$, $(200, 200)$, $(400, 100)$ and $(800, -240)$ m. In this simulation, all the obstacles are assumed to be circular in shape. The static obstacles are located at $(-200, 0)$, $(50, -250)$, $(0, -300)$, $(-250, 400)$, $(700, 150)$, $(100, 300)$, $(350, -300)$ and $(460, 200)$ m with radii of 100, 50, 50, 75, 50, 60, 40, 50, 50, and 40 m, respectively. And two dynamic obstacles are moving from their respective initial positions, i.e. $(300, 200)$ m and $(-420, 400)$ m with velocity vectors of $[-2 \ 0]$ and $[0 \ -2.5]$ m/sec, respectively.

Three robots are considered to follow these waypoints. The robots are with the maximum linear speed limit of 12 m/sec and the maximum and the minimum angular speeds of 0.5 rad/sec and -0.5 rad/sec, respectively. The virtual robot is

5.8 Simulation Results

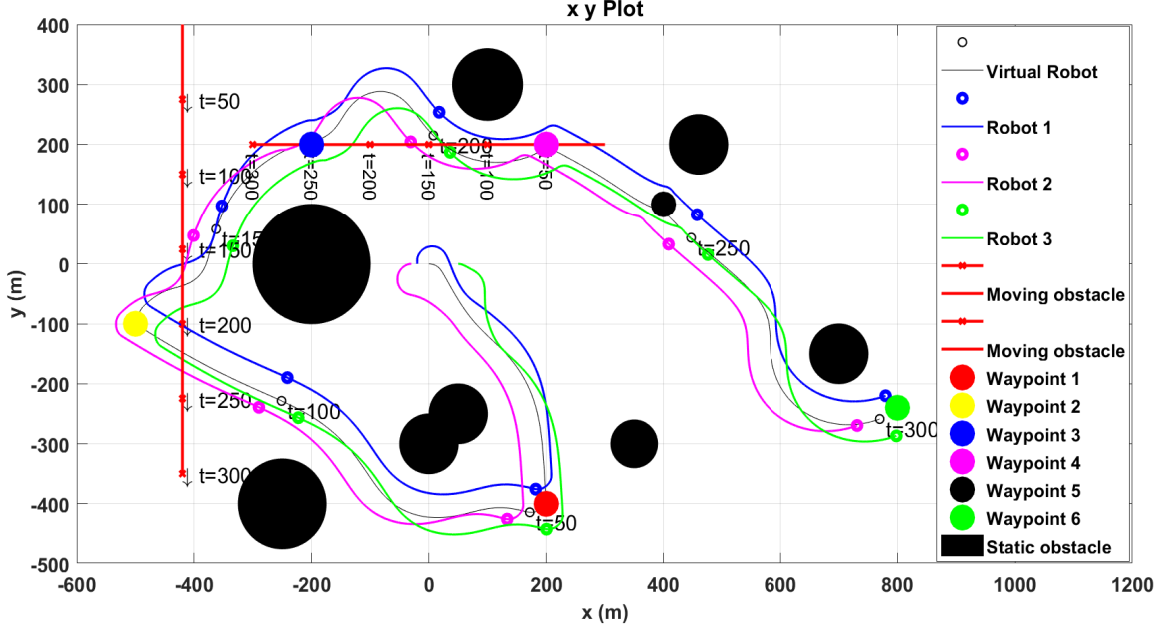


Figure 5.21: Waypoint following by the formation of three non-holonomic robots in an obstacle filled environment. Snapshots of each robot and moving obstacles at 7 instants are shown with time gap of 50 sec

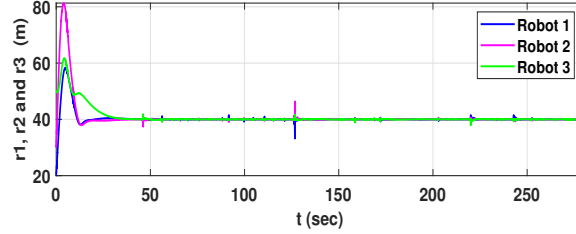


Figure 5.22: Distances of all robots from the virtual robot

considered as a constant speed vehicle and it has the angular speed limit which is the same as that of the actual robots. The virtual robot follows the desired waypoints in the given order from the initial position, $(0, 0)$ m. The control parameters used for the numerical simulation are given in Table 5.1. Three actual robots are making formation with the virtual robot from initial positions, $(-20, 0)$, $(-30, 0)$ and $(50, 0)$ m, respectively. The desired formation distance of each robot from the virtual robot is 30 m. Fig. 5.21 shows the trajectories of all three actual robots and the virtual robot. It shows the status of the position

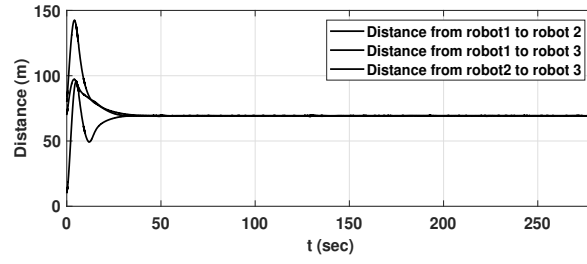
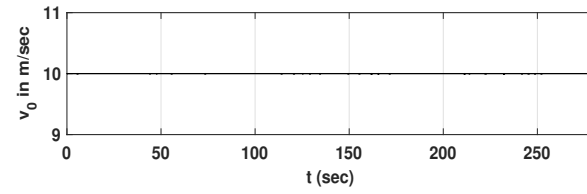
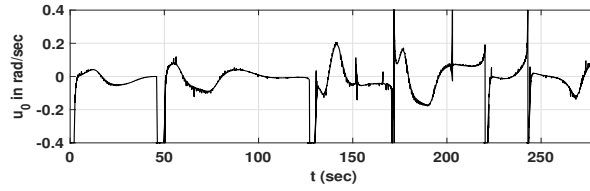


Figure 5.23: Distances of all robots from the other robots

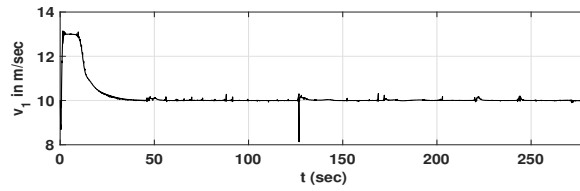


(a) Linear speed in m/sec

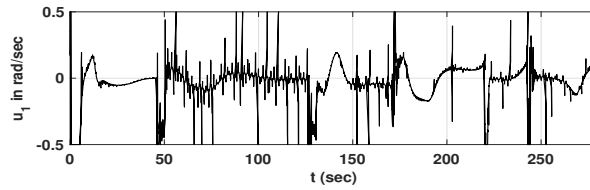


(b) Angular speed in rad/sec

Figure 5.24: Control inputs of virtual robot



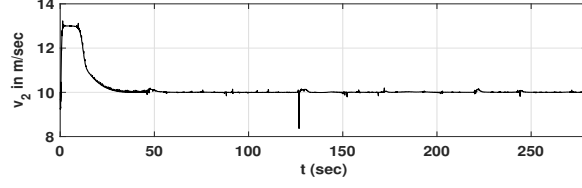
(a) Linear speed in m/sec



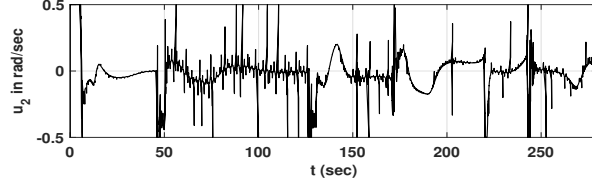
(b) Angular speed in rad/sec

Figure 5.25: Control inputs of robot 1

5.8 Simulation Results

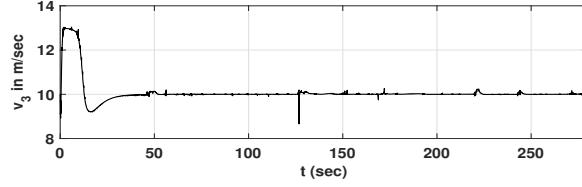


(a) Linear speed in m/sec

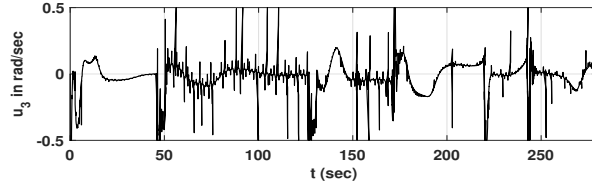


(b) Angular speed in rad/sec

Figure 5.26: Control inputs of robot 2



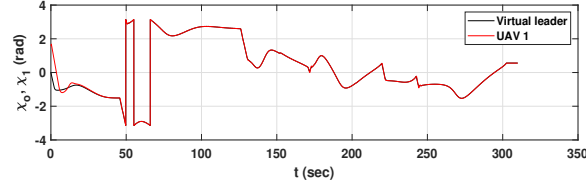
(a) Linear speed in m/sec



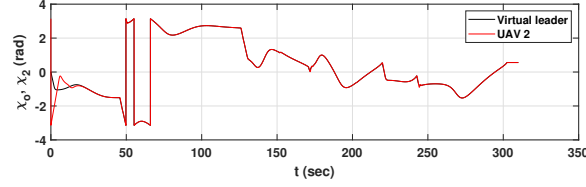
(b) Angular speed in rad/sec

Figure 5.27: Control inputs of robot 3

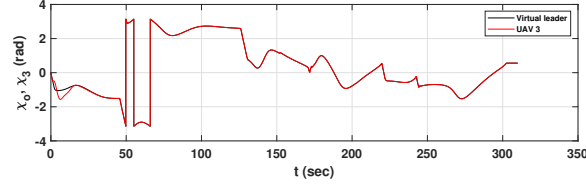
of each robot and the position of moving obstacles at seven regular intervals of time. Moreover, it shows that all the robots follow the waypoints while avoiding obstacles. The profiles of distance of each vehicle from the virtual robot are depicted in Fig. 5.22. From this figure one can see that all three robots converge to a formation circle with bounded errors. From Fig. 5.22 and Fig. 5.23, we can also notice that all the vehicles maintain an almost constant distance from the virtual robot, and the neighbor vehicles, which means vehicles are not colliding with each other throughout the operation. Fig. 5.24, Fig. 5.25, Fig. 5.26 and



(a) Heading angles in rad



(b) Heading angles in rad



(c) Heading angles in rad

Figure 5.28: Heading angle comparison

Fig. 5.27 depict linear and angular speed profiles of the virtual robot and the actual robots respectively. It can be observed from these plots that all the control inputs are within the specified bounds. In Fig. 5.28, the heading angles of all the robots are compared with the heading angle of the virtual robot and it shows that the heading angles of all the vehicles are approximately the same throughout the mission.

Table 5.1: Control parameters

Parameter	Value	Parameter	Value
r_{min}	20 m	k_1	0.25
β_1	2	k_3	1.5
β_2	2		
k_4	$2/r_o$		

5.9 Summary

This work presents a novel vector field guidance algorithm without any singularity for a waypoint following problem by a swarm/group of robots with nonholonomic and safety constraints, in an environment filled with both static and dynamic obstacles. The magnitude of the velocity in the guided vectors is independent of the position or size of the obstacles. From any initial poses the balanced formation is achieved with asymptotic convergence. The proposed algorithm has also been compared with the Barycentric coordinate-based control approach (BCB) for a three-vehicle system to illustrate the efficacy of the proposed approach. This work can further be extended to three-dimensional waypoint following problems by n number of Unmanned Aerial Vehicles (UAVs).

Conclusion

The thesis concludes as follows:

1. A fast convergence C^2 continuous paths have been developed using LVGF approach for 2D waypoint following. Then the algorithm is successfully extended to 3D waypoint following scenarios. Experiments have been conducted on a Quadrotor platform to explore a given series of waypoints to establish its efficacy.
2. A novel guidance algorithm for UAVs to track standoff targets has been proposed, utilizing the traditional vector field guidance framework. The practical realizability of the algorithm on microcontrollers has been demonstrated by testing it on a wheeled mobile robot.
3. A singularity-free novel algorithm has also been proposed for waypoint following by a swarm of nonholonomic vehicles with safety constraints in an environment filled with static and dynamic obstacles.

6.1 Future Scopes

The possible directions of future work are given below:

1. The LGVF approach can further be extended to handle more complex situations, such as navigating through strong winds while following 3D waypoints.
2. The obstacle avoidance algorithm proposed in this study could be expanded to handle different shapes of obstacles.
3. Additionally, the formation control algorithm could be extended to accommodate different topologies of UAVs, which would be an interesting area of future research.

References

- [1] M. Erdelj, E. Natalizio, K. R. Chowdhury, and I. F. Akyildiz, “Help from the sky: Leveraging uavs for disaster management,” *IEEE Pervasive Computing*, vol. 16, no. 1, pp. 24–32, 2017.
- [2] E. Bahja, P. R. Hersh, and R. Kinicki, “Forest fire risk reduction in albania: Using drone systems to improve current methods of fire,” 2019.
- [3] S. J. Kim and G. J. Lim, “Drone-aided border surveillance with an electrification line battery charging system,” *Journal of Intelligent & Robotic Systems*, vol. 92, no. 3-4, pp. 657–670, 2018.
- [4] Z. Zhou, C. Zhang, C. Xu, F. Xiong, Y. Zhang, and T. Umer, “Energy-efficient industrial internet of uavs for power line inspection in smart grid,” *IEEE Transactions on Industrial Informatics*, vol. 14, no. 6, pp. 2705–2714, 2018.
- [5] G. Hoareau, J. J. Liebenberg, J. G. Musial, and T. R. Whitman, “Package transport by unmanned aerial vehicles,” Aug. 15 2017, uS Patent 9,731,821.
- [6] F. Garcia-Ruiz, S. Sankaran, J. M. Maja, W. S. Lee, J. Rasmussen, and R. Ehsani, “Comparison of two aerial imaging platforms for identification of Huanglongbing-infected citrus trees,” *Computers and Electronics in Agriculture*, vol. 91, pp. 106–115, 2013.
- [7] S. Hota and D. Ghose, “Waypoint-based trajectory planning of fixed-wing mavs in 3d space,” *Journal of Intelligent & Robotic Systems*, vol. 86, no. 1, pp. 95–113, 2017.
- [8] L. E. Dubins, “On curves of minimal length with a constraint on average curvature, and with prescribed initial and terminal positions and tangents,” *American Journal of mathematics*, vol. 79, no. 3, pp. 497–516, 1957.
- [9] S. M. LaValle, *Planning algorithms*. Cambridge university press, 2006.
- [10] P. Sujit, S. Saripalli, and J. B. Sousa, “Unmanned aerial vehicle path following: A survey and analysis of algorithms for fixed-wing unmanned aerial vehicles,” *IEEE Control Systems Magazine*, vol. 34, no. 1, pp. 42–59, 2014.
- [11] R. Rysdyk, “Uav path following for constant line-of-sight,” in *2nd AIAA "Unmanned Unlimited" Conf. and Workshop & Exhibit, San Diego, California*, 2003, p. 6626.
- [12] S. Park, J. Deyst, and J. P. How, “Performance and lyapunov stability of a nonlinear path following guidance method,” *Journal of guidance, control, and dynamics*, vol. 30, no. 6, pp. 1718–1728, 2007.
- [13] A. Ratnoo, P. Sujit, and M. Kothari, “Adaptive optimal path following for high wind flights,” *IFAC Proceedings Volumes*, vol. 44, no. 1, pp. 12 985–12 990, 2011.

-
- [14] J. Osborne and R. Rysdyk, "Waypoint guidance for small uavs in wind," in *Infotech@Aerospace, Arlington, Virginia*, 2005, p. 6951.
 - [15] M. Kothari, I. Postlethwaite, and D.-W. Gu, "Uav path following in windy urban environments," *Journal of Intelligent & Robotic Systems*, vol. 74, no. 3-4, pp. 1013–1028, 2014.
 - [16] T. S. Andersen and R. Kristiansen, "Quaternion path-following in three dimensions for a fixed-wing uav using quaternion blending," in *2018 IEEE Conference on Control Technology and Applications (CCTA), Copenhagen, Denmark*. IEEE, 2018, pp. 1597–1602.
 - [17] K. Tanaka, M. Tanaka, Y. Takahashi, A. Iwase, and H. O. Wang, "3-d flight path tracking control for unmanned aerial vehicles under wind environments," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 12, pp. 11 621–11 634, 2019.
 - [18] D. Lawrence, "Lyapunov vector fields for uav flock coordination," in *2nd AIAA "Unmanned Unlimited" Conf. and Workshop & Exhibit*, 2003, p. 6575.
 - [19] T. Harinarayana, S. Hota, and R. Kushwaha, "Vector field guidance for standoff target tracking," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, vol. 236, no. 14, pp. 2963–2973, 2022.
 - [20] A. A. Pothén and A. Ratnoo, "Curvature-constrained lyapunov vector field for standoff target tracking," *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 10, pp. 2729–2736, 2017.
 - [21] D. Lawrence, E. Frew, and W. Pisano, "Lyapunov vector fields for autonomous uav flight control," in *AIAA Guidance, Navigation and Control Conference and Exhibit*, 2007, p. 6317.
 - [22] V. M. Goncalves, L. C. Pimenta, C. A. Maia, B. C. Dutra, and G. A. Pereira, "Vector fields for robot navigation along time-varying curves in n -dimensions," *IEEE Transactions on Robotics*, vol. 26, no. 4, pp. 647–659, 2010.
 - [23] Y. A. Kapitanyuk, A. V. Proskurnikov, and M. Cao, "A guiding vector-field algorithm for path-following control of nonholonomic mobile robots," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 4, pp. 1372–1385, 2017.
 - [24] W. Yao, H. G. de Marina, B. Lin, and M. Cao, "Singularity-free guiding vector field for robot navigation," *IEEE Transactions on Robotics*, vol. 37, no. 4, pp. 1206–1221, 2021.
 - [25] W. Yao, B. Lin, B. D. Anderson, and M. Cao, "Guiding vector fields for following occluded paths," *IEEE Transactions on Automatic Control*, 2022.
 - [26] D. R. Nelson, D. B. Barber, T. W. McLain, and R. W. Beard, "Vector field path following for miniature air vehicles," *IEEE Transactions on Robotics*, vol. 23, no. 3, pp. 519–529, 2007.
 - [27] S. Griffiths, "Vector field approach for curved path following for miniature aerial vehicles," in *AIAA guidance, navigation, and control conference and exhibit, Keystone, colorado*, 2006, p. 6467.
 - [28] Y. Liang, Y. Jia, Z. Wang, and F. Matsuno, "Combined vector field approach for planar curved path following with fixed-wing uavs," in *2015 American Control Conference (ACC), Chicago, IL, USA*. IEEE, 2015, pp. 5980–5985.
 - [29] Y. Liang and Y. Jia, "Combined vector field approach for 2d and 3d arbitrary twice differentiable curved path following with constrained uavs," *Journal of Intelligent & Robotic Systems*, vol. 83, no. 1, pp. 133–160, 2016.

-
- [30] A. M. Rezende, V. M. Gonçalves, G. V. Raffo, and L. C. Pimenta, "Robust fixed-wing uav guidance with circulating artificial vector fields," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Madrid, Spain*. IEEE, 2018, pp. 5892–5899.
- [31] W. Yao and M. Cao, "Path following control in 3d for nonholonomic robots using a vector field," *arXiv preprint arXiv:1911.02304*, 2019.
- [32] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.
- [33] R. J. Szczerba, P. Galkowski, I. S. Glicktein, and N. Ternullo, "Robust algorithm for real-time route planning," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 36, no. 3, pp. 869–878, 2000.
- [34] D. M. Coleman and J. T. Wunderlich, "O3: An optimal and opportunistic path planner (with obstacle avoidance) using voronoi polygons," in *2008 10th IEEE International Workshop on Advanced Motion Control, Trento, Italy*. IEEE, 2008, pp. 371–376.
- [35] B. Banerjee and B. Chandrasekaran, "A framework of voronoi diagram for planning multiple paths in free space," *Journal of Experimental & Theoretical Artificial Intelligence*, vol. 25, no. 4, pp. 457–475, 2013.
- [36] A. Richards and J. P. How, "Aircraft trajectory planning with collision avoidance using mixed integer linear programming," in *American Control Conference, Anchorage, AK, USA*, vol. 3. IEEE, 2002, pp. 1936–1941.
- [37] J. Chen, F. Ye, and T. Jiang, "Path planning under obstacle-avoidance constraints based on ant colony optimization algorithm," in *2017 IEEE 17th International Conference on Communication Technology (ICCT), Chengdu, China*. IEEE, 2017, pp. 1434–1438.
- [38] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," in *Autonomous robot vehicles*. Springer, 1986, pp. 396–404.
- [39] J.-O. Kim and P. Khosla, "Real-time obstacle avoidance using harmonic potential functions," 1992.
- [40] S. R. Munasinghe, C. Oh, J.-J. Lee, and O. Khatib, "Obstacle avoidance using velocity dipole field method," *ICCAS*, pp. 1657–1661, 2005.
- [41] Z. Li and T. Bui, "Robot path planning using fluid model," *Journal of Intelligent and Robotic Systems*, vol. 21, no. 1, pp. 29–50, 1998.
- [42] C. I. Connolly, J. B. Burns, and R. Weiss, "Path planning using laplace's equation," in *Proceedings., IEEE International Conference on Robotics and Automation, Cincinnati, OH, USA*. IEEE, 1990, pp. 2102–2106.
- [43] S. Waydo and R. M. Murray, "Vehicle motion planning using stream functions," in *2003 IEEE International Conference on Robotics and Automation (Cat. No. 03CH37422)*, vol. 2. IEEE, 2003, pp. 2484–2491.
- [44] P. Yao, H. Wang, and Z. Su, "Uav feasible path planning based on disturbed fluid and trajectory propagation," *Chinese Journal of Aeronautics*, vol. 28, no. 4, pp. 1163–1177, 2015.

-
- [45] J. P. Wilhelm and G. Clem, "Vector field uav guidance for path following and obstacle avoidance with minimal deviation," *Journal of Guidance, Control, and Dynamics*, vol. 42, no. 8, pp. 1848–1856, 2019.
 - [46] Y. Zhu, T. Moleski, and J. Wilhelm, "Evaluation of decay functions for vector field based obstacle avoidance," in *AIAA Scitech 2019 Forum, Athens, Ohio*, 2019, p. 1058.
 - [47] A. Marchidan and E. Bakolas, "Collision avoidance for an unmanned aerial vehicle in the presence of static and moving obstacles," *Journal of Guidance, Control, and Dynamics*, vol. 43, no. 1, pp. 96–110, 2020.
 - [48] M. Braquet and E. Bakolas, "Vector field-based collision avoidance for moving obstacles with time-varying elliptical shape," *IFAC-PapersOnLine, Austin, Texas*, vol. 55, no. 37, pp. 587–592, 2022.
 - [49] R. Rysdyk, "Unmanned aerial vehicle path following for target observation in wind," *Journal of guidance, control, and dynamics*, vol. 29, no. 5, pp. 1092–1100, 2006.
 - [50] M. Quigley, M. A. Goodrich, S. Griffiths, A. Eldredge, and R. W. Beard, "Target acquisition, localization, and surveillance using a fixed-wing mini-uav and gimbaled camera," in *Proceedings of the 2005 IEEE international conference on robotics and automation, Barcelona, Spain*. IEEE, 2005, pp. 2600–2605.
 - [51] E. W. Frew, D. A. Lawrence, and S. Morris, "Coordinated standoff tracking of moving targets using lyapunov guidance vector fields," *Journal of guidance, control, and dynamics*, vol. 31, no. 2, pp. 290–306, 2008.
 - [52] H. Chen, K. Chang, and C. S. Agate, "Tracking with uav using tangent-plus-lyapunov vector field guidance," in *2009 12th International Conference on Information Fusion, Seattle, WA, USA*. IEEE, 2009, pp. 363–372.
 - [53] L. Ma, C. Cao, N. Hovakimyan, V. Dobrokhodov, and I. Kaminer, "Adaptive vision-based guidance law with guaranteed performance bounds," *Journal of guidance, control, and dynamics*, vol. 33, no. 3, pp. 834–852, 2010.
 - [54] S. Park, "Circling over a target with relative side bearing," *Journal of Guidance, Control, and Dynamics*, vol. 39, pp. 1454–1458, 2016.
 - [55] H. Oh, S. Kim, A. Tsourdos, and B. A. White, "Decentralised standoff tracking of moving targets using adaptive sliding mode control for uavs," *Journal of Intelligent & Robotic Systems*, vol. 76, no. 1, pp. 169–183, 2014.
 - [56] S. Sun, H. Wang, J. Liu, and Y. He, "Fast lyapunov vector field guidance for standoff target tracking based on offline search," *IEEE Access*, 2019.
 - [57] S. Hota and D. Ghose, "A modified dubins method for optimal path planning of a miniature air vehicle converging to a straight line path," in *2009 American Control Conference, St. Louis, MO, USA*. IEEE, 2009, pp. 2397–2402.
 - [58] A. Ranjan and S. Hota, "Time-optimal convergence of fixed-wing uavs to a circular path in steady uniform wind," *International Conference on Theoretical, Applied, Computational and Experimental Mechanics, Kharagpur*, 2017.
 - [59] L. Consolini, F. Morbidi, D. Prattichizzo, and M. Tosques, "Leader–follower formation control of nonholonomic mobile robots with input constraints," *Automatica*, vol. 44, no. 5, pp. 1343–1349, 2008.

-
- [60] B. Yun, B. M. Chen, K. Y. Lum, and T. H. Lee, "Design and implementation of a leader-follower cooperative control system for unmanned helicopters," *Journal of Control Theory and Applications*, vol. 8, no. 1, pp. 61–68, 2010.
 - [61] Q. Ouyang, Z. Wu, Y. Cong, and Z. Wang, "Formation control of unmanned aerial vehicle swarms: A comprehensive review," *Asian Journal of Control*, 2022.
 - [62] A. Gopalabhatla, T. Harinarayana, and S. Hota, "Lyapunov vector field based guidance algorithm for standoff target tracking by formation flight," in *2021 International Conference on Unmanned Aircraft Systems (ICUAS)*, Athens, Greece. IEEE, 2021, pp. 129–136.
 - [63] Z. Sun and Y. Xia, "Receding horizon tracking control of unicycle-type robots based on virtual structure," *International Journal of Robust and Nonlinear Control*, vol. 26, no. 17, pp. 3900–3918, 2016.
 - [64] H. G. Tanner, A. Jadbabaie, and G. J. Pappas, "Stable flocking of mobile agents, part i: Fixed topology," in *42nd IEEE International Conference on Decision and Control, Maui, HI, USA (IEEE Cat. No. 03CH37475)*, vol. 2. IEEE, 2003, pp. 2010–2015.
 - [65] T. Harinarayana and S. Hota, "Coordinated standoff target tracking by multiple uavs in obstacle-filled environments," in *2021 Seventh Indian Control Conference (ICC)*, Mumbai, India. IEEE, 2021, pp. 111–116.
 - [66] R. Dutta, L. Sun, and D. Pack, "A decentralized formation and network connectivity tracking controller for multiple unmanned systems," *IEEE Transactions on Control Systems Technology*, vol. 26, no. 6, pp. 2206–2213, 2017.
 - [67] R. Sharma, M. Kothari, C. N. Taylor, and I. Postlethwaite, "Cooperative target-capturing with inaccurate target information," in *Proceedings of the 2010 American Control Conference, Baltimore, MD, USA*. IEEE, 2010, pp. 5520–5525.
 - [68] H. Chen, X. Wang, L. Shen, Z. Li, Z. Liu, and Y. Yu, "Formation reconfiguration for fixed-wing uavs," *Journal of Intelligent & Robotic Systems*, vol. 102, no. 1, pp. 1–14, 2021.
 - [69] P. A. Sperandio Giacomini and E. M. Hemerly, "Reconfiguration between longitudinal and circular formations for multi-uav systems by using segments," *Journal of Intelligent & Robotic Systems*, vol. 78, no. 2, pp. 339–355, 2015.
 - [70] Z. Lin, L. Wang, Z. Han, and M. Fu, "A graph laplacian approach to coordinate-free formation stabilization for directed networks," *IEEE Transactions on Automatic Control*, vol. 61, no. 5, pp. 1269–1280, 2015.
 - [71] J. Seo, Y. Kim, S. Kim, and A. Tsourdos, "Collision avoidance strategies for unmanned aerial vehicles in formation flight," *IEEE Transactions on aerospace and electronic systems*, vol. 53, no. 6, pp. 2718–2734, 2017.
 - [72] T. S. Andersen, G. Bouzidi, and R. Kristiansen, "Curvature continuous path generation for uavs based on generalized expo-rational b-splines," in *2018 International Conference on Unmanned Aircraft Systems (ICUAS)*, Dallas, TX, USA. IEEE, 2018, pp. 715–720.
 - [73] N. K. Singh and S. Hota, "Three-dimensional waypoint following for fixed-wing unmanned aerial vehicles in obstacle-filled environments," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, vol. 234, no. 3, pp. 640–654, 2020.
 - [74] C.-H. Kim, K.-M. Jeong, and T.-w. Jeong, "Semi-autonomous navigation of an unmanned ground vehicle for bird expellant in an airport," in *2012 12th International Conference on Control, Automation and Systems*, Jeju, Korea (South). IEEE, 2012, pp. 2063–2067.

-
- [75] A. Ravankar, A. A. Ravankar, Y. Kobayashi, and T. Emaru, "Shp: smooth hypocycloidal paths with collision-free and decoupled multi-robot path planning," *International Journal of Advanced Robotic Systems*, vol. 13, no. 3, p. 133, 2016.
 - [76] A. A. Neto, D. G. Macharet, and M. F. Campos, "Feasible rrt-based path planning using seventh order bézier curves," in *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems, Taipei, Taiwan*. IEEE, 2010, pp. 1445–1450.
 - [77] T. Maekawa, T. Noda, S. Tamura, T. Ozaki, and K.-i. Machida, "Curvature continuous path generation for autonomous vehicle using b-spline curves," *Computer-Aided Design*, vol. 42, no. 4, pp. 350–359, 2010.
 - [78] H. Belaidi, A. Hentout, B. Bouzouia, H. Bentarzi, and A. Belaidi, "Nurbs trajectory generation and following by an autonomous mobile robot navigating in 3d environment," in *The 4th Annual IEEE International Conference on Cyber Technology in Automation, Control and Intelligent, Hong Kong, China*. IEEE, 2014, pp. 168–173.
 - [79] A. Lekkas, A. R. Dahl, M. Breivik, and T. I. Fossen, "Continuous-curvature path generation using fermat's spiral," *Modeling, Identification and Control*, vol. 33, no. 4, pp. 183–198, 2013.
 - [80] S. Zhao, D. V. Dimarogonas, Z. Sun, and D. Bauso, "A general approach to coordination control of mobile agents with motion constraints," *IEEE Transactions on Automatic Control*, vol. 63, no. 5, pp. 1509–1516, 2017.
 - [81] K. Fathian, T. H. Summers, and N. R. Gans, "Distributed formation control and navigation of fixed-wing uavs at constant altitude," in *2018 International Conference on Unmanned Aircraft Systems (ICUAS), Dallas, TX, USA*. IEEE, 2018, pp. 300–307.
 - [82] J. Wang and M. Xin, "Integrated optimal formation control of multiple unmanned aerial vehicles," *IEEE Transactions on Control Systems Technology*, vol. 21, no. 5, pp. 1731–1744, 2012.
 - [83] P. Yao, H. Wang, and Z. Su, "Cooperative path planning with applications to target tracking and obstacle avoidance for multi-uavs," *Aerospace science and technology*, vol. 54, pp. 10–22, 2016.
 - [84] S. Hota and D. Ghose, "Curvature-constrained trajectory generation for waypoint following for miniature air vehicle," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, vol. 228, no. 11, pp. 2066–2082, 2014.
 - [85] "Aviones," <http://aviones.sourceforge.net>, 2007.
 - [86] R. Beard and T. McLain, "Navigation, guidance, and control of small unmanned aircraft," *Guidance, and Control of Small Unmanned Aircraft, Brigham Young University*, 2009.
 - [87] S. Hota and D. Ghose, "Time-optimal convergence to a rectilinear path in the presence of wind," *Journal of Intelligent & Robotic Systems*, vol. 74, pp. 791–815, 2014.
 - [88] A. U. F. Simulator, *Software Package, MAGICC Lab, Brigham Young University*, 2007. [Online]. Available: <http://aviones.sourceforge.net>
 - [89] S. Hota and D. Ghose, "Optimal trajectory planning for unmanned aerial vehicles in three-dimensional space," *Journal of aircraft*, vol. 51, no. 2, pp. 681–688, 2014.
 - [90] —, "Time-optimal convergence to a rectilinear path in the presence of wind," *Journal of Intelligent and Robotic Systems*, vol. 74, no. 4, pp. 791–815, 2014.

-
- [91] *High-performance Robot for Indoor Labs*, 2018. [Online]. Available: <https://www.quanser.com/2017/04/QBot-2-for-QUARC-Datasheet>
- [92] K. Fathian, S. Safaoui, T. H. Summers, and N. R. Gans, “Robust distributed planar formation control for higher order holonomic and nonholonomic agents,” *IEEE Transactions on Robotics*, vol. 37, no. 1, pp. 185–205, 2020.

Publications from this Thesis

Journals

1. T. Harinarayana, S. Hota and R. Kushwaha, "Vector field guidance for stand-off target tracking," *Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering*, Vol. 236, No. 14, pp. 2963–2973, 2022.

Conferences

1. T. Harinarayana, S. V. Krishnan, S. Hota and R. Kushwaha "A Lyapunov Guidance Vector Field Based Continuous Curvature Path Generation for Waypoint Following of UAVs," In *2021 International Conference on Unmanned Aircraft Systems (ICUAS)*, 2021, pp. 498–506.
2. A. Gopalabhatla, T. Harinarayana and S. Hota, "Lyapunov Vector Field Based Guidance Algorithm for Standoff Target Tracking by Formation Flight," In *2021 International Conference on Unmanned Aircraft Systems (ICUAS)*, 2021, pp. 129–136.
3. T. Harinarayana and S. Hota, "Coordinated Standoff Target Tracking by Multiple UAVs in Obstacle-filled Environments," In *2021 Seventh Indian Control Conference (ICC)*, 2021, pp. 111–116.
4. T. Harinarayana, N. K. Singh and S. Hota, "Lyapunov vector field guidance for straight line path convergence," In *International Conference, Aeronautical Society of India 2020 on Aviation Technology - Current Scenario*, 2020.

Author's Biography

Tammineni Harinarayana is a Ph.D. candidate from Department of Aerospace Engineering, Indian Institute of Technology Kharagpur.

Contact Information

Permanent Address:

Jadupalli (Village),
Srikakulam (District),
PIN-532216,
India.

Mobile: +91-9491990653

Email: tammineni.harinarayana28@iitkgp.ac.in

Research Interests

Unmanned Vehicles, Path Planning, Guidance and Control, Optimization.